# Physiology MCQ Bank

1st Professional MBBS — Corrected Answers with Reasons

**How to read:** each True/False statement is marked **[ T ]** (true) or **[ F ]** (false). The italic line below gives the reason. For single-best-answer questions (Q11–20 in newer papers) the correct option is shown in green with the reason.

**†** after a question title means the original option wording or answer was faulty (typo, two correct options, or a wrong key) and has been corrected here.

Answers follow standard medical biochemistry textbook facts. Where a statement is debatable, the more accepted textbook position is used. Please verify against your own textbook/teacher if in doubt.

# November 2025 — Paper I

### Q1. Cell membrane †

a) Maintains the intracellular environment **[ T ]**
*Reason: Cell membrane separates ICF from ECF and controls what enters/leaves.*

b) Contains receptor proteins and ion channels **[ T ]**
*Reason: Integral proteins form receptors, channels, carriers and pumps.*

c) Separates intracellular fluid from extracellular fluid **[ T ]**
*Reason: Cell membrane is the boundary between ICF and ECF.*

d) Is composed of inorganic substances **[ F ]**
*Reason: Membrane is lipid (phospholipid, cholesterol), protein and carbohydrate — all organic.*

e) Is freely permeable to electrolytes **[ F ]**
*Reason: The lipid bilayer is impermeable to ions; they cross via channels/pumps.*

### Q2. Mitochondria

a) Have a membrane similar to the cell membrane **[ T ]**
*Reason: Mitochondria have an inner and outer lipid membrane.*

b) Are the chief site of protein synthesis **[ F ]**
*Reason: Ribosomes (rough ER) synthesise proteins; mitochondria make ATP.*

c) Are the chief site for ATP generation **[ T ]**
*Reason: Mitochondria generate most ATP by oxidative phosphorylation.*

d) Are absent near the membranes of actively secreting cells **[ F ]**
*Reason: Mitochondria are abundant where ATP is needed (secretory/transport cells).*

e) Have their own DNA **[ T ]**
*Reason: Mitochondria have their own circular DNA.*

### Q3. Negative feedback means

a) Center receiving negative information from periphery **[ F ]**
*Reason: Negative feedback: response opposes the initiating stimulus.*

b) Center sending negative signal to appropriate organs **[ T ]**

c) A reversible phenomenon **[ T ]**

d) Opposite to initiating stimuli **[ T ]**
*Reason: Negative feedback opposes the change.*

e) Acceleration of initiating stimuli **[ F ]**
*Reason: Acceleration of the stimulus is positive feedback.*

### Q4. $Na^+$-$K^+$ pump

a) Constant pumping activity is carried out in all cells **[ T ]**
*Reason: Pump works continuously in all cells.*

b) Cell volume is maintained **[ T ]**
*Reason: $Na^+$-$K^+$ pump prevents cell swelling.*

c) $Na^+$ is actively transported into the cell **[ F ]**
*Reason: $Na^+$-$K^+$ pump moves 3 $Na^+$ OUT and 2 $K^+$ IN.*

d) $K^+$ is actively transported out of the cell **[ F ]**
*Reason: $Na^+$-$K^+$ pump moves $K^+$ INTO the cell.*

e) Equal number of $Na^+$ and $K^+$ are transported **[ F ]**
*Reason: 3 $Na^+$ out for 2 $K^+$ in.*

### Q5. Enteric nervous system

a) Means the special nervous system for GIT **[ T ]**
*Reason: The ENS (myenteric + submucosal plexus) is the intrinsic gut nervous system.*

b) Causes GIT secretion **[ T ]**
*Reason: Submucosal plexus controls secretion and local blood flow.*

c) Helps in gut motility **[ T ]**
*Reason: Myenteric plexus controls motility.*

d) Is under autonomic control **[ F ]**
*Reason: ENS works independently; ANS only modulates it.*

e) Is stimulated by stretching of gut wall **[ T ]**
*Reason: Stretch triggers the peristaltic reflex via ENS.*

### Q6. Erythropoiesis is regulated by

a) Rate of erythrocyte destruction **[ F ]**
*Reason: EPO (driven by hypoxia) regulates erythropoiesis, not destruction rate.*

b) Erythropoietin **[ T ]**
*Reason: EPO from kidney stimulates RBC production.*

c) Vitamin $B_{12}$ **[ F ]**
*Reason: $B_{12}$ is needed for maturation (DNA synthesis), not regulation.*

d) Androgen **[ T ]**
*Reason: Androgens stimulate EPO and erythropoiesis.*

e) Blood pressure **[ F ]**
*Reason: Blood pressure does not regulate erythropoiesis.*

### Q7. Plasma albumin

a) Contributes 80% of plasma colloid osmotic pressure **[ T ]**
*Reason: Albumin contributes ~80% of plasma oncotic pressure.*

b) Is negatively charged at normal blood pH **[ T ]**
*Reason: Albumin is negatively charged at pH 7.4.*

c) Carries $CO_2$ in blood **[ T ]**
*Reason: Plasma proteins (carbamino) carry a small fraction of $CO_2$.*

d) Is filtered freely at renal glomerulus **[ F ]**
*Reason: Albumin is too large (69 kDa) and negatively charged → not filtered.*

e) Lacks essential amino acids **[ F ]**
*Reason: Albumin contains essential amino acids.*

### Q8. A person with group A (+ve) blood

a) Has anti-B antibody in the plasma **[ T ]**
*Reason: Group A has A antigen and anti-B antibody.*

b) May have the genotype AB **[ F ]**
*Reason: Group A genotype is AA or AO; AB is a different group.*

c) Can be a donor for a person with AB or A **[ T ]**
*Reason: Group A can donate to A and AB.*

d) Has A agglutinogen in RBC membrane **[ T ]**
*Reason: Group A RBCs carry A agglutinogen.*

e) Has D agglutinogen in RBC membrane **[ T ]**
*Reason: '+ve' means Rh D antigen present on RBC.*

### Q9. Platelets

a) Have a role in blood coagulation **[ T ]**
*Reason: Platelets form the plug and support coagulation.*

b) Liberate 5-HT **[ T ]**
*Reason: Dense granules release serotonin, ADP, $Ca^{2+}$.*

c) Form plug by sticking to each other **[ T ]**
*Reason: Platelet aggregation forms the platelet plug.*

d) Secrete ADP **[ T ]**
*Reason: Dense granules release ADP.*

e) Provide phospholipid for the intrinsic pathway **[ T ]**
*Reason: Platelet phospholipid (PF3) is needed for intrinsic/common pathways.*

### Q10. Tests for bleeding disorder are

a) Coagulation time **[ T ]**
*Reason: CT tests the intrinsic/common pathways.*

b) Bleeding time **[ T ]**
*Reason: BT tests platelet function.*

c) Partial thromboplastin time **[ T ]**
*Reason: PTT tests the intrinsic pathway.*

d) Platelet count **[ T ]**
*Reason: Low count (thrombocytopenia) causes bleeding.*

e) RBC count **[ F ]**
*Reason: RBC count is not a test for bleeding disorder.*

**Q11. Properties of cardiac muscle are all EXCEPT**

a) Refractory period
b) Conductivity
c) Auto-excitability
**d) Summation** ✓

**Answer: d) Summation**

*Reason: Cardiac muscle cannot show summation/tetanus because of the long refractory period.*

**Q12. Regarding SA node cells, all are true EXCEPT**

a) Innervated by the vagus nerve
**b) Found in both atria** ✓
c) The normal pacemaker of the heart
d) Inherently leaky to $Na^+$ ions

**Answer: b) Found in both atria**

*Reason: SA node lies in the right atrial wall at the SVC opening.*

**Q13. First heart sound is**

**a) Produced at the beginning of ventricular systole** ✓
b) Caused by opening of atrioventricular valves
c) Caused by closure of aortic and pulmonary valves
d) Produced at the beginning of ventricular diastole

**Answer: a) Produced at the beginning of ventricular systole**

*Reason: S1 is produced at the start of ventricular systole.*

**Q14. QRS complex is †**

**a) Due to atrial repolarization and ventricular depolarization** ✓
b) Due to atrial depolarization
c) 0.12 to 0.4 second in duration
d) Due to ventricular repolarization

**Answer: a) Due to atrial repolarization and ventricular depolarization**

*Reason: QRS = ventricular depolarisation (atrial repolarisation is hidden in it).*

**Q15. Regarding surfactant, all are true EXCEPT**

a) Is secreted by type-II pneumocytes
b) Increases the lung compliance
**c) Increases surface tension** ✓
d) Prevents collapse of the lungs

**Answer: c) Increases surface tension**

*Reason: Surfactant DEcreases surface tension.*

**Q16. Lung defense mechanisms include †**

**a) Macrophages** ✓
b) Serotonin
c) Negative intrapleural pressure
d) Pulmonary stretch receptors

**Answer: a) Macrophages**

*Reason: Alveolar macrophages phagocytose particles.*

**Q17. The oxy-Hb dissociation curve is shifted to the right by**

**a) Increased 2,3-DPG** ✓
b) Decreased $PCO_2$
c) Increased pH
d) Increased PO2

**Answer: a) Increased 2,3-DPG**

*Reason: ↑2,3-DPG, ↑$H^+$, ↑$PCO_2$, ↑temperature shift ODC right.*

**Q18. The brush border enzyme is**

**a) Aminopeptidase** ✓
b) Nuclease
c) Collagenase
d) Ribonuclease

**Answer: a) Aminopeptidase**

*Reason: Brush-border enzymes: aminopeptidase, dipeptidases, disaccharidases.*

**Q19. Intrinsic factor**

**a) Is secreted by oxyntic cells** ✓
b) Forms a protective barrier
c) Prevents digestion of stomach wall
d) Activates brush border enzymes

**Answer: a) Is secreted by oxyntic cells**

*Reason: Intrinsic factor is secreted by parietal (oxyntic) cells.*

**Q20. Regarding bile salts, all are true EXCEPT**

a) Are derived from cholesterol
b) Are essential for fat digestion
c) Form micelles
**d) Are absorbed mainly in the upper small gut** ✓

**Answer: d) Are absorbed mainly in the upper small gut**

*Reason: Bile salts are absorbed in the terminal ileum.*

# November 2025 — Paper II

### Q1. Aquaporins †

a) Are protein channels **[ T ]**
*Reason: Aquaporins are water-channel proteins.*

b) Are membrane lipids **[ F ]**
*Reason: Aquaporins are proteins, not lipids.*

c) Form water channels in renal tubules **[ T ]**
*Reason: AQP-1 (PCT), AQP-2 (ADH, collecting duct).*

d) Type 4 is present in salivary gland **[ F ]**
*Reason: AQP-5 is in salivary/lacrimal glands; AQP-4 is in brain and basolateral collecting duct.*

e) Type 5 is present in brain **[ F ]**
*Reason: AQP-4 is the brain aquaporin; AQP-5 is in salivary glands.*

### Q2. Proximal convoluted tubules secrete

a) Bicarbonate ion **[ F ]**
*Reason: PCT reabsorbs bicarbonate (via $H^+$ secretion), it does not secrete it.*

b) Bile salts **[ F ]**
*Reason: Bile salts are the most important component of bile for fat digestion.*

c) Hydrogen ion **[ T ]**
*Reason: PCT secretes $H^+$ ($Na^+$-$H^+$ exchange).*

d) Potassium ion **[ F ]**
*Reason: PCT reabsorbs $K^+$; late DCT/CD secrete $K^+$.*

e) Oxalate **[ T ]**
*Reason: PCT secretes organic anions (oxalate, urate, PAH).*

### Q3. The pituitary gland

a) Secretes thyroid stimulating hormone **[ T ]**
*Reason: Thyrotrophs of anterior pituitary secrete TSH.*

b) Secretes ADH **[ F ]**
*Reason: ADH is made in the hypothalamus and released from the posterior pituitary.*

c) Has the optic chiasma anteriorly **[ T ]**
*Reason: Optic chiasma lies above/in front of the pituitary stalk.*

d) Secretion is regulated by hypothalamus **[ T ]**
*Reason: Hypothalamic releasing/inhibiting hormones control it.*

e) Is endodermal in origin **[ F ]**
*Reason: Pituitary derives from oral ectoderm (Rathke's pouch) and neuroectoderm.*

### Q4. Thyroid stimulating hormone

a) Stimulates synthesis of $T_3$ and $T_4$ **[ T ]**
*Reason: TSH stimulates synthesis/release of $T_3$, $T_4$.*

b) Is biochemically an amine **[ F ]**
*Reason: TSH is a glycoprotein.*

c) Is released from the anterior pituitary **[ T ]**
*Reason: TSH is secreted by anterior pituitary thyrotrophs.*

d) Can be estimated by ELISA **[ T ]**
*Reason: TSH is measured by immunoassay.*

e) Is high in thyrotoxicosis **[ F ]**
*Reason: TSH is suppressed in thyrotoxicosis.*

### Q5. Hormones that act on the female breast

a) Aldosterone **[ F ]**
*Reason: Aldosterone retains $Na^+$ and water.*

b) Adrenal glucocorticoids **[ T ]**
*Reason: Glucocorticoids contribute (permissive) to breast development with GH, insulin, thyroxine.*

c) Growth hormone **[ T ]**
*Reason: GH contributes to breast development.*

d) Cortisol **[ T ]**
*Reason: Cortisol (glucocorticoid) supports breast development.*

e) Estrogen **[ T ]**
*Reason: Estrogen causes ductal growth.*

**Q6. Knee jerk**

a) Is a monosynaptic reflex **[ T ]**
b) Is a superficial reflex **[ F ]**
*Reason: Knee jerk is a deep tendon (stretch) reflex.*
c) Receptor is the patellar tendon **[ F ]**
*Reason: Receptor is the muscle spindle.*
d) Is exaggerated in LMN lesion **[ F ]**
*Reason: Knee jerk is lost/reduced in LMN lesion.*
e) Response is extension **[ T ]**

**Q7. Slow adapting receptors are**

a) Pain receptors **[ T ]**
b) Receptors in the vestibular apparatus **[ T ]**
c) Thermoreceptors **[ T ]**
d) Baroreceptors **[ T ]**
e) Chemoreceptors **[ T ]**

**Q8. In Wallerian degeneration**

a) Chromatolysis occurs **[ F ]**
*Reason: Chromatolysis is in the cell body (retrograde reaction), not Wallerian.*
b) Axon breaks into short lengths **[ T ]**
*Reason: Wallerian degeneration: axon breaks up, myelin → fat droplets.*
c) The neurofibrils ultimately disappear **[ T ]**
*Reason: Distal axon degenerates completely.*
d) Mitotic division of Schwann cells does not occur **[ F ]**
*Reason: Schwann cells proliferate in Wallerian degeneration.*
e) Myelin sheath disintegrates into fat droplets **[ T ]**
*Reason: Myelin breaks into fat droplets.*

**Q9. Abnormalities of taste are**

a) Ageusia **[ T ]**
*Reason: Ageusia = loss of taste.*
b) Presbycusis **[ F ]**
*Reason: Presbycusis = age-related hearing loss.*
c) Hypogeusia **[ T ]**
*Reason: Hypogeusia = decreased taste.*
d) Hyposmia **[ F ]**
*Reason: Hyposmia = decreased smell.*
e) Dysgeusia **[ T ]**
*Reason: Dysgeusia = distorted taste.*

**Q10. The auditory sensation involves the following**

a) Labyrinth **[ T ]**
*Reason: Cochlea (hearing) lies in the labyrinth.*
b) Saccule **[ F ]**
*Reason: Saccule belongs to the vestibular (equilibrium) system.*
c) Lateral geniculate body **[ F ]**
*Reason: Auditory relay is the medial geniculate body; lateral is visual.*
d) Spiral ganglion **[ T ]**
*Reason: Spiral ganglion cells form the cochlear nerve.*
e) Scala media **[ T ]**
*Reason: Scala media contains the organ of Corti.*

**Q11. Norepinephrine excites mainly**

**a) α adrenergic receptors** ✓
b) β1 adrenergic receptors
c) β2 adrenergic receptors
d) β3 adrenergic receptors

**Answer: a) α adrenergic receptors**
*Reason: Norepinephrine acts mainly on α-receptors (and β1 in heart).*

**Q12. The substance that is mostly reabsorbed by the kidney is**

**a) Amino acids** ✓
b) Bicarbonate ions
c) Chloride ions
d) Sodium ions

**Answer: a) Amino acids**
*Reason: Amino acids are ~100% reabsorbed.*

**Q13. The most potent stimulus for growth hormone secretion**

a) Increased blood free fatty acid
**b) Decreased blood glucose** ✓
c) Ghrelin hormone
d) Exercise

**Answer: b) Decreased blood glucose**
*Reason: Hypoglycaemia (and fasting, exercise, deep sleep, ghrelin) stimulates GH; high FFA/glucose inhibit it.*

**Q14. The principal estrogen secreted by the ovaries is**

**a) β-estradiol** ✓
b) Estriol
c) Estrone
d) Progestin

**Answer: a) β-estradiol**
*Reason: 17β-estradiol is the most potent and principal ovarian estrogen.*

**Q15. The most potent amino acids that potentiate insulin secretion**

**a) Arginine, Lysine** ✓
b) Leucine, Lysine
c) Cysteine, Arginine
d) Tryptophan, Tyrosine

**Answer: a) Arginine, Lysine**
*Reason: Arginine and lysine are the most potent amino-acid insulin secretagogues.*

**Q16. The adrenal steroid is degraded mainly in the**

a) Gall bladder
**b) Liver** ✓
c) Kidney
d) Pancreas

**Answer: b) Liver**
*Reason: Adrenal steroids are inactivated by conjugation in the liver.*

**Q17. The major source of heat production is**

a) Ingestion of food
**b) Muscular exertion** ✓
c) Norepinephrine
d) Thyroid hormone

**Answer: b) Muscular exertion**
*Reason: Skeletal muscle (exertion) is the biggest source of heat.*

**Q18. Neurotransmitter that almost always causes excitation**

a) Acetylcholine
b) GABA
**c) Glutamate** ✓
d) Serotonin

**Answer: c) Glutamate**
*Reason: Glutamate is the main excitatory CNS transmitter; GABA/glycine inhibit.*

**Q19. The most sensitive touch receptor is**

a) Free nerve ending
**b) Meissner's corpuscles** ✓
c) Merkel's discs
d) Pacinian corpuscles

**Answer: b) Meissner's corpuscles**

*Reason: Meissner's corpuscles are the most sensitive touch receptors of the skin.*

**Q20. The most important articulator of speech is**

a) Lips
b) Mouth
c) Soft palate
**d) Tongue** ✓

**Answer: d) Tongue**

*Reason: The tongue is the most important articulator.*

# May 2025 — Paper I

### Q1. Cell membrane

a) Has stable composition throughout the cell cycle **[ F ]**
*Reason: Membrane composition changes during the cell cycle.*

b) Is freely permeable to ions but not to protein **[ F ]**
*Reason: The bilayer blocks ions and water-soluble substances; it passes lipid-soluble ones.*

c) Is freely permeable to fat-soluble substances **[ T ]**
*Reason: Lipid-soluble substances (O2, $CO_2$, alcohol, steroids) dissolve in the bilayer.*

d) Contains relatively fewer proteins than lipid **[ F ]**
*Reason: By weight membrane is ~55% protein, ~25% phospholipid, ~13% cholesterol.*

e) Transports glucose by a co-transport mechanism **[ T ]**
*Reason: Glucose enters by $Na^+$–glucose co-transport (SGLT) in gut/renal tubule.*

### Q2. Gap junctions

a) Are present in the apical region of cell **[ F ]**
*Reason: Gap junctions lie laterally/at intercalated discs, not apical.*

b) Are abundant in AV node **[ F ]**
*Reason: AV node has few gap junctions, hence slow conduction.*

c) Have transmembrane protein connections **[ T ]**
*Reason: Connexons join adjacent cells.*

d) Permit rapid propagation of impulse **[ T ]**
*Reason: Gap junctions allow rapid spread of excitation.*

e) Prevent movement of ions **[ F ]**
*Reason: Gap junctions ALLOW ion/small molecule movement.*

### Q3. Macrocytic anemia develops in †

a) Vitamin $B_{12}$ deficiency **[ T ]**
*Reason: $B_{12}$ deficiency → macrocytic (megaloblastic) anaemia.*

b) Folic acid deficiency **[ T ]**
*Reason: Folate deficiency → megaloblastic anaemia.*

c) Vitamin C deficiency **[ F ]**
*Reason: Vitamin C deficiency → scurvy (usually normocytic/mild anaemia).*

d) Iron overload **[ F ]**
*Reason: Iron overload does not cause macrocytic anaemia.*

e) Pernicious anemia **[ T ]**

### Q4. Erythropoiesis is regulated by

a) Rate of erythrocyte destruction **[ F ]**
*Reason: EPO (driven by hypoxia) regulates erythropoiesis, not destruction rate.*

b) Erythropoietin **[ T ]**
*Reason: EPO from kidney stimulates RBC production.*

c) Vitamin $B_{12}$ **[ F ]**
*Reason: $B_{12}$ is needed for maturation (DNA synthesis), not regulation.*

d) Androgen **[ T ]**
*Reason: Androgens stimulate EPO and erythropoiesis.*

e) Blood pressure **[ F ]**
*Reason: Blood pressure does not regulate erythropoiesis.*

### Q5. Plasma albumin

a) Contributes 80% of plasma colloid osmotic pressure **[ T ]**
*Reason: Albumin contributes ~80% of plasma oncotic pressure.*

b) Is negatively charged at normal blood pH **[ T ]**
*Reason: Albumin is negatively charged at pH 7.4.*

c) Carries $CO_2$ in blood **[ T ]**
*Reason: Plasma proteins (carbamino) carry a small fraction of $CO_2$.*

d) Is filtered freely at renal glomerulus **[ F ]**
*Reason: Albumin is too large (69 kDa) and negatively charged → not filtered.*

e) Lacks essential amino acids **[ F ]**
*Reason: Albumin contains essential amino acids.*

**Q6. First heart sound**

a) Is prolonged and loud **[ T ]**
*Reason: S1 (lub) is longer, lower pitched and louder than S2.*

b) Is best heard over aortic area **[ F ]**
*Reason: S1 is best at mitral/tricuspid (apex); S2 at aortic/pulmonary.*

c) Is best heard over mitral area **[ T ]**
*Reason: S1 is best heard at the mitral (apex) area.*

d) Is produced due to closure of AV valves **[ T ]**
*Reason: S1 is due to closure of AV valves at start of systole.*

e) Marks the clinical beginning of diastole **[ F ]**
*Reason: S1 marks the start of systole; S2 the start of diastole.*

**Q7. Obstructive airways disease (COPD) is similar to restrictive lung disease (RLD) in that it reduces**

a) Vital capacity (VC) **[ T ]**

b) Forced expiratory volume in one second (FEV1) **[ T ]**

c) The ratio FEV1/VC **[ F ]**
*Reason: FEV1/VC falls in obstruction, normal/high in restriction.*

d) Residual volume **[ F ]**
*Reason: Residual volume prevents total collapse.*

e) Peak expiratory flow rate to the same degree **[ F ]**
*Reason: PEFR falls more in obstruction.*

**Q8. GIT movement primarily responsible for propulsion of food along the digestive tract**

a) Segmentation contraction **[ F ]**
*Reason: Propulsion is by peristalsis; segmentation mixes.*

b) Tonic contraction **[ F ]**
*Reason: Tonic contraction maintains tone, not propulsion.*

c) Peristalsis **[ T ]**
*Reason: Peristalsis is the propulsive movement of the gut.*

d) Haustration **[ F ]**
*Reason: Haustration is a mixing movement of the colon.*

e) Pendular movement **[ F ]**
*Reason: Pendular movements mix in small intestine.*

**Q9. Sinoatrial node cells are †**

a) Found in both atria **[ F ]**
*Reason: SA node lies in the right atrial wall at the SVC opening.*

b) Innervated by the vagus **[ T ]**
*Reason: Right vagus mainly supplies the SA node.*

c) Unable to generate impulses when completely denervated **[ F ]**
*Reason: SA node is autorhythmic; denervated rate ≈ 100/min.*

d) Connected to the AV node by fine bundles of Purkinje tissue **[ F ]**
*Reason: SA and AV nodes are connected by atrial internodal fibres, not Purkinje fibres.*

e) Able to generate impulses because their membrane potential is unstable **[ T ]**
*Reason: Unstable RMP gives SA node automaticity.*

**Q10. A person with group A (+ve) blood**

a) Has anti-B antibody in plasma **[ T ]**

b) May have the genotype AB **[ F ]**
*Reason: Group A genotype is AA or AO; AB is a different group.*

c) Can be a donor for a person with AB or A **[ T ]**
*Reason: Group A can donate to A and AB.*

d) Has A agglutinogen on RBC membrane **[ T ]**
*Reason: Group A RBCs carry A agglutinogen.*

e) Has B agglutinogen on RBC membrane **[ F ]**
*Reason: Group A has A agglutinogen, not B.*

**Q11. Which part of the ECG corresponds to ventricular repolarization?**

a) The P wave
b) The QRS duration
**c) The T wave** ✓
d) The U wave

**Answer: c) The T wave**
*Reason: T wave = ventricular repolarisation; QRS = depolarisation; P = atrial.*

**Q12. The second heart sound is caused by**

**a) Closure of the aortic and pulmonary valves** ✓
b) Vibrations in the ventricular wall during systole
c) Ventricular filling
d) Closure of the mitral and tricuspid valves

**Answer: a) Closure of the aortic and pulmonary valves**
*Reason: S2 is caused by closure of the aortic and pulmonary valves.*

**Q13. When the radius of the resistance vessels is increased, which of the following is increased?**

a) Systolic blood pressure
b) Diastolic blood pressure
c) Viscosity of the blood
**d) Capillary blood flow** ✓

**Answer: d) Capillary blood flow**
*Reason: Resistance ∝ $1/r^4$; larger radius → less resistance → more flow.*

**Q14. Most of the iron in the body is present in**

**a) Hemoglobin** ✓
b) Myoglobin
c) Ferritin
d) Transferrin

**Answer: a) Hemoglobin**
*Reason: ~65% of body iron is in haemoglobin.*

**Q15. A person whose blood group is 'B positive' has which of the following?**

**a) The rhesus D antigen and the B antigen on their RBC and anti-A agglutinin** ✓
b) The rhesus antigen and the B antigen on their RBC and anti-B agglutinin
c) The rhesus D antigen and the A antigen on their RBC and anti-A agglutinin
d) No rhesus D antigen and the B antigen on their RBC and anti-A agglutinin

**Answer: a) The rhesus D antigen and the B antigen on their RBC and anti-A agglutinin**
*Reason: B+ = B antigen + D antigen; plasma anti-A.*

**Q16. What substance is the product of the second step in the blood clotting process?**

**a) Thrombin** ✓
b) Prothrombin
c) Prothrombin activator
d) Fibrin

**Answer: a) Thrombin**
*Reason: Step 2 converts prothrombin to thrombin.*

**Q17. How does homeostasis return the body to a healthy state after stressful stimuli? By producing**

**a) Responses that oppose the stress** ✓
b) Learned behavior
c) Reflex action
d) Buffering mechanism

**Answer: a) Responses that oppose the stress**
*Reason: Negative feedback produces responses opposing the change.*

**Q18. Defecation reflex is integrated by**

a) Brain stem
b) Cerebral cortex
**c) Spinal cord** ✓
d) Basal ganglia

**Answer: c) Spinal cord**

*Reason: Defecation reflex is integrated in the sacral spinal cord.*

**Q19. What ensures that the lungs expand as the chest wall expands?**

a) A secreted surfactant
**b) Negative pressure between the pleurae** ✓
c) Serous liquid secreted by the pleura
d) The elastic recoil of the alveolar tissue

**Answer: b) Negative pressure between the pleurae**

*Reason: Negative intrapleural pressure couples lungs to the chest wall.*

**Q20. Which structure has the thickest wall?**

a) The aorta
b) The interventricular septum
**c) The left ventricle** ✓
d) The right ventricle

**Answer: c) The left ventricle**

*Reason: The left ventricle generates highest pressure and has the thickest wall.*

# May 2025 — Paper II

### Q1. Active $Na^+$ transport occurs in

a) Cortical collecting tubule **[ T ]**
*Reason: Active $Na^+$ reabsorption occurs in cortical collecting duct (principal cells).*

b) Distal convoluted tubule **[ T ]**
*Reason: DCT actively reabsorbs $Na^+$ (thiazide-sensitive NCC).*

c) Proximal convoluted tubule **[ T ]**
*Reason: PCT reabsorbs $Na^+$ actively.*

d) Thin descending limb of LH **[ F ]**
*Reason: Thin descending limb has no active $Na^+$ transport.*

e) Thin ascending limb of LH **[ F ]**
*Reason: Thin ascending limb is passive to NaCl.*

### Q2. Indicators used to measure ECF

a) Antipyrine **[ F ]**
*Reason: Antipyrine measures TBW.*

b) Evans blue dye **[ F ]**
*Reason: Evans blue measures plasma volume.*

c) 22Na **[ T ]**
*Reason: 22Na distributes in ECF.*

d) Inulin **[ T ]**
*Reason: Inulin measures ECF (and GFR).*

e) Thiosulfate **[ T ]**
*Reason: Thiosulfate/inulin/sucrose/22Na measure ECF.*

### Q3. Hormones that use cAMP as second messenger

a) Calcitonin **[ T ]**
*Reason: Calcitonin is not a physiological gastrin stimulus.*

b) Glucagon **[ T ]**
*Reason: Glucagon acts through cAMP.*

c) Oxytocin **[ F ]**
*Reason: Oxytocin acts through IP3/$Ca^{2+}$.*

d) Secretin **[ T ]**
*Reason: Secretin inhibits gastric motility/emptying and stimulates $HCO_3^-$ secretion.*

e) Thyroid releasing hormone **[ F ]**
*Reason: TRH acts via PLC–IP3.*

### Q4. Thyroid stimulating hormone

a) Is biochemically an amine **[ F ]**
*Reason: TSH is a glycoprotein.*

b) Can be estimated by ELISA **[ T ]**
*Reason: TSH is measured by immunoassay.*

c) Is high in thyrotoxicosis **[ F ]**
*Reason: TSH is suppressed in thyrotoxicosis.*

d) Is released from the anterior pituitary **[ T ]**
*Reason: TSH is secreted by anterior pituitary thyrotrophs.*

e) Stimulates synthesis of $T_3$ and $T_4$ **[ T ]**
*Reason: TSH stimulates synthesis/release of $T_3$, $T_4$.*

### Q5. Specific effects of testosterone on pelvis

a) Causes a broad ovoid-shaped pelvis **[ F ]**
*Reason: Estrogen causes a broad pelvis; testosterone a narrow, rough, strong pelvis.*

b) Causes pelvic bones to be rougher **[ T ]**
*Reason: Testosterone thickens, roughens and strengthens bones.*

c) Increases strength of pelvis **[ T ]**
*Reason: Testosterone adds bone matrix.*

d) Narrows the pelvic outlet **[ T ]**

e) Shortens the pelvis **[ F ]**
*Reason: Testosterone lengthens the pelvis.*

**Q6. Neuronal circuits in the spinal cord can cause**

a) Feeding reflex **[ F ]**
*Reason: Feeding is a hypothalamic/brainstem function.*

b) Reaction to pain **[ T ]**
*Reason: Withdrawal reflex is spinal.*

c) Reaction to pleasure **[ F ]**
*Reason: Pleasure reaction is a limbic/higher function.*

d) Reflexes that control GIT movement **[ T ]**
*Reason: Spinal reflexes control gut movement (defecation).*

e) Walking movements **[ T ]**
*Reason: Walking patterns are generated in spinal cord circuits.*

**Q7. Tactile senses are**

a) Pain **[ F ]**
*Reason: Pain is not a tactile (touch) sense.*

b) Pressure **[ T ]**

c) Temperature **[ F ]**
*Reason: Temperature is detected by thermoreceptors (not a touch sense); body temperature is regulated by the hypothalamus.*

d) Touch **[ T ]**

e) Vibration **[ T ]**
*Reason: Vibration is carried in the dorsal column.*

**Q8. Somatostatin acts to**

a) Depress the secretion of insulin **[ T ]**
*Reason: Somatostatin inhibits insulin and glucagon.*

b) Decrease motility of stomach **[ T ]**
*Reason: Somatostatin decreases GI motility.*

c) Decrease secretion of GIT **[ T ]**
*Reason: Somatostatin decreases GI secretions.*

d) Increase secretion of glucagon **[ F ]**
*Reason: Somatostatin inhibits glucagon.*

e) Increase absorption of GIT **[ F ]**
*Reason: Somatostatin reduces GI absorption.*

**Q9. Pain-producing chemical substances are**

a) Bradykinin **[ T ]**

b) Endorphins **[ F ]**
*Reason: Endorphins relieve pain; they are not pain-producing.*

c) Enkephalins **[ F ]**
*Reason: Enkephalins are analgesic peptides.*

d) Potassium ions **[ T ]**
*Reason: PCT reabsorbs $K^+$; late DCT/CD secrete $K^+$.*

e) Proteolytic enzymes **[ T ]**

**Q10. During accommodation for near vision**

a) Curvature of cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*

b) Curvature of lens increases **[ T ]**
*Reason: Ciliary muscle contracts → lens becomes more convex.*

c) Depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*

d) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*

e) Visual axes of the two eyes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

**Q11. Post-menopausal hot flush is mainly due to**

a) Decreased androgen
**b) Decreased estrogen** ✓
c) Increased progesterone
d) Increased FSH

**Answer: b) Decreased estrogen**

*Reason: Oestrogen withdrawal causes hot flushes.*

**Q12. The body's most powerful sodium-retaining hormone is †**

**a) Angiotensin II** ✓
b) Antidiuretic hormone
c) Atrial natriuretic peptide
d) Parathyroid hormone

**Answer: a) Angiotensin II**

*Reason: Aldosterone is the most powerful; of the options, angiotensin II acts via aldosterone.*

**Q13. The most important effect of glucagon in high concentration is**

**a) Activate adipose cell lipase** ✓
b) Enhances the strength of the heart
c) Inhibit storage of triglyceride in liver
d) Increases blood flow in tissues

**Answer: a) Activate adipose cell lipase**

*Reason: High-dose glucagon activates lipase, making FFA available.*

**Q14. The most likely method of heat loss from skin is**

a) Conduction
b) Convection
c) Evaporation
**d) Radiation** ✓

**Answer: d) Radiation**

*Reason: Radiation is ~60% of heat loss in a resting nude person.*

**Q15. One of the most important clinical causes of edema is †**

a) Allergic reaction
b) Anemia
**c) Nephrotic syndrome** ✓
d) Pregnancy

**Answer: c) Nephrotic syndrome**

*Reason: Nephrotic syndrome → hypoalbuminaemia → ↓oncotic pressure → edema.*

**Q16. Highest concentration of voltage-gated sodium channels is present in †**

a) Axon terminal
b) Dendrites
c) Initial segment of axon
**d) Node of Ranvier** ✓

**Answer: d) Node of Ranvier**

*Reason: Nodes of Ranvier have the highest $Na^+$-channel density (~2000/μm²).*

**Q17. Corticospinal tract fibers mostly originate from**

**a) Primary motor cortex** ✓
b) Premotor areas
c) Somatosensory areas
d) Supplementary motor areas

**Answer: a) Primary motor cortex**

*Reason: ~30% from primary motor cortex, 30% premotor/supplementary, 40% somatosensory; primary motor cortex is the chief origin.*

**Q18. Blood–cerebrospinal fluid barrier is highly permeable to**

a) Chloride
b) Plasma proteins
c) Sodium
**d) Water** ✓

**Answer: d) Water**

*Reason: Water, $CO_2$ and O2 cross the BCSF barrier freely.*

**Q19. Aqueous humor secretion begins mainly with the active transport of**

a) Divalent cations of alkali metals group
b) Potassium cations of the KCl
c) Sodium anions of the NaCl
**d) Sodium cations of the NaCl** ✓

**Answer: d) Sodium cations of the NaCl**

*Reason: Active $Na^+$ transport into ciliary epithelium draws water.*

**Q20. A major part of flavor of food is actually contributed by the sense of**

a) Audition
b) Gustation
**c) Olfaction** ✓
d) Trigeminal sense

**Answer: c) Olfaction**

*Reason: Smell contributes most of flavour.*

# November 2024 — Paper I

### Q1. Regarding action potential

a) Produced by threshold stimuli **[ T ]**
*Reason: AP is all-or-none; occurs at threshold.*

b) Depolarization phase is caused by influx of $K^+$ **[ F ]**
*Reason: Depolarisation is due to $Na^+$ (or $Ca^{2+}$) influx; $K^+$ efflux repolarises.*

c) Influx of $Ca^{2+}$ can produce depolarization **[ T ]**
*Reason: $Ca^{2+}$ influx causes depolarisation in cardiac/smooth muscle.*

d) Consists of either depolarization or repolarization **[ F ]**
*Reason: An action potential has both depolarisation and repolarisation.*

e) Depolarization is mostly due to $Na^+$ influx **[ T ]**
*Reason: Nerve/skeletal AP depolarisation is due to $Na^+$ influx.*

### Q2. Maturation of RBC means

a) Cells are smaller in size **[ T ]**
*Reason: RBC size decreases with maturation.*

b) Appearance of Hb and increase in its concentration **[ T ]**
*Reason: Hb accumulates as RBC mature.*

c) Disappearance of nucleus **[ T ]**
*Reason: Nucleus is extruded.*

d) Cytoplasm transforming to basophilic **[ F ]**
*Reason: Cytoplasm turns acidophilic (as Hb accumulates).*

e) Persistence of organelles **[ F ]**
*Reason: Organelles disappear in mature RBC.*

### Q3. A person with A (+ve) blood group has

a) Anti-A agglutinin in plasma **[ F ]**
*Reason: Group A plasma has anti-B, not anti-A.*

b) A agglutinogen on RBC membrane **[ T ]**
*Reason: Group A RBCs carry A agglutinogen.*

c) D-Rh factor on RBC membrane **[ T ]**
*Reason: Rh-positive RBC carry the D antigen.*

d) D-Rh factor present on RBC membrane **[ T ]**
*Reason: Rh-positive RBC carry the D antigen.*

e) Anti-D agglutinin in plasma **[ F ]**
*Reason: Rh-positive persons do not make anti-D.*

### Q4. Plasma albumin

a) Contributes more to plasma colloid osmotic pressure **[ T ]**
*Reason: Albumin gives most of the colloid osmotic pressure.*

b) Filters freely through glomerular membrane **[ F ]**
*Reason: Albumin does not filter through the glomerulus.*

c) Is positively charged at the normal pH of blood **[ F ]**
*Reason: Albumin carries negative charge at normal pH.*

d) Carries carbon dioxide in blood **[ T ]**

e) Lacks the essential amino acids **[ F ]**
*Reason: Albumin contains essential amino acids.*

### Q5. In normal lung

a) Minute respiratory volume is always 6 L/min **[ F ]**
*Reason: Minute volume varies with activity.*

b) The air remaining in lung after quiet expiration is FRC **[ T ]**
*Reason: FRC = ERV + RV ≈ 2.3 L.*

c) Vital capacity equals inspiratory capacity plus residual volume **[ F ]**
*Reason: VC = IC + ERV (not RV).*

d) All volumes and capacities are 20–25% more in male **[ T ]**
*Reason: Women have 20–25% smaller volumes than men.*

e) Functional residual capacity is about 1200 mL **[ F ]**
*Reason: FRC ≈ 2300 mL; RV ≈ 1200 mL.*

### Q6. Dead space air

a) Is not useful for gas exchange **[ T ]**
*Reason: Dead space air does not take part in exchange.*

b) Is expired lastly **[ F ]**
*Reason: Dead space air is expired FIRST.*

c) In young adult man is about 250 mL **[ F ]**
*Reason: Anatomical dead space ≈ 150 mL.*

d) Saturates inspired air with $H_2O$ vapour **[ T ]**
*Reason: Dead space humidifies and warms inspired air.*

e) Is disadvantageous for removing the expiratory gas from lung **[ F ]**
*Reason: Dead space is wasted ventilation.*

### Q7. In ECG

a) QRS complex is due to ventricular repolarization **[ F ]**
*Reason: QRS = ventricular depolarisation.*

b) S-T segment indicates ventricle is depolarized **[ T ]**
*Reason: S-T segment = ventricles fully depolarised (plateau).*

c) P wave is absent in SA nodal block **[ T ]**
*Reason: SA block → no atrial depolarisation → P wave absent.*

d) Normal P-R interval is about 0.30 sec **[ F ]**
*Reason: Normal PR = 0.12–0.20 s.*

e) P-R interval lengthens due to parasympathetic **[ T ]**
*Reason: Vagal stimulation slows AV conduction, lengthening PR.*

### Q8. Regarding saliva secretion

a) Is increased by sympathetic stimulation **[ T ]**
*Reason: Saliva is stimulated by both divisions; parasympathetic gives watery saliva.*

b) Can be affected by higher center **[ T ]**
*Reason: Saliva secretion is affected by appetite, emotion.*

c) Increased during nervousness **[ F ]**
*Reason: Nervousness inhibits saliva (dry mouth).*

d) Is primarily under hormonal control **[ F ]**
*Reason: Salivary secretion is purely nervous.*

e) Has antiseptic action **[ T ]**
*Reason: Lysozyme, IgA and thiocyanate make saliva antiseptic.*

### Q9. Hormones that inhibit gastric emptying are

a) CCK **[ T ]**
*Reason: CCK delays gastric emptying, stimulates gallbladder contraction and enzyme secretion.*

b) Motilin **[ F ]**
*Reason: Motilin stimulates gastric/intestinal motility (MMC).*

c) Secretin **[ T ]**
*Reason: Secretin inhibits gastric motility/emptying and stimulates $HCO_3^-$ secretion.*

d) Gastrin **[ F ]**
*Reason: Gastrin stimulates acid secretion and mildly motility; it does not inhibit emptying.*

e) GIP **[ T ]**
*Reason: GIP inhibits gastric motility and acid; stimulates insulin.*

### Q10. From the upper small intestine maximum absorption of

a) Glucose **[ T ]**
*Reason: Glucose, amino acids, vitamins, iron are absorbed in upper small intestine.*

b) Amino acid **[ T ]**
*Reason: Amino acids are completely reabsorbed in PCT.*

c) Water-soluble vitamins **[ T ]**

d) Long chain fatty acid **[ T ]**

e) $Fe^{2+}$ **[ T ]**

**Q11. Plasma membrane is mainly composed of**

a) Carbohydrate
b) Protein
c) Cholesterol
**d) Phospholipid** ✓
e) Phospholipid and cholesterol

**Answer: d) Phospholipid**

*Reason: The membrane is a phospholipid bilayer with embedded protein and cholesterol.*

**Q12. Which of the following promotes rouleaux formation?**

a) Albumin
b) Prealbumin
c) Globulin
**d) Fibrinogen** ✓
e) Prothrombin

**Answer: d) Fibrinogen**

*Reason: Fibrinogen (and globulins) promote rouleaux and ESR.*

**Q13. Oxy-Hb dissociation curve is shifted to the right by**

**a) Increased 2,3-DPG** ✓
b) Decreased $PCO_2$
c) Increased pH
d) Increased PO2
e) Decreased $H^+$ concentration

**Answer: a) Increased 2,3-DPG**

*Reason: ↑2,3-DPG, ↑$H^+$, ↑$PCO_2$, ↑temperature shift ODC right.*

**Q14. Lymphocytes**

a) Form 1–2% of the WBC
b) Have important phagocytic function
**c) Can be transformed by stimulus to plasma cells** ✓
d) Are nonmotile
e) Develop in thymus only

**Answer: c) Can be transformed by stimulus to plasma cells**

*Reason: B lymphocytes transform into plasma cells.*

**Q15. Anemia due to maturation defect is**

a) Hemolytic anemia
b) Microcytic anemia
c) Normocytic anemia
**d) Pernicious anemia** ✓
e) Blood loss anemia

**Answer: d) Pernicious anemia**

*Reason: Pernicious anaemia ($B_{12}$ deficiency) is a maturation defect → macrocytic.*

**Q16. Peripheral cell membrane proteins act as †**

a) Pumps
b) Adhesion molecules
c) Channels
d) Carriers
**e) Enzymes** ✓

**Answer: e) Enzymes**

*Reason: Some membrane proteins are enzymes.*

**Q17. Gastrocolic reflex is due to release of**

a) Serotonin
**b) Gastrin** ✓
c) Somatostatin
d) Ach
e) VIP

**Answer: b) Gastrin**

*Reason: Gastrin stimulates acid secretion and mildly motility; it does not inhibit emptying.*

**Q18. Main aim of BP regulating mechanism is to maintain the**

a) Systolic BP
b) Diastolic BP
**c) Mean BP** ✓
d) Pulse pressure
e) Both systolic and pulse pressure

**Answer: c) Mean BP**

*Reason: Mean BP is the effective perfusion pressure.*

**Q19. Hypoxia is**

a) Complete absence of O2 at tissue level
b) Lack of O2 in blood
c) Lack of O2 and accumulation of $CO_2$ in the tissue
**d) Lack of O2 at tissue level** ✓
e) Lack of PO2 in inspired air

**Answer: d) Lack of O2 at tissue level**

*Reason: Hypoxia = reduced O2 at tissue level.*

**Q20. Most sensitive index to assess severity of obstructive lung diseases is**

a) Forced vital capacity
**b) FEV1** ✓
c) PEFR
d) Total lung capacity
e) Residual volume

**Answer: b) FEV1**

*Reason: FEV1 (and FEV1/FVC) is the best index of airway obstruction.*

# November 2024 — Paper II

### Q1. Factors that favor glomerular filtration

a) Glomerular filtration coefficient **[ T ]**
*Reason: Higher Kf increases GFR.*

b) Bowman's capsule hydrostatic pressure **[ F ]**
*Reason: Capsular pressure opposes filtration.*

c) Systemic arterial pressure **[ T ]**
*Reason: Higher arterial pressure raises glomerular pressure.*

d) Afferent arteriolar resistance **[ F ]**
*Reason: Raised afferent resistance LOWERS GFR.*

e) Glomerular hydrostatic pressure **[ T ]**
*Reason: Glomerular capillary hydrostatic pressure drives filtration.*

### Q2. Proteins and polypeptide hormones are

a) Aldosterone **[ F ]**
*Reason: Aldosterone retains $Na^+$ and water.*

b) Estrogen **[ F ]**
*Reason: Estrogen causes ductal growth.*

c) Growth hormone **[ T ]**
*Reason: GH contributes to breast development.*

d) Parathormone **[ T ]**

e) Oxytocin **[ T ]**
*Reason: Oxytocin acts through IP3/$Ca^{2+}$.*

### Q3. Regarding insulin

a) Secretion may increase in $K^+$-depleted patient **[ F ]**
*Reason: $K^+$ depletion decreases insulin secretion.*

b) Activates hormone-sensitive lipase **[ F ]**
*Reason: Insulin inhibits lipase.*

c) Is a hormone associated with energy abundance **[ T ]**
*Reason: Insulin signals energy abundance.*

d) IGF-1 and GH promote normal growth of bones and tissues **[ T ]**
*Reason: IGF-1 and GH promote growth.*

e) Deficiency makes the person more prone to infection **[ T ]**
*Reason: Hyperglycaemia/insulin deficiency impairs immunity.*

### Q4. Regarding cortisol

a) A steroid hormone **[ T ]**
*Reason: GH is a protein hormone.*

b) Decreased protein synthesis (extrahepatic) **[ T ]**
*Reason: Cortisol is catabolic in muscle.*

c) Decreased utilization of free fatty acids **[ F ]**
*Reason: Cortisol increases FFA mobilisation and use.*

d) Promotes wound healing **[ F ]**
*Reason: Cortisol impairs wound healing.*

e) Increased blood volume **[ F ]**
*Reason: Cortisol has minor mineralocorticoid action.*

### Q5. Hormones are cleared from plasma by

a) Metabolic destruction by the tissues **[ T ]**
*Reason: Hormones are cleared by degradation, binding, excretion.*

b) Binding with the tissues **[ T ]**

c) Excretion by liver into the bile **[ T ]**

d) Excretion by the kidney into the urine **[ T ]**

e) Excretion by the GIT into the stool **[ F ]**
*Reason: GIT excretion is minor.*

**Q6. Brain stem controls**

a) Equilibrium **[ T ]**
b) Vision **[ F ]**
*Reason: Vision is not controlled by the brainstem.*
c) Cardiovascular system **[ T ]**
d) Reflexes **[ T ]**
e) Respiration **[ T ]**

**Q7. Sensations transmitted by anterolateral system are**

a) Vibration **[ F ]**
*Reason: Vibration is carried in the dorsal column.*
b) Warmth and cold **[ T ]**
c) Fine touch **[ F ]**
*Reason: Fine touch is a dorsal-column modality.*
d) Crude touch **[ T ]**
e) Tickle and itch **[ T ]**

**Q8. Regarding spermatogenesis †**

a) Begins at an average age of 13 years **[ T ]**
*Reason: Spermatogenesis begins at puberty (~13 years).*
b) Decreased markedly in old age **[ F ]**
*Reason: Declines gradually; does not stop.*
c) Entire period is about 34 days **[ F ]**
*Reason: Spermatogenesis takes ~74 days.*
d) Occurs in seminiferous tubules of testes **[ T ]**
*Reason: Spermatogenesis occurs in seminiferous tubules.*
e) Luteinizing hormone promotes spermatogenesis **[ F ]**
*Reason: FSH (and testosterone) promote spermatogenesis; LH stimulates testosterone.*

**Q9. Norepinephrine**

a) Is released in circulating blood **[ T ]**
b) Is secreted mainly at synapses **[ T ]**
c) Is metabolized rapidly **[ F ]**
*Reason: NE is removed slowly (hence longer action).*
d) Has a more prolonged action than acetylcholine **[ T ]**
e) Probably activates inhibitory receptors **[ F ]**
*Reason: NE acts mainly on excitatory receptors.*

**Q10. The pigment layer of retina**

a) Contains the cell bodies of rods and cones **[ F ]**
*Reason: Pigment epithelium is the outermost layer; cell bodies are in nuclear layers.*
b) Is the innermost layer of the retina **[ F ]**
*Reason: Pigment epithelium is the outermost layer.*
c) Stores large quantities of vitamin A **[ T ]**
*Reason: Pigment epithelium stores vitamin A (retinol).*
d) Removes waste from the photoreceptors **[ T ]**
*Reason: RPE phagocytoses shed photoreceptor discs.*
e) Prevents light reflection throughout the globe of eyeball **[ T ]**
*Reason: Melanin absorbs stray light.*

**Q11. The most important factor that influences potassium secretion by the cortical collecting duct is**

a) Activation of renin–angiotensin system
**b) High potassium diet** ✓
c) Increased plasma ADH concentration
d) Increased plasma aldosterone concentration
e) Reabsorption of sodium ion by cortical collecting duct

**Answer: b) High potassium diet**
*Reason: High ECF $K^+$ (diet) is the main stimulus of $K^+$ secretion by principal cells.*

**Q12. Which of the following is the most important effect of growth hormone?**

**a) Decreased glucose uptake by the tissue** ✓
b) Increased glucose production by the liver
c) Increased insulin secretion
d) Decreased glycolysis
e) Increased gluconeogenesis

**Answer: a) Decreased glucose uptake by the tissue**

*Reason: GH is anti-insulin: ↓ glucose uptake.*

**Q13. Which basal ganglia fiber carries more clinical importance?**

a) Corticostriate fiber
b) Thalamostriate fiber
**c) Nigrostriatal fiber** ✓
d) Striatopallidal fiber
e) Strionigral fiber

**Answer: c) Nigrostriatal fiber**

*Reason: Nigrostriatal dopaminergic fibres degenerate in Parkinson's disease.*

**Q14. Which provides most of the damping function?**

a) Cerebral cortex
b) Muscle spindle
**c) Cerebellum** ✓
d) Basal ganglia
e) Spinal cord

**Answer: c) Cerebellum**

*Reason: Cerebellum damps/smooths movement (prevents overshoot).*

**Q15. Which has the most potent glucocorticoid activity (natural hormone)?**

a) Aldosterone
**b) Cortisol** ✓
c) Corticosterone
d) Cortisone
e) 9α-fludrocortisone

**Answer: b) Cortisol**

*Reason: Cortisol (glucocorticoid) supports breast development.*

**Q16. The autonomic nervous system is activated mainly by centers located in**

a) Basal ganglia
b) Cerebral cortex
c) Cerebellum
**d) Spinal cord** ✓
e) Thalamus

**Answer: d) Spinal cord**

*Reason: ANS is activated mainly from spinal cord, brainstem, hypothalamus.*

**Q17. The most important cause of insulin resistance is**

a) Excess glucocorticoids
b) Fatty liver disease
**c) Obesity** ✓
d) Pregnancy
e) Polycystic ovary disease

**Answer: c) Obesity**

*Reason: Obesity is the commonest cause of insulin resistance.*

**Q18. Which has the slowest conduction velocity?**

a) Aβ fiber
b) Aα fiber
c) Aμ fiber
d) B fiber
**e) C fiber** ✓

**Answer: e) C fiber**

*Reason: C fibres (0.5–2 m/s) conduct slowest.*

**Q19. The majority of heat loss from the body occurs by**

a) Evaporation
b) Conduction
**c) Radiation** ✓
d) Convection
e) Evaporation and conduction

**Answer: c) Radiation**

*Reason: Radiation accounts for ~60% of heat loss.*

**Q20. Which one of the following is the most potent bitter substance?**

a) Caffeine
b) Magnesium sulphate
c) Nicotine
d) Saccharin
**e) Strychnine hydrochloride** ✓

**Answer: e) Strychnine hydrochloride**

*Reason: Strychnine is among the most potent bitter substances.*

# May 2024 — Paper I

### Q1. Facilitated diffusion †

a) Of glucose can increase 10 to 20 fold by insulin **[ T ]**
*Reason: Insulin increases glucose facilitated diffusion 10–20 fold (GLUT-4).*

b) Is a passive process **[ T ]**
*Reason: Diffusion/facilitated diffusion needs no ATP.*

c) Is always a carrier-mediated transport process **[ T ]**
*Reason: Facilitated diffusion uses carriers.*

d) Is an energy-dependent process **[ F ]**
*Reason: Facilitated diffusion is passive.*

e) Rate increases proportionately with the concentration of diffusing substance **[ F ]**
*Reason: Facilitated diffusion saturates at Vmax.*

### Q2. Following statements regarding platelets are correct

a) Are non-nucleated cells **[ T ]**
*Reason: Platelets are cell fragments of megakaryocytes — no nucleus.*

b) Plug forms platforms for deposition of fibrin thread **[ T ]**
*Reason: Platelet plug supports fibrin deposition.*

c) Show adhesiveness when they come in contact with collagen **[ T ]**
*Reason: Platelets adhere to exposed collagen.*

d) Do not have contractile power **[ F ]**
*Reason: Platelets contain actin/myosin/thrombosthenin (clot retraction).*

e) Plug formation is facilitated by ADP **[ T ]**
*Reason: ADP makes platelets sticky.*

### Q3. P-R interval in ECG

a) Indicates conduction through A-V node **[ T ]**
*Reason: PR interval = AV conduction time.*

b) Is about 0.8 sec **[ F ]**
*Reason: PR is 0.12–0.20 s; 0.8 s is the whole cardiac cycle.*

c) Decreases in bradycardia **[ F ]**
*Reason: PR lengthens slightly in bradycardia.*

d) 0.25–0.45 sec indicates 2nd degree heart block **[ F ]**
*Reason: PR > 0.2 s = 1st-degree block; 2nd degree has dropped beats.*

e) Prolonged in hypokalemia **[ F ]**
*Reason: Hypokalaemia mainly causes ST depression, flat T, U waves.*

### Q4. Fetal hemoglobin

a) Has high affinity to O2 **[ T ]**
*Reason: HbF has high O2 affinity.*

b) Can bind 2,3-DPG less avidly **[ T ]**
*Reason: HbF binds 2,3-DPG poorly, so its curve is left-shifted.*

c) Consists of 2α and 2γ chains **[ T ]**
*Reason: HbF = 2α2γ; HbA = 2α2β.*

d) Has lower P50 than adult Hb **[ T ]**
*Reason: HbF has lower P50 (left shift).*

e) Is normally replaced by adult Hb after birth **[ T ]**
*Reason: HbF is replaced by HbA in the first months after birth.*

### Q5. Following statements regarding WBC are correct

a) Life span of granulocytes after release in blood is 4 to 8 days **[ F ]**
*Reason: Granulocytes live 4–8 hours in blood, a few days in tissue.*

b) Stored in bone marrow until they are needed **[ T ]**
*Reason: Mature granulocytes are stored in marrow.*

c) Phagocytosis is the most important function of eosinophils **[ F ]**
*Reason: Eosinophils mainly attack parasites/allergy; neutrophils/macrophages phagocytose.*

d) Plasma cells function mainly in immunity **[ T ]**
*Reason: Plasma cells secrete antibodies.*

e) Lymphocytes normally constitute about 10% of total WBC **[ F ]**
*Reason: Lymphocytes are 20–40% of WBC.*

### Q6. The absolute refractory period in the ventricles

a) Is the period when the ventricles are completely inexcitable **[ T ]**
*Reason: Absolute refractory period = no stimulus can excite.*

b) Corresponds to the period of ventricular depolarization **[ T ]**
*Reason: ARP covers depolarisation/plateau.*

c) Corresponds to the period of ventricular contraction **[ T ]**
*Reason: ARP lasts through ventricular contraction.*

d) Decreases during parasympathetic stimulation of the heart **[ F ]**
*Reason: Slower rate lengthens (not shortens) refractory period.*

e) Is shorter than atrial muscle **[ F ]**
*Reason: Ventricular ARP (0.25–0.3 s) is longer than atrial (0.15 s).*

### Q7. During ventricular systole

a) 110 mL blood is present in ventricle before ejection **[ T ]**
*Reason: End-diastolic volume ≈ 110–120 mL.*

b) Intraventricular pressure rises gradually **[ F ]**
*Reason: Pressure rises rapidly in isovolumetric contraction.*

c) Atrial pressure wave 'c' appears **[ T ]**
*Reason: 'c' wave = bulging of AV valve into atrium at ventricular systole.*

d) Produces second heart sound **[ F ]**
*Reason: S2 occurs at the END of systole.*

e) Pumps blood into pulmonary vein **[ F ]**
*Reason: Right ventricle pumps into the pulmonary ARTERY.*

### Q8. The compliance of the lung

a) Is affected only by elasticity of elastin in connective tissue **[ F ]**
*Reason: Compliance depends on elastin, collagen AND surface tension.*

b) Is greater in standing than in recumbent position **[ T ]**
*Reason: Lung compliance is lower in recumbent posture (blood engorgement).*

c) Is increased by the surface tension of the fluid lining alveoli **[ F ]**
*Reason: Surface tension DEcreases compliance.*

d) Is increased by surfactant **[ T ]**
*Reason: Surfactant raises compliance.*

e) Is low in obesity **[ T ]**
*Reason: Obesity reduces chest wall compliance.*

### Q9. Regarding local hormones of GIT

a) Gastrin stimulates growth of the gastric mucosa **[ T ]**
*Reason: Gastrin is trophic to gastric mucosa.*

b) Acid in antrum stimulates gastrin secretion **[ F ]**
*Reason: Antral acid inhibits gastrin release.*

c) Migrating motor complex is initiated by motilin **[ T ]**
*Reason: MMC is initiated by motilin in fasting.*

d) Somatostatin stimulates the secretion of secretin **[ F ]**
*Reason: Somatostatin inhibits secretin, gastrin, CCK.*

e) CCK increases the secretion of bicarbonate by the duct cells of the pancreas **[ F ]**
*Reason: Secretin stimulates ductal $HCO_3^-$; CCK stimulates enzyme (acinar) secretion.*

### Q10. Peristalsis

a) Occurs due to stimulation of myenteric plexus **[ T ]**
*Reason: Peristalsis is initiated by the myenteric plexus.*

b) Is a propulsive movement **[ T ]**
*Reason: Peristalsis propels contents aorally→anally.*

c) Occurs at a rate of 20 cm/sec in large intestine **[ F ]**
*Reason: Peristalsis moves at 2–25 cm/s in small intestine; much slower in colon.*

d) Is initiated by distention of gut wall **[ T ]**
*Reason: Distension initiates the peristaltic reflex.*

e) Does not follow the law of the gut **[ F ]**
*Reason: Peristalsis follows the law of the gut (myenteric reflex + anal direction).*

**Q11. The main contribution for generating RMP is**

a) Sodium diffusion inside the cell
**b) Potassium diffusion outside the cell** ✓
c) Impermeant protein molecules
d) $Na^+$-$K^+$ pump

**Answer: b) Potassium diffusion outside the cell**
*Reason: $K^+$ diffusion (–94 mV) is the main contributor to RMP; $Na^+$–$K^+$ pump adds ~ –4 mV.*

**Q12. The most lethal effect of mismatched blood transfusion is**

a) Circulatory shock
**b) Acute renal shut down** ✓
c) Hemoglobinemia
d) Jaundice

**Answer: b) Acute renal shut down**
*Reason: Haemolysis → haemoglobin in tubules → acute renal shutdown.*

**Q13. Ejection fraction of heart is**

**a) A valuable index of ventricular function** ✓
b) The percentage of ESV ejected per stroke
c) Reduced in exercise
d) Increased in myocardial infarction

**Answer: a) A valuable index of ventricular function**
*Reason: EF = SV/EDV ≈ 60–65%; falls in failure.*

**Q14. Automaticity is the most important property of**

a) AV node
b) Purkinje fiber
**c) SA node** ✓
d) Junctional tissue

**Answer: c) SA node**
*Reason: SA node has the highest automaticity.*

**Q15. Regarding lung function tests**

a) Normal FEV1 is 70% of FVC
b) FEV1 is decreased more than FVC in a patient with lung fibrosis
**c) FEV1 is dependent on expiratory effort** ✓
d) In obstructive airway disease FVC and FEV1/FVC both fall markedly

**Answer: c) FEV1 is dependent on expiratory effort**
*Reason: FEV1 depends on expiratory effort; normal FEV1/FVC ≈ 80%.*

**Q16. The more powerful effect of angiotensin-II on blood pressure regulation is**

**a) Vasoconstriction** ✓
b) Salt and water retention
c) Stimulate thirst
d) Stimulate ADH

**Answer: a) Vasoconstriction**
*Reason: Vasoconstriction raises resistance (radius).*

**Q17. Inhibition of platelet aggregation in the intact endothelium is by**

a) Serotonin
b) Thromboxane A2
**c) Prostacyclin** ✓
d) Von Willebrand factor

**Answer: c) Prostacyclin**
*Reason: Prostacyclin ($PGI_2$) and NO from endothelium inhibit platelet aggregation.*

**Q18. The trigger of the intrinsic pathway is the activation of**

a) Factor X
b) Factor V
**c) Factor XII** ✓
d) Factor II

**Answer: c) Factor XII**

*Reason: Contact activation of factor XII triggers the intrinsic pathway.*

**Q19. The diaphragm †**

a) Moves up to cause inspiration
**b) Moves as much as 7 cm with deep breathing** ✓
c) Always contracts as a single unit
d) Is a muscle of expiration only

**Answer: b) Moves as much as 7 cm with deep breathing**

*Reason: Diaphragm descends 1.5 cm in quiet breathing and up to 7 cm in deep breathing.*

**Q20. Which abolishes automatic respiration? Destruction of**

**a) Pre-Bötzinger complex** ✓
b) Ventral group respiratory neurons
c) Dorsal group respiratory neurons
d) Section at the inferior border of the pons

**Answer: a) Pre-Bötzinger complex**

*Reason: Pre-Bötzinger complex is the primary rhythm generator.*

# May 2024 — Paper II

### Q1. Renal tubule reabsorbs normally

a) All filtered $HCO_3^-$ in respiratory alkalosis **[ F ]**
*Reason: In respiratory alkalosis kidneys excrete $HCO_3^-$.*

b) A large volume of water relative to excretion **[ T ]**
*Reason: Reabsorption ≫ excretion.*

c) By passive mechanism only **[ F ]**
*Reason: Active and passive mechanisms both operate.*

d) All filtered amino acids **[ T ]**
*Reason: Amino acids are completely reabsorbed in PCT.*

e) More potassium than chloride **[ F ]**
*Reason: ~90% $K^+$ is reabsorbed but $Cl^-$ ~99%; $Cl^-$ exceeds $K^+$.*

### Q2. Proteinuria

a) Results in decreased plasma volume **[ T ]**
*Reason: Proteinuria → hypoalbuminaemia → low plasma oncotic pressure.*

b) Reduces oncotic pressure in blood **[ T ]**
*Reason: Albumin loss lowers plasma oncotic pressure.*

c) Is due to increased permeability of tubular membrane **[ F ]**
*Reason: Proteinuria is due to glomerular membrane damage.*

d) Is a benign condition **[ F ]**
*Reason: Persistent proteinuria suggests renal disease.*

e) Is due to increased tubular secretion **[ F ]**
*Reason: Protein is not secreted by tubules.*

### Q3. Renal blood flow

a) Is less than 20% of cardiac output **[ F ]**
*Reason: Kidneys receive ~20–25% of cardiac output.*

b) Is 1.2 to 1.3 L/min **[ T ]**
*Reason: RBF ≈ 1.1–1.2 L/min (≈ 22% of CO).*

c) Can be measured by applying Fick principle **[ T ]**
*Reason: Fick principle with PAH measures renal plasma flow.*

d) Can be measured by infusing p-aminohippuric acid **[ T ]**
*Reason: PAH clearance ≈ RPF.*

e) Is regulated by dopamine made in kidney **[ T ]**
*Reason: Renal dopamine increases RBF.*

### Q4. Adrenogenital syndrome

a) Occurs due to excessive secretion of cortisol **[ F ]**
*Reason: Adrenogenital syndrome: enzyme block (21-hydroxylase) → low cortisol, high ACTH, high androgens.*

b) Occurs due to excessive secretion of androgen **[ T ]**
*Reason: Low cortisol → ↑ACTH → androgen excess.*

c) Causes precocious puberty in male **[ T ]**
*Reason: Androgen excess → precocious puberty in boys.*

d) Causes excess secretion of ACTH **[ T ]**
*Reason: Low cortisol → ↑ACTH.*

e) Is known as congenital adrenal hyperplasia **[ T ]**
*Reason: Adrenogenital syndrome = CAH.*

### Q5. The fovea centralis

a) Lies where the visual axis impinges on the retina **[ T ]**
*Reason: Fovea lies on the visual axis.*

b) Is not crossed by any major blood vessels **[ T ]**
*Reason: Fovea is avascular.*

c) Is the thickest part of the retina **[ F ]**

d) Offers higher visual acuity than other parts of retina **[ T ]**
*Reason: Fovea has the highest acuity.*

e) Lies on the nasal side of the optic disc **[ F ]**
*Reason: Fovea lies temporal to the optic disc.*

**Q6. Posterior column damage in the spinal cord may impair**

a) Vibration sense **[ T ]**
b) Pain sensation **[ F ]**
*Reason: Posterior column carries fine touch, vibration, proprioception — not pain.*
c) The flexor plantar response to stimulation of the sole **[ F ]**
*Reason: Plantar reflex tests pyramidal tract, not the dorsal column.*
d) Tactile sensation **[ T ]**
e) The ability to stand steadily with eyes closed **[ T ]**

**Q7. Basal ganglia**

a) Work with the corticospinal tract to control complex motor activity **[ T ]**
b) Are a part of the accessory motor system **[ T ]**
c) Are responsible for coordinating movement and balance **[ F ]**
*Reason: Cerebellum coordinates movement and balance.*
d) Regulate muscle tone **[ T ]**
e) Are also known as the 'little brain' **[ F ]**
*Reason: The cerebellum is the 'little brain'.*

**Q8. Destruction of the anterior pituitary gland causes**

a) Amenorrhoea **[ T ]**
*Reason: Hypopituitarism → loss of FSH/LH → amenorrhoea.*
b) Diabetes insipidus **[ F ]**
*Reason: GH excess can cause diabetes mellitus (diabetogenic), not DI.*
c) Diabetes mellitus **[ F ]**
d) Impaired ability to cope with stress **[ T ]**
*Reason: ACTH loss → low cortisol.*
e) Impaired calcium ion concentration regulation **[ F ]**
*Reason: Calcium regulation is by PTH, not anterior pituitary.*

**Q9. Spermatogenesis is regulated by**

a) Cortisol **[ F ]**
*Reason: Cortisol (glucocorticoid) supports breast development.*
b) Follicle stimulating hormone **[ T ]**
c) Luteinizing hormone **[ T ]**
d) Testosterone **[ T ]**
*Reason: Steroid hormones act via intracellular receptors (gene transcription).*
e) Inhibin **[ T ]**
*Reason: Sertoli cells secrete inhibin.*

**Q10. Hypoglycemic coma differs from hyperglycemic coma in that there is more likelihood of a**

a) Rapid loss of consciousness **[ T ]**
*Reason: Hypoglycaemic coma is rapid.*
b) Weak pulse **[ F ]**
*Reason: Hypoglycaemic coma: bounding pulse.*
c) Normal blood pH **[ T ]**
*Reason: No ketoacidosis.*
d) Glucose-free urine **[ T ]**
*Reason: Glucose is absent.*
e) High acetone level in urine **[ F ]**
*Reason: Ketones are absent in hypoglycaemia.*

**Q11. During strenuous sports activity a young male has heart rate 160/min, moderately dilated pupil and profuse sweating. His BP is 160/60 mmHg. The most likely cause of decreased diastolic pressure is**

a) Generalized sympathetic activity
b) Alpha adrenergic effect
**c) Increased production of metabolites** ✓
d) Beta adrenergic effect
e) Cholinergic activity

**Answer: c) Increased production of metabolites**
*Reason: Metabolites cause vasodilatation in active muscle → fall in diastolic pressure.*

**Q12. Regarding severe damage in Wernicke's area †**

a) The person cannot hear sounds
b) Can recognize different words
c) Unable to arrange the words
d) Person may be able to read words from printed page
**e) Unable to recognize the thought that is conveyed ✓**

**Answer: e) Unable to recognize the thought that is conveyed**

*Reason: Wernicke's area damage → fluent speech but poor comprehension (cannot interpret thought).*

**Q13. In presence of ADH the greatest fraction of filtered water is absorbed in the**

a) Distal tubule
b) Loop of Henle
**c) Proximal tubule ✓**
d) Cortical collecting duct
e) Late distal tubule

**Answer: c) Proximal tubule**

*Reason: ~65% of filtered water is reabsorbed in PCT.*

**Q14. Most important hormone in the acute recovery from hypoglycemia**

a) Growth hormone
**b) Glucagon and epinephrine ✓**
c) Thyroid hormone
d) Cortisol
e) Insulin

**Answer: b) Glucagon and epinephrine**

*Reason: Glucagon and epinephrine act within minutes.*

**Q15. Extrapyramidal tract mainly controls**

a) The limb movement
**b) The tone, posture and equilibrium ✓**
c) Swinging of arm
d) Reflex activities
e) Autonomic visceral activities

**Answer: b) The tone, posture and equilibrium**

*Reason: Extrapyramidal tracts control tone, posture and equilibrium.*

**Q16. Which one is the most correct statement for menstrual cycle in women?**

a) It occurs with uterine mucosal shedding
b) Periodical vaginal bleeding
c) Periodic preparation for fertilization and pregnancy
**d) Periodic secretion of estrogen and progesterone ✓**
e) Regular cyclical changes in ovary

**Answer: d) Periodic secretion of estrogen and progesterone**

*Reason: The menstrual cycle is the monthly rhythmic change in secretion of estrogen/progesterone.*

**Q17. A 7-year-old boy is evaluated for short stature. His growth hormone level is normal, but IGF-1 is reduced. His growth failure is likely due to a defect in**

a) GHRH released from hypothalamus
b) GHRH receptor
c) Androgen synthesis
d) Testosterone synthesis
**e) GH receptor ✓**

**Answer: e) GH receptor**

*Reason: GH normal/high but IGF-1 low = GH receptor defect.*

**Q18. Major neurotransmitters and neuromodulators include amino acids are**

a) Serotonin
b) Substance P and opioids
**c) Glutamate, GABA, glycine** ✓
d) Dopamine, serotonin
e) Nitric oxide and carbon monoxide

**Answer: c) Glutamate, GABA, glycine**

*Reason: Glutamate, GABA, glycine are amino-acid transmitters.*

**Q19. Activation of a sensory nerve from the muscle spindle causes contraction of the extensor muscle and relaxation of the flexor muscle. The relaxation of the flexor muscle is an example of**

a) Negative feedback inhibition
b) Presynaptic inhibition
**c) Postsynaptic inhibition** ✓
d) Renshaw cell inhibition
e) Indirect inhibition

**Answer: c) Postsynaptic inhibition**

*Reason: Reciprocal inhibition of antagonists is postsynaptic inhibition (inhibitory interneuron).*

**Q20. Defect in fine and skilled voluntary motor movements is caused by damage in**

**a) Lateral corticospinal tract** ✓
b) Rubrospinal tract and tectospinal tract
c) Lateral and anterior corticospinal tract
d) Lateral corticospinal tract and rubrospinal tract
e) Reticulospinal tract and vestibulospinal tract

**Answer: a) Lateral corticospinal tract**

*Reason: Lateral corticospinal tract controls fine skilled distal movements.*

# November 2023 — Paper I

### Q1. Mitochondria

a) Have a membrane similar to the cell membrane **[ T ]**
*Reason: Mitochondria have an inner and outer lipid membrane.*

b) Are the chief site of protein synthesis **[ F ]**
*Reason: Ribosomes (rough ER) synthesise proteins; mitochondria make ATP.*

c) Are the chief site for ATP generation **[ T ]**
*Reason: Mitochondria generate most ATP by oxidative phosphorylation.*

d) Are absent near the membranes of actively secreting cells **[ F ]**
*Reason: Mitochondria are abundant where ATP is needed (secretory/transport cells).*

e) Have their own DNA **[ T ]**
*Reason: Mitochondria have their own circular DNA.*

### Q2. Regarding RMP †

a) RMP of large nerve fiber is –70 mV **[ F ]**
*Reason: Large nerve fibre RMP ≈ –90 mV.*

b) Excitable tissues have less negative RMP **[ F ]**
*Reason: Excitable cells have a more negative RMP than non-excitable cells.*

c) RBC have less negative RMP **[ T ]**
*Reason: RBC RMP is only about –10 mV.*

d) RMP of neuron is –90 mV **[ F ]**
*Reason: Neuron RMP ≈ –65 to –70 mV; large fibre –90 mV.*

e) Non-excitable tissues have more negative RMP **[ F ]**
*Reason: Non-excitable cells have less negative RMP.*

### Q3. Erythrocytes

a) Are responsible for the major part of blood viscosity **[ T ]**
*Reason: RBC are the major determinant of viscosity.*

b) Contain the enzyme carbonic anhydrase **[ T ]**
*Reason: RBC carbonic anhydrase speeds $CO_2 + H_2O \rightleftharpoons H_2CO_3$.*

c) Metabolize glucose to produce $CO_2$ and $H_2O$ **[ F ]**
*Reason: RBC metabolise glucose anaerobically (lactate); they have no mitochondria.*

d) Swell to bursting point when suspended in 0.9% saline **[ F ]**
*Reason: 0.9% saline is isotonic; RBC burst in hypotonic fluid.*

e) Have rigid walls **[ F ]**
*Reason: RBC membrane is flexible, allowing passage through capillaries.*

### Q4. Sinoatrial node cells are

a) Found in both atria **[ F ]**
*Reason: SA node lies in the right atrial wall at the SVC opening.*

b) Innervated by the vagus **[ T ]**
*Reason: Right vagus mainly supplies the SA node.*

c) Unable to generate impulses when completely denervated **[ F ]**
*Reason: SA node is autorhythmic; denervated rate ≈ 100/min.*

d) Connected to the AV node by the bundles of Purkinje tissue **[ F ]**
*Reason: SA and AV nodes are connected by atrial internodal fibres, not Purkinje fibres.*

e) Stable with regard to their membrane potential **[ F ]**
*Reason: SA node membrane potential is unstable (pacemaker potential).*

### Q5. A person with group A (+ve) blood

a) Has anti-B antibody in the plasma **[ T ]**
*Reason: Group A has A antigen and anti-B antibody.*

b) May have the genotype AB **[ F ]**
*Reason: Group A genotype is AA or AO; AB is a different group.*

c) Can be a donor for a person with AB or A **[ T ]**
*Reason: Group A can donate to A and AB.*

d) Has A agglutinogen in RBC membrane **[ T ]**
*Reason: Group A RBCs carry A agglutinogen.*

e) Has D agglutinogen in RBC membrane **[ T ]**
*Reason: '+ve' means Rh D antigen present on RBC.*

**Q6. Acclimatization to altitude causes**

a) A small increase in P50 **[ T ]**
*Reason: Acclimatisation raises 2,3-DPG → P50 increases slightly.*

b) Decreased erythropoietin secretion **[ F ]**
*Reason: Hypoxia INCREASES EPO.*

c) Increase in number of mitochondria **[ T ]**
*Reason: Acclimatisation increases mitochondria.*

d) Decrease in myoglobin **[ F ]**
*Reason: Myoglobin increases at altitude.*

e) Decrease in tissue content of cytochrome oxidase **[ F ]**
*Reason: Tissue cytochrome oxidase increases.*

**Q7. Tachycardia is**

a) Positive chronotropic effect on heart **[ T ]**
*Reason: Tachycardia = increased rate (chronotropic).*

b) More common in athletes **[ F ]**
*Reason: Athletes have bradycardia.*

c) Positive inotropic effect of heart **[ F ]**
*Reason: Rate effect is chronotropic; inotropic = force.*

d) Controlled by β blocker **[ T ]**
*Reason: β-blockers slow the heart.*

e) Common during sleep **[ F ]**
*Reason: HR falls during sleep.*

**Q8. Factors increasing heart rate are**

a) Release of norepinephrine **[ T ]**
*Reason: NE raises HR via β1.*

b) Release of acetylcholine **[ F ]**
*Reason: ACh slows HR.*

c) Release of $Ca^{2+}$ **[ T ]**
*Reason: $Ca^{2+}$ influx increases pacemaker slope/force.*

d) Increased $Na^{+}$ **[ T ]**
*Reason: Increased $Na^{+}$ leak raises rate.*

e) Decreased $K^{+}$ **[ F ]**
*Reason: Low $K^{+}$ effect on HR is small; high $K^{+}$ slows/arrests the heart.*

**Q9. The respiratory center is**

a) In the hypothalamus **[ F ]**
*Reason: Respiratory centres are in medulla/pons.*

b) In the lower part of brainstem **[ T ]**
*Reason: Medulla and pons house respiratory centres.*

c) Not involved in the vomiting reflex **[ F ]**
*Reason: Vomiting involves respiratory/vomiting centres in the medulla.*

d) Influenced by higher centers **[ T ]**
*Reason: Bronchomotor tone is under higher control.*

e) Involved in the swallowing reflex **[ T ]**
*Reason: Respiration is inhibited during swallowing.*

**Q10. Enterohepatic circulation**

a) Means recirculation of bile salt **[ T ]**
*Reason: Enterohepatic circulation recycles bile salts.*

b) Causes secretion of bile **[ T ]**
*Reason: Returning bile salts stimulate bile secretion (choleretic).*

c) Helps in conservation of bile salts **[ T ]**
*Reason: ~95% reabsorbed.*

d) Helps in excretion of bile acid **[ T ]**
*Reason: Recirculation conserves, and the net loss is excreted.*

e) Stimulates bile salt production in the liver **[ F ]**
*Reason: Returning bile salts INHIBIT hepatic synthesis (feedback).*

**Q11. The most important function of cell membrane is to**

**a) Act as a selective barrier to the passage of molecules** ✓
b) Anchor cells to adjacent cells
c) Maintain the physical integrity of the cell
d) Protect the cell from some harmful chemicals

**Answer: a) Act as a selective barrier to the passage of molecules**
*Reason: The membrane is a selective barrier.*

**Q12. Extracellular fluid contains mainly large amounts of**

**a) Bicarbonate ions** ✓
b) Calcium ions
c) Magnesium ions
d) Potassium ions

**Answer: a) Bicarbonate ions**
*Reason: PCT reabsorbs bicarbonate (via $H^+$ secretion), it does not secrete it.*

**Q13. Most important lung defense mechanism includes**

**a) Macrophages** ✓
b) Negative intrapleural pressure
c) Serotonin
d) Surfactant

**Answer: a) Macrophages**
*Reason: Alveolar macrophages phagocytose particles.*

**Q14. Very high rise of ESR occurs in**

a) Chronic infection
b) Hemorrhage
c) Pregnancy
**d) Tuberculosis** ✓

**Answer: d) Tuberculosis**
*Reason: TB/chronic infection and myeloma raise ESR the most.*

**Q15. Most important factor that increases cardiac output is †**

a) Eating
**b) Excitement** ✓
c) High environment temperature
d) Pregnancy

**Answer: b) Excitement**
*Reason: Excitement raises CO/GH.*

**Q16. Most common index for lung function**

**a) FVC** ✓
b) FRC
c) RV
d) VC

**Answer: a) FVC**
*Reason: FVC is the commonest simple lung test.*

**Q17. Most important factor of hepatic duct bile is**

a) Alkaline phosphatase
**b) Bile salts** ✓
c) Cholesterol
d) Fatty acids

**Answer: b) Bile salts**
*Reason: Bile salts are the most important component of bile for fat digestion.*

**Q18. Principal physiologic action of gastrin**

a) Contraction at gastroesophageal junction
**b) Increase gastric motility** ✓
c) Stimulation of growth of mucosa of the stomach
d) Stimulate insulin secretion

**Answer: b) Increase gastric motility**

*Reason: Gastrin mildly increases motility; its chief action is acid secretion.*

**Q19. The most important muscle involved in forced expiration**

a) Anterior serrati
b) Internal intercostal
**c) Rectus abdominis** ✓
d) Scaleni

**Answer: c) Rectus abdominis**

*Reason: Abdominal muscles (rectus abdominis) are the most powerful forced-expiratory muscles.*

**Q20. Most important component of surfactant is**

a) Carbohydrate
b) Neutral lipid
**c) Phosphatidylcholine** ✓
d) Phosphatidylglycerol

**Answer: c) Phosphatidylcholine**

*Reason: Dipalmitoyl phosphatidylcholine (DPPC) is the main surfactant lipid.*

# November 2023 — Paper II

### Q1. Renal tubular reabsorption

a) Is highly selective **[ T ]**
*Reason: Reabsorption is selective.*

b) Is very large relative to excretion **[ T ]**
*Reason: Reabsorption of water and solutes is far greater than excretion.*

c) Is only by passive mechanism **[ F ]**
*Reason: Active transport ($Na^+$, glucose) and passive both occur.*

d) Of water is an active process **[ F ]**
*Reason: Water follows solutes passively by osmosis.*

e) Of water from DCT is obligatory **[ F ]**
*Reason: DCT/collecting duct water reabsorption is facultative (ADH).*

### Q2. Renin is

a) An enzyme **[ T ]**
*Reason: Renin is a proteolytic enzyme.*

b) Secreted by cells of macula densa **[ F ]**
*Reason: Macula densa senses NaCl in DCT; JG cells secrete renin.*

c) Secreted in response to low sodium concentration in DCT **[ T ]**
*Reason: Low NaCl at macula densa stimulates renin release.*

d) Secreted in response to low potassium concentration in DCT **[ F ]**
*Reason: Renin is not triggered by low $K^+$ in DCT.*

e) Secreted by juxtaglomerular cells **[ T ]**
*Reason: JG cells secrete renin.*

### Q3. Volume of urine is regulated by

a) Intake of water **[ T ]**
*Reason: Water intake lowers osmolarity → less ADH → more urine.*

b) ADH **[ T ]**
*Reason: ADH concentrates urine.*

c) PTH **[ F ]**
*Reason: PTH regulates $Ca^{2+}$, not urine volume.*

d) Aldosterone **[ T ]**
*Reason: Aldosterone retains $Na^+$ and water.*

e) TmG **[ F ]**
*Reason: TmG is the glucose transport maximum.*

### Q4. Thyroid hormone

a) Secretion is increased by cold environment **[ T ]**
*Reason: Cold → TRH → TSH ↑.*

b) Is stored extracellularly **[ T ]**
*Reason: Thyroglobulin is stored in follicular lumen (extracellular).*

c) Has effects on brain development **[ T ]**
*Reason: Thyroid hormone is vital for neuronal development.*

d) Secretion is increased by sympathetic stimulation **[ F ]**
*Reason: Saliva is stimulated by both divisions; parasympathetic gives watery saliva.*

e) Secretion is stimulated by hot environment **[ F ]**
*Reason: Heat lowers TSH.*

### Q5. The following hormones use cAMP as a second messenger

a) Testosterone **[ F ]**
*Reason: Steroid hormones act via intracellular receptors (gene transcription).*

b) Estrogen **[ F ]**
*Reason: Estrogen causes ductal growth.*

c) ADH **[ T ]**
*Reason: ADH concentrates urine.*

d) Glucagon **[ T ]**
*Reason: Glucagon acts through cAMP.*

e) TSH **[ T ]**
*Reason: TSH acts via cAMP.*

**Q6. Basal ganglia dysfunctions are characterized by**

a) Intention tremor **[ F ]**
*Reason: Intention tremor is cerebellar; basal ganglia give resting tremor.*
b) Extreme rigidity **[ T ]**
c) Past pointing **[ F ]**
*Reason: Past pointing (dysmetria) is cerebellar.*
d) Ataxia **[ F ]**
*Reason: Ataxia is a cerebellar sign.*
e) Resting involuntary tremor **[ T ]**

**Q7. Stretch reflex**

a) Is a monosynaptic reflex **[ T ]**
b) Is a polysynaptic reflex **[ F ]**
*Reason: Stretch (knee jerk) reflex is monosynaptic.*
c) Receptor is muscle spindle **[ T ]**
d) Response is relaxation of muscle **[ F ]**
*Reason: Stretch reflex causes contraction.*
e) Neurotransmitter is glutamate **[ T ]**

**Q8. Synapse**

a) For signal transmission in CNS almost all are electrical type **[ F ]**
*Reason: Most CNS synapses are chemical.*
b) Always transmits the signals in one direction **[ T ]**
*Reason: Chemical synapses are one-way.*
c) Often transmits signals in either direction **[ F ]**
*Reason: Chemical synapses conduct one way.*
d) Develops fatigue, a protective mechanism **[ T ]**
*Reason: Synaptic fatigue is a protective mechanism.*
e) Delay time determines whether a reflex pathway is mono- or polysynaptic **[ T ]**
*Reason: Synaptic delay (0.5 ms) estimates number of synapses.*

**Q9. Olfactory epithelium**

a) Contains unipolar nerve cells **[ F ]**
*Reason: Olfactory receptor neurons are bipolar.*
b) Has receptors for smell **[ T ]**
*Reason: Olfactory epithelium holds smell receptors.*
c) Is derived from CNS **[ F ]**
*Reason: Olfactory epithelium arises from olfactory placode.*
d) Lies over cribriform plate **[ T ]**
*Reason: Olfactory epithelium lies in roof of nose under cribriform plate.*
e) Contains cilia **[ T ]**
*Reason: Olfactory neurons have cilia.*

**Q10. Refractive power of eyeball is**

a) Less in myopia **[ F ]**
*Reason: Myopia: refractive power too high for the eyeball length.*
b) More in hypermetropia **[ F ]**
*Reason: Hypermetropia: refractive power too low.*
c) Expressed in diopters **[ T ]**
*Reason: Refractive power is measured in dioptres.*
d) Maximum in cornea **[ T ]**
*Reason: Cornea gives ~43 of 59 D.*
e) Related with focal length **[ T ]**
*Reason: Power = 1/focal length (m).*

**Q11. Which one of the following is most likely to directly stimulate the release of renin?**

a) Angiotensin II
b) Increased blood pressure
**c) Decreased renal perfusion** ✓
d) Increased aldosterone secretion
e) Sodium concentration in macula densa

**Answer: c) Decreased renal perfusion**

*Reason: Low renal perfusion pressure directly stimulates JG baroreceptors.*

**Q12. Secretion of which of the hormones is directly stimulated by extracellular fluid volume expansion?**

a) Antidiuretic hormone
b) Aldosterone
**c) Atrial natriuretic peptide** ✓
d) Renin
e) Prostaglandins

**Answer: c) Atrial natriuretic peptide**

*Reason: Volume expansion stretches atria → ANP.*

**Q13. Secretion of cortisol is controlled almost entirely by**

a) Corticotropin-releasing factor
b) Mental stress
**c) Adrenocorticotropic hormone** ✓
d) Increased angiotensin II
e) Increased potassium concentration

**Answer: c) Adrenocorticotropic hormone**

*Reason: ACTH controls cortisol secretion.*

**Q14. Insulin secretion is enhanced by**

a) Growth hormone
b) Decreased level of blood amino acid
c) Increased level of serum albumin
d) Obesity
**e) Increased blood glucose level** ✓

**Answer: e) Increased blood glucose level**

*Reason: Blood glucose is the main stimulus.*

**Q15. Cutting which structure causes blindness in the temporal fields of the left and right eyes**

a) Optic nerve of right side
**b) Center of optic chiasma** ✓
c) Lateral side of optic chiasma
d) Optic tract of left side
e) Visual cortex

**Answer: b) Center of optic chiasma**

*Reason: Chiasmal midline lesion → bitemporal hemianopia.*

**Q16. Human chorionic gonadotropin hormone**

a) Is a steroid hormone
b) Acts directly on the uterus
**c) Is detected in the urine as an early sign of pregnancy** ✓
d) Level remains always the same in the whole pregnancy
e) Is responsible for the onset of labor

**Answer: c) Is detected in the urine as an early sign of pregnancy**

*Reason: hCG in urine is the basis of pregnancy tests.*

**Q17. Sensory pathway involves**

a) Corticospinal tract
b) Anterior horn cells
c) Broadman area 1,2,3
**d) Dorsal root ganglia** ✓
e) Corticorubral tract

**Answer: d) Dorsal root ganglia**

*Reason: First-order sensory neurons have cell bodies in the dorsal root ganglia.*

**Q18. Upper motor neuron lesion produces**

a) Flaccid paralysis
b) Extensive sensory loss
c) Plantar flexion
**d) Increased deep reflexes response** ✓
e) Loss of deep reflexes

**Answer: d) Increased deep reflexes response**

*Reason: UMN lesion → spastic paralysis with exaggerated reflexes.*

**Q19. During puberty which event occurs first in girls?**

a) Adrenarche
b) Active attitude
**c) Thelarche** ✓
d) Menarche
e) Pubarche

**Answer: c) Thelarche**

*Reason: Breast budding (thelarche) is the first sign of puberty in girls.*

**Q20. The most potent stimulus for ADH secretion is**

a) Decreased blood volume
**b) Increased osmolarity** ✓
c) Decreased blood pressure
d) Renin

**Answer: b) Increased osmolarity**

*Reason: Increased plasma osmolarity is the most potent ADH stimulus.*

# May 2023 — Paper I (New Curriculum)

### Q1. Pacemaker potential is

a) More slope with vagal stimulation **[ F ]**
*Reason: Vagal stimulation DEcreases the slope (more hyperpolarised, slower rate).*

b) Due to $K^+$ efflux **[ F ]**
*Reason: Pacemaker potential is due to decreased $K^+$ efflux, $Na^+$ leak and T-type $Ca^{2+}$ entry.*

c) Due to $Ca^{2+}$ channel opening **[ T ]**
*Reason: T-type $Ca^{2+}$ channels contribute to late pacemaker depolarisation.*

d) Due to $K^+$/$Ca^{2+}$ channel opening **[ T ]**
*Reason: Funny current and $Ca^{2+}$ channels shape the pacemaker potential.*

e) Shorter in sympathetic stimulation **[ T ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*

### Q2. Endocytosis is initiated by

a) Coated pits **[ T ]**
*Reason: Receptor-mediated endocytosis begins at clathrin-coated pits.*

b) Clathrin-coated vesicles **[ T ]**
*Reason: Clathrin coats vesicles in receptor-mediated endocytosis.*

c) Non-clathrin-coated vesicles **[ T ]**
*Reason: Clathrin coats vesicles in receptor-mediated endocytosis.*

d) Lysosomes **[ F ]**
*Reason: Lysosomes digest the content after endocytosis; they do not initiate it.*

e) Endosomes **[ T ]**
*Reason: Endocytosis: pit → vesicle → endosome.*

### Q3. The factors that inhibit gastric emptying are

a) CCK **[ T ]**
*Reason: CCK delays gastric emptying, stimulates gallbladder contraction and enzyme secretion.*

b) Increased acidity of duodenal chyme **[ T ]**
*Reason: Acid in duodenum inhibits emptying.*

c) Gastrocolic reflex **[ F ]**
*Reason: Gastrocolic reflex = food in stomach → colonic mass movement; enterogastric reflex inhibits emptying.*

d) Gastrin **[ F ]**
*Reason: Gastrin stimulates acid secretion and mildly motility; it does not inhibit emptying.*

e) Enterogastric reflex **[ T ]**
*Reason: Enterogastric reflex slows gastric emptying.*

### Q4. A shift of the oxygen dissociation curve of blood to the left

a) Decreases the O2 content of blood at a given PO2 **[ F ]**
*Reason: Left shift INCREASES O2 content at a given PO2 (higher affinity).*

b) Impairs O2 delivery to the tissues at the normal tissue PO2 **[ T ]**
*Reason: Left shift holds O2 tighter, impairing unloading.*

c) Occurs in blood perfusing cold extremities **[ T ]**
*Reason: Cold lowers P50 → left shift.*

d) Occurs in blood stored for several weeks **[ T ]**
*Reason: Stored blood loses 2,3-DPG → left shift.*

e) Is characteristic of fetal blood when compared with adult blood **[ T ]**
*Reason: HbF → left shift.*

### Q5. A person regarding respiratory system

a) Breathes 12–18 times/minute **[ T ]**
*Reason: Normal RR 12–18/min.*

b) Ventilates about 400 mL of air/breath **[ T ]**
*Reason: Tidal volume ≈ 500 mL (400–500).*

c) Breathes 6–8 L/minute **[ T ]**
*Reason: Minute ventilation 6–8 L/min.*

d) Inhales about 250 mL of O2/min **[ T ]**
*Reason: O2 consumption ≈ 250 mL/min.*

e) Exhales about 150 mL of $CO_2$/min **[ F ]**
*Reason: $CO_2$ output ≈ 200 mL/min.*

**Q6. At high altitude there is an increased**

a) Pulmonary ventilation **[ T ]**
*Reason: Hypoxia stimulates ventilation via carotid bodies.*

b) Alveolar $H_2O$ vapour pressure **[ F ]**
*Reason: Alveolar water vapour pressure is constant (47 mmHg).*

c) Arterial PO2 **[ F ]**
*Reason: Altitude lowers arterial PO2.*

d) Arterial pH **[ T ]**
*Reason: Hyperventilation → respiratory alkalosis → pH rises.*

e) Cerebral blood flow **[ T ]**
*Reason: Hypoxia dilates cerebral vessels.*

**Q7. Regarding the conductive system of the heart**

a) It is formed from modified nervous system **[ F ]**
*Reason: Conductive system is modified cardiac muscle.*

b) The SA node is the natural cardiac pacemaker **[ T ]**
*Reason: SA node is the natural pacemaker.*

c) Vagal stimulation results in increased rate of SA node **[ F ]**
*Reason: Vagal stimulation SLOWS SA node.*

d) AV nodal delay is pathological **[ F ]**
*Reason: AV nodal delay (0.1 s) is physiological, letting atria empty first.*

e) Conduction velocity is fastest in the Purkinje system **[ T ]**
*Reason: Purkinje fibres conduct at 1.5–4 m/s, the fastest.*

**Q8. During the cardiac cycle †**

a) The majority of the filling is due to atrial systole **[ F ]**
*Reason: ~80% of filling is passive; atrial systole gives 20%.*

b) Atrial systole starts after the P wave of ECG **[ T ]**

c) The mitral valve closes at the beginning of ventricular systole **[ T ]**
*Reason: S1 is produced at the start of ventricular systole.*

d) At the end of diastole volume is 120 mL **[ T ]**

e) Right atrial systole precedes left atrial systole **[ T ]**

**Q9. Regarding spirometry and respiratory volume**

a) FEV1 is greater than 3.5 L in the adult **[ T ]**
*Reason: Normal FEV1 ≈ 4 L (80% of FVC).*

b) Residual volume 1.2 L may be determined by spirometry **[ F ]**
*Reason: RV cannot be measured by spirometry (needs helium dilution).*

c) ERV is measured by active not passive effort **[ T ]**
*Reason: ERV is measured by active forced expiration.*

d) Alveolar exchange is 150 mL/breath **[ F ]**
*Reason: Alveolar ventilation per breath = 500 – 150 = 350 mL.*

e) With each breath 500 mL of air moves in and out of lungs **[ T ]**
*Reason: Tidal volume ≈ 500 mL.*

**Q10. Following statements regarding platelets are correct**

a) Are formed in the bone marrow **[ T ]**
*Reason: Platelets are formed from megakaryocytes in bone marrow.*

b) Are nucleated **[ F ]**
*Reason: Platelets are non-nucleated.*

c) Show adhesiveness coming in contact with collagen **[ T ]**
*Reason: Platelets adhere to exposed collagen.*

d) Do not have contractile power **[ F ]**
*Reason: Platelets contain actin/myosin/thrombosthenin (clot retraction).*

e) Help in clot retraction **[ T ]**
*Reason: Platelet thrombosthenin causes clot retraction.*

**Q11. What is the purpose of the $Na^+$-$K^+$ pump? †**

a) Perform endocytosis
b) Perform bulk transport through the plasma membrane
**c) Produce a concentration gradient for $Na^+$** ✓
d) Produce a concentration gradient for $Ca^{2+}$

**Answer: c) Produce a concentration gradient for $Na^+$**

*Reason: Pump creates the $Na^+$ gradient.*

**Q12. WBCs which are the most active phagocytic cells**

a) Lymphocytes and macrophages
b) Neutrophils and eosinophils
**c) Neutrophils and monocytes** ✓
d) Eosinophils and lymphocytes

**Answer: c) Neutrophils and monocytes**

*Reason: Neutrophils and monocytes/macrophages are the main phagocytes.*

**Q13. Most essential factor for erythropoiesis is**

a) Testosterone
b) Thyroxine
c) ACTH
**d) Erythropoietin** ✓

**Answer: d) Erythropoietin**

*Reason: EPO from kidney stimulates RBC production.*

**Q14. Which of the following is the first one to activate platelets?**

a) ADP
b) ATP
c) Thrombin
**d) Collagen** ✓

**Answer: d) Collagen**

*Reason: Collagen exposure initiates platelet adhesion/activation.*

**Q15. Prolonged bleeding time is seen in**

a) Hemophilia
b) Anemia
**c) Von Willebrand disease** ✓
d) Liver disease

**Answer: c) Von Willebrand disease**

*Reason: vWD impairs platelet adhesion → prolonged BT.*

**Q16. Site of extramedullary thrombopoiesis**

**a) Liver** ✓
b) Femur
c) Pelvic bone
d) Spine

**Answer: a) Liver**

*Reason: Liver and spleen are extramedullary haemopoietic sites.*

**Q17. Non-nucleated cell of the following erythroid series is**

**a) Reticulocyte** ✓
b) Normoblast
c) Basophilic erythroblast
d) Pro-erythroblast

**Answer: a) Reticulocyte**

*Reason: Reticulocyte has no nucleus.*

**Q18. Which of the following will not be screened before blood transfusion?**

**a) Hepatitis-A** ✓
b) Hepatitis-B
c) Hepatitis-C
d) Syphilis

**Answer: a) Hepatitis-A**

*Reason: Hepatitis A is not blood-borne.*

**Q19. Which one is the potent stimulus for the carotid body?**

a) Increased $PCO_2$
b) Increased $H^+$ concentration in arterial blood
**c) When arterial blood PO2 < 60 mmHg** ✓
d) When arterial PO2 < 100 mmHg

**Answer: c) When arterial blood PO2 < 60 mmHg**

*Reason: Carotid body is stimulated when PaO2 < 60 mmHg.*

**Q20. At which condition is oxygen therapy 100% effective? †**

a) In anemic hypoxia
**b) In atmospheric hypoxia** ✓
c) Histotoxic hypoxia
d) In hypoxia caused by tissue edema

**Answer: b) In atmospheric hypoxia**

*Reason: Hypoxic hypoxia responds fully to O2 therapy.*

# May 2023 — Paper II (New Curriculum)

### Q1. Proximal tubule

a) Reabsorbs 67% of $Na^+$ and $H_2O$ **[ T ]**
*Reason: PCT reabsorbs ~65% of filtered $Na^+$ and water.*

b) Is the site of glomerulo-tubular balance **[ T ]**
*Reason: PCT absorbs a constant fraction of GFR.*

c) Fluid is slightly iso-osmotic **[ T ]**
*Reason: PCT fluid stays iso-osmotic.*

d) Reabsorbs 50% of filtered glucose **[ F ]**
*Reason: PCT reabsorbs 100% of filtered glucose below threshold.*

e) Secretes $H^+$ and $Cl^-$ **[ F ]**
*Reason: PCT secretes $H^+$; $Cl^-$ is passively reabsorbed.*

### Q2. The renal clearance of

a) Bicarbonate is similar to that of glucose **[ T ]**
*Reason: Both are completely reabsorbed at normal plasma levels → clearance ≈ 0.*

b) PAH is nearer to 600 mL/min **[ T ]**
*Reason: PAH clearance ≈ RPF (~625 mL/min).*

c) Glucose is more than threshold **[ F ]**
*Reason: Glucose clearance is zero below threshold.*

d) Inulin is the same as GFR **[ T ]**
*Reason: Inulin is filtered, not reabsorbed or secreted.*

e) Creatinine is lower than urea **[ F ]**
*Reason: Creatinine clearance (≈ GFR) exceeds urea clearance (≈ 60–70).*

### Q3. GFR is

a) Relatively constant **[ T ]**
*Reason: Autoregulation keeps GFR constant.*

b) About 20% of the renal plasma flow **[ T ]**
*Reason: Filtration fraction ≈ 20%.*

c) Increased by angiotensin II stimulation **[ T ]**
*Reason: Ang II constricts efferent arteriole → maintains GFR.*

d) Altered by mesangial cell contraction **[ T ]**
*Reason: Mesangial contraction lowers Kf and GFR.*

e) 125 L/min **[ F ]**
*Reason: GFR is 125 mL/min (≈ 180 L/day).*

### Q4. Administration of ADH

a) Decreases water loss by lungs **[ F ]**
*Reason: ADH acts on renal collecting ducts.*

b) Decreases water loss by kidney **[ T ]**

c) Increases perspiration **[ F ]**
*Reason: ADH does not affect sweating.*

d) Decreases reabsorption of $Na^+$ by PCT **[ F ]**
*Reason: ADH does not act on PCT.*

e) Increases reabsorption of water by distal tubule **[ T ]**
*Reason: ADH acts on late DCT and collecting duct.*

### Q5. The pituitary gland

a) Lies superior to the sphenoid bone **[ T ]**
*Reason: Pituitary sits in sella turcica of the sphenoid bone.*

b) Is all ectodermal **[ T ]**
*Reason: Anterior pituitary from Rathke's pouch (ectoderm), posterior from neuroectoderm.*

c) Has the optic chiasma anteriorly **[ T ]**
*Reason: Optic chiasma lies above/in front of the pituitary stalk.*

d) Is supplied from the internal carotid artery **[ T ]**
*Reason: Pituitary is supplied by branches of the internal carotid.*

e) Lies inferior to the cavernous sinus **[ F ]**
*Reason: Pituitary lies between the cavernous sinuses (medial to them).*

**Q6. Impaired growth hormone secretion**

a) May cause delayed puberty in children **[ T ]**
*Reason: GH deficiency delays growth and puberty.*

b) Leads to mental retardation **[ F ]**
*Reason: GH deficiency does not impair intelligence (thyroid does).*

c) Can be treated by thyroxine **[ F ]**
*Reason: GH deficiency needs GH.*

d) Leads to disproportion in growth of upper and lower segment **[ F ]**
*Reason: GH deficiency gives proportionate dwarfism.*

e) Causes short stature in a young man **[ T ]**
*Reason: GH deficiency → short stature.*

**Q7. Aldosterone on renal tubule normally reabsorbs**

a) All filtered glucose **[ F ]**
*Reason: Aldosterone does not reabsorb glucose; PCT does.*

b) All filtered plasma proteins **[ F ]**
*Reason: Liver SYNTHESISES plasma proteins; it does not store them.*

c) About 5–10% of filtered water **[ T ]**

d) More $K^+$ ion than $Cl^-$ ion **[ F ]**

e) Sodium, $Cl^-$ ion and water **[ T ]**

**Q8. The cerebellum**

a) Is essential for finely coordinated movements **[ T ]**

b) Does not modify the discharge of spinal motor neurons **[ F ]**
*Reason: Cerebellum modifies spinal motor neuron discharge via descending tracts.*

c) Has afferent input from the motor cortex **[ T ]**
*Reason: Cerebellum receives corticopontocerebellar input from motor cortex.*

d) Has afferent input from the muscle proprioceptors **[ T ]**
*Reason: Spinocerebellar tracts carry proprioception.*

e) Has no afferent input from the vestibular system **[ F ]**
*Reason: Flocculonodular lobe receives vestibular input.*

**Q9. Posterior column damage in the spinal cord may impair**

a) Vibration sense **[ T ]**

b) Pain sensation **[ F ]**
*Reason: Posterior column carries fine touch, vibration, proprioception — not pain.*

c) The flexor plantar response to stimulation of the sole **[ F ]**
*Reason: Plantar reflex tests pyramidal tract, not the dorsal column.*

d) Tactile sensation **[ T ]**

e) The ability to stand steadily with eyes closed **[ T ]**

**Q10. During accommodation for near vision**

a) The curvature of the cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*

b) Chromatic and spherical aberration is decreased **[ T ]**
*Reason: Small pupil reduces aberrations.*

c) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*

d) The depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*

e) The visual axes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

**Q11. Abnormalities of taste are**

**a) Ageusia** ✓

b) Anosmia

c) Presbyacusis

d) Otitis media

**Answer: a) Ageusia**
*Reason: Ageusia = loss of taste.*

**Q12. Field of vision is more in**

a) Nasally
**b) Temporally** ✓
c) Upper quadrant
d) Lower quadrant

**Answer: b) Temporally**
*Reason: Temporal visual field is widest (nose does not obstruct).*

**Q13. Growth hormone secretion is mostly increased in**

a) Hypoglycemia
b) Severe protein deficiency
**c) During the first hours of sleep** ✓
d) Excitement

**Answer: c) During the first hours of sleep**
*Reason: GH peaks during slow-wave sleep.*

**Q14. Thermogenic hormones are**

a) Somatomedin C
b) PTH
c) Estrogen
**d) Thyroxine** ✓

**Answer: d) Thyroxine**
*Reason: Thyroxine increases BMR and heat production.*

**Q15. Brain stem controls**

a) Temperature
**b) Equilibrium** ✓
c) Respiratory volumes
d) Muscle tone only

**Answer: b) Equilibrium**
*Reason: Brainstem (vestibular nuclei) controls equilibrium, respiration, CVS.*

**Q16. Placenta does NOT secrete**

a) Estrogen
b) HCG
**c) Prolactin** ✓
d) Progesterone

**Answer: c) Prolactin**
*Reason: Placenta secretes hPL, not prolactin.*

**Q17. Knee jerk is**

a) Lost in cerebellar lesion
**b) Caused by stretching of the quadriceps femoris** ✓
c) A superficial reflex
d) A polysynaptic reflex

**Answer: b) Caused by stretching of the quadriceps femoris**
*Reason: Knee jerk is the monosynaptic stretch reflex of the quadriceps.*

**Q18. Regarding pain, which is incorrect?**

a) Is protective in nature
b) Is regulated by multiple systems
c) Receptors are free nerve endings
**d) Receptors are quickly adapted** ✓

**Answer: d) Receptors are quickly adapted**
*Reason: Pain receptors adapt very little (slowly).*

**Q19. Basal ganglia lesion does NOT produce**

**a) Past pointing** ✓
b) Tremor
c) Rigidity
d) Hypokinesia

**Answer: a) Past pointing**

*Reason: Past pointing is a cerebellar sign.*

**Q20. In presence of ADH, maximum water is reabsorbed in**

a) Distal tubules
**b) PCT** ✓
c) Loop of Henle
d) Cortical collecting ducts

**Answer: b) PCT**

*Reason: PCT reabsorbs the greatest fraction (~65%) of filtered water.*

# May 2023 — Paper I (Old Curriculum)

### Q1. $Na^+$-$K^+$ pump

a) Is a passive transport **[ F ]**
*Reason: $Na^+$-$K^+$ pump is primary active transport using ATP.*
b) Maintains cell volume **[ T ]**
c) Is present in all cell membranes **[ T ]**
*Reason: All cells have the $Na^+$-$K^+$ pump.*
d) Generates energy for co-transport **[ T ]**
*Reason: The $Na^+$ gradient made by the pump drives secondary active transport.*
e) Requires no carrier protein **[ F ]**
*Reason: Pump is a carrier protein ($Na^+$-$K^+$ ATPase).*

### Q2. Lipid bilayer of cell membrane

a) Is fluid in nature **[ T ]**
b) Prevents passage of $Ca^{2+}$ **[ T ]**
c) Is composed of phospholipid and cholesterol **[ T ]**
d) Has hydrophilic phosphate ends **[ T ]**
e) Is coated internally with glycocalyx **[ F ]**
*Reason: Glycocalyx coats the outside of the membrane.*

### Q3. Contractile protein

a) Myosin forms thin filament **[ F ]**
*Reason: Actin forms thin filaments; myosin forms thick.*
b) Troponin C is for binding of $Ca^{2+}$ **[ T ]**
c) Spectrin maintains biconcavity of RBC **[ T ]**
d) Is associated with muscle contraction **[ T ]**
e) Tropomyosin is a component of myosin filament **[ F ]**
*Reason: Tropomyosin is on the thin (actin) filament.*

### Q4. Skeletal muscle

a) Membrane is sarcolemma **[ T ]**
b) Has well developed T-tubules **[ T ]**
c) Is involuntary in action **[ F ]**
*Reason: Skeletal muscle is voluntary.*
d) Is innervated by autonomic nerve **[ F ]**
*Reason: Skeletal muscle is somatic-nerve supplied.*
e) Sarcoplasmic reticulum stores $Ca^{2+}$ for contraction **[ T ]**

### Q5. Neutrophils

a) Are phagocytic cells **[ T ]**
*Reason: Neutrophils are professional phagocytes.*
b) Have a life span of 4–8 hours in circulating blood **[ T ]**
*Reason: Neutrophils live 4–8 h in circulation.*
c) Count increases in allergic disorders **[ F ]**
*Reason: Eosinophils (not neutrophils) increase in allergy.*
d) Secrete platelet activating factor **[ F ]**
*Reason: PAF comes from endothelium, platelets, macrophages — not neutrophils' main role.*
e) Play a main role in immune system **[ T ]**
*Reason: Lymphocytes are the main immune cells; neutrophils are innate.*

### Q6. Haemolytic disease of the newborn

a) Affects babies of Rh-ve mothers **[ T ]**
*Reason: Rh-negative mother with Rh-positive fetus → anti-D.*
b) Causes jaundice **[ T ]**
c) Can be treated by exchange transfusion **[ T ]**
*Reason: Exchange transfusion treats severe HDN.*
d) Causes severe anemia **[ T ]**
*Reason: Haemolysis causes anaemia in HDN.*
e) Is an ABO incompatibility **[ F ]**
*Reason: HDN is Rh incompatibility (ABO is milder).*

**Q7. Blood coagulation requires**

a) Heparin **[ F ]**
*Reason: Heparin is an anticoagulant.*

b) Fibrinolysis **[ F ]**
*Reason: Fibrinolysis dissolves clots; it is not needed for coagulation.*

c) Ionized calcium **[ T ]**
*Reason: $Ca^{2+}$ is factor IV in coagulation.*

d) An adequate intake of vitamin K **[ T ]**

e) Anti-hemophilic factors **[ T ]**
*Reason: Factor VIII is required in the intrinsic pathway.*

**Q8. Microcytic hypochromic RBC are found in**

a) Acute blood loss **[ F ]**
*Reason: Acute blood loss gives normocytic anaemia.*

b) Folic acid deficiency **[ F ]**
*Reason: Folate deficiency → megaloblastic anaemia.*

c) Thalassemia **[ T ]**

d) Aplasia of bone marrow **[ F ]**
*Reason: Aplastic anaemia is normocytic.*

e) Vitamin $B_{12}$ deficiency **[ F ]**
*Reason: $B_{12}$ deficiency → macrocytic (megaloblastic) anaemia.*

**Q9. Regarding O negative blood group**

a) A and B antigens on RBC membrane **[ F ]**
*Reason: O group has no A or B antigens.*

b) Has no antibody to ABO system in blood **[ F ]**
*Reason: O group has both anti-A and anti-B.*

c) Has no antibody to Rh system **[ T ]**
*Reason: Rh antibodies form only after exposure.*

d) 2nd part of Landsteiner's law is not obeyed here **[ F ]**
*Reason: O group obeys Landsteiner's law (has both antibodies).*

e) Referred to as universal donor **[ T ]**
*Reason: O-negative RBC lack A, B, D antigens.*

**Q10. End diastolic volume is**

a) The volume remaining in the ventricle after ventricular systole **[ F ]**
*Reason: That is end-systolic volume; EDV is volume at the end of diastole.*

b) About 110–120 mL **[ T ]**

c) Proportional to venous return **[ T ]**

d) Proportional to ejection fraction **[ F ]**
*Reason: EDV is not proportional to EF.*

e) Inversely related to stroke volume **[ F ]**
*Reason: Larger EDV → larger SV (Starling).*

**Q11. Following statements are correct**

a) Arterioles are resistance vessels **[ T ]**
*Reason: Arterioles are the main resistance vessels.*

b) BP = CO × SV **[ F ]**
*Reason: BP = CO × TPR.*

c) Veins are capacitance vessels **[ T ]**
*Reason: Veins hold ~60–65% of blood.*

d) Epinephrine is a vasoconstrictor **[ T ]**
*Reason: Epinephrine constricts skin vessels; dilates muscle vessels.*

e) Velocity profile of laminar flow is parabolic **[ T ]**
*Reason: Laminar flow has a parabolic velocity profile.*

### Q12. Following statements are true

a) Mean BP is the maximum BP **[ F ]**
*Reason: Mean BP ≈ diastolic + 1/3 pulse pressure.*

b) Systolic BP is calculated pressure **[ F ]**
*Reason: Systolic BP is measured; mean BP is calculated.*

c) BP can be measured by Fick's principle **[ F ]**
*Reason: Fick's principle measures cardiac output/blood flow, not BP.*

d) In hyperthyroidism pulse pressure is increased **[ T ]**
*Reason: Hyperthyroidism → ↑ systolic, ↓ diastolic → wide pulse pressure.*

e) Increased right atrial pressure decreases VR **[ T ]**
*Reason: Raised RAP reduces venous return.*

### Q13. Ejection fraction of the heart is

a) A valuable index of ventricular function **[ T ]**
*Reason: EF = SV/EDV ≈ 60–65%; falls in failure.*

b) The percentage of ESV ejected per stroke **[ F ]**
*Reason: EF is the percentage of EDV (not ESV) ejected.*

c) Reduced in exercise **[ F ]**
*Reason: EF increases in exercise.*

d) Reduced in myocardial infarction **[ T ]**
*Reason: EF falls with myocardial damage.*

e) About 65% **[ T ]**
*Reason: Normal EF ≈ 60–65%.*

### Q14. P-R interval

a) Indicates conduction through A-V node **[ T ]**
*Reason: PR interval = AV conduction time.*

b) Is about 0.8 sec **[ F ]**
*Reason: PR is 0.12–0.20 s; 0.8 s is the whole cardiac cycle.*

c) Decreases in bradycardia **[ F ]**
*Reason: PR lengthens slightly in bradycardia.*

d) 0.25–0.45 sec indicates 2nd degree heart block **[ F ]**
*Reason: PR > 0.2 s = 1st-degree block; 2nd degree has dropped beats.*

e) Prolonged in hypokalemia **[ F ]**
*Reason: Hypokalaemia mainly causes ST depression, flat T, U waves.*

### Q15. In obstructive airway disease

a) Residual volume is decreased **[ F ]**
*Reason: Residual volume increases in obstructive disease.*

b) FEV1 is markedly reduced **[ T ]**

c) Total lung capacity is markedly increased **[ T ]**

d) FEV1/FVC ratio is normal **[ F ]**
*Reason: FEV1/FVC is reduced in obstruction.*

e) Only airway tube is involved **[ T ]**

### Q16. Non-respiratory functions of lungs

a) Provide immunity by IgA **[ T ]**

b) Secrete prostaglandin **[ T ]**

c) Transport O2 and $CO_2$ **[ F ]**
*Reason: That is the respiratory function.*

d) Conversion of angiotensin-I to angiotensin-II **[ T ]**

e) Secrete renin **[ F ]**
*Reason: Lungs make ACE, not renin.*

### Q17. The residual volume

a) Is 3–4 L on average in young adults **[ F ]**
*Reason: RV ≈ 1.2 L.*

b) Is smaller in old than in young **[ F ]**
*Reason: Residual volume increases with age.*

c) Is measured directly using a spirometer **[ F ]**
*Reason: RV needs helium dilution/plethysmography.*

d) Is the volume remaining after quiet expiration **[ F ]**
*Reason: That is FRC.*

e) Is increased in COPD patients **[ T ]**
*Reason: Emphysema loses elastic recoil → compliance increases.*

### Q18. During exercise

a) P50 decreases **[ F ]**
*Reason: Exercise raises temperature, $CO_2$, $H^+$ → P50 increases.*

b) pH increases **[ F ]**
*Reason: Exercise lowers pH (lactate, $CO_2$).*

c) Rate of respiration increases **[ T ]**
*Reason: Exercise increases rate and depth.*

d) Tissue PO2 increases **[ F ]**
*Reason: Tissue PO2 falls in exercise.*

e) Oxyhemoglobin dissociation curve shifts to the right **[ T ]**
*Reason: Exercise shifts ODC right (Bohr effect).*

### Q19. The rate of blood flow is

a) Directly proportional to pressure differences **[ T ]**
*Reason: Flow ∝ pressure difference (Ohm's law).*

b) Regulated by vasoconstrictor agents **[ T ]**
*Reason: Vasomotor agents regulate flow.*

c) Not related to viscosity **[ F ]**
*Reason: Flow ∝ 1/viscosity.*

d) Inversely proportional to the resistance **[ T ]**
*Reason: Flow = ΔP/R.*

e) Inversely proportional to velocity of blood flow **[ F ]**
*Reason: Flow = velocity × area; they are directly related.*

### Q20. Regarding gastric juice

a) Secretion increases by enterokinase **[ F ]**
*Reason: Enterokinase is intestinal.*

b) Contains the major enzyme for CHO digestion **[ F ]**
*Reason: Salivary/pancreatic amylase digest carbohydrate; gastric juice has none.*

c) Contains proteolytic enzyme **[ T ]**

d) Gastrin causes increased HCl acid secretion **[ T ]**

e) Contains disaccharidase **[ F ]**
*Reason: Disaccharidases are brush-border enzymes.*

# May 2023 — Paper II (Old Curriculum)

**Q1. Posterior column damage in the spinal cord may impair**

a) Vibration sense **[ T ]**
b) Pain sensation **[ F ]**
*Reason: Posterior column carries fine touch, vibration, proprioception — not pain.*
c) The flexor plantar response to stimulation of the sole **[ F ]**
*Reason: Plantar reflex tests pyramidal tract, not the dorsal column.*
d) Tactile sensation **[ T ]**
e) The ability to stand steadily with eyes closed **[ T ]**

**Q2. Parasympathetic stimulation results in**

a) Pupillary constriction **[ T ]**
b) Decreased heart rate **[ T ]**
c) Contraction of bronchial smooth muscle **[ T ]**
d) Secretion of gastric juice **[ T ]**
e) Defecation **[ T ]**

**Q3. During accommodation for near vision**

a) The curvature of the cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*
b) Chromatic and spherical aberration is decreased **[ T ]**
*Reason: Small pupil reduces aberrations.*
c) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*
d) The depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*
e) The visual axes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

**Q4. The pituitary gland**

a) Lies superior to the sphenoid bone **[ T ]**
*Reason: Pituitary sits in sella turcica of the sphenoid bone.*
b) Is all ectodermal **[ T ]**
*Reason: Anterior pituitary from Rathke's pouch (ectoderm), posterior from neuroectoderm.*
c) Has the optic chiasma anteriorly **[ T ]**
*Reason: Optic chiasma lies above/in front of the pituitary stalk.*
d) Is supplied from the internal carotid artery **[ T ]**
*Reason: Pituitary is supplied by branches of the internal carotid.*
e) Lies inferior to the cavernous sinus **[ F ]**
*Reason: Pituitary lies between the cavernous sinuses (medial to them).*

**Q5. The loop of Henle**

a) Is responsible for high medullary osmolarity **[ T ]**
*Reason: Loop of Henle counter-current multiplier builds the gradient.*
b) Returns fluid of low osmolarity to the DCT **[ T ]**
*Reason: Fluid leaving the loop is hypo-osmotic.*
c) Has a high permeability to water in thick ascending limb **[ F ]**
*Reason: Thick ascending limb is impermeable to water.*
d) Actively pumps ions in the descending limb **[ F ]**
*Reason: Descending limb is passive.*
e) Actively transports $Na^+$ in the thick ascending limb **[ T ]**
*Reason: NKCC2 transports $Na^+$-$K^+$-2Cl- in the thick ascending limb.*

### Q6. The cerebellum

a) Is essential for finely coordinated movements **[ T ]**
b) Does not modify the discharge of spinal motor neurons **[ F ]**
*Reason: Cerebellum modifies spinal motor neuron discharge via descending tracts.*
c) Has afferent input from the motor cortex **[ T ]**
*Reason: Cerebellum receives corticopontocerebellar input from motor cortex.*
d) Has afferent input from the muscle proprioceptors **[ T ]**
*Reason: Spinocerebellar tracts carry proprioception.*
e) Has no afferent input from the vestibular system **[ F ]**
*Reason: Flocculonodular lobe receives vestibular input.*

### Q7. Parathormone

a) Is produced by the cells of the thyroid gland **[ F ]**
*Reason: PTH is made by chief cells of parathyroids.*
b) Decreases urinary excretion of calcium **[ T ]**
c) Blood level increases when serum calcium falls **[ T ]**
d) Depresses activity of the anterior pituitary **[ F ]**
*Reason: PTH does not affect the anterior pituitary.*
e) Is independent of magnesium level **[ F ]**
*Reason: Low $Mg^{2+}$ impairs PTH secretion and action.*

### Q8. During normal menstrual cycle

a) Oestrogen is predominant during the proliferative phase **[ T ]**
*Reason: Proliferative phase = estrogen.*
b) Menstrual blood loss averages 150 mL **[ F ]**
*Reason: Menstrual loss averages 35–50 mL.*
c) Cervical mucus is more viscous at the time of ovulation **[ F ]**
*Reason: Cervical mucus is thin, watery and elastic (ferning) at ovulation.*
d) Menstruation is preceded by a rise in plasma progesterone **[ F ]**
*Reason: Menstruation follows a FALL in progesterone.*
e) Menstruation is associated with dilatation of spiral arteries **[ F ]**
*Reason: Menstruation follows spiral artery constriction/ischaemia.*

### Q9. Human chorionic gonadotrophin (HCG)

a) Has the biological properties of LH **[ T ]**
*Reason: hCG acts on the LH receptor.*
b) Is secreted from ovary **[ F ]**
*Reason: hCG comes from syncytiotrophoblast.*
c) Is responsible for the onset of parturition **[ F ]**
*Reason: hCG is not related to parturition.*
d) Is detectable before implantation of the ovum **[ F ]**
*Reason: hCG secretion begins after implantation.*
e) Is significantly elevated in pre-eclampsia **[ F ]**
*Reason: hCG is not a diagnostic/significantly elevated marker in pre-eclampsia.*

### Q10. Progesterone

a) Is natriuretic **[ T ]**
*Reason: Progesterone is natriuretic (blocks aldosterone).*
b) Requires estrogen priming **[ T ]**
*Reason: Progesterone receptors are induced by estrogen.*
c) Is synthesized in the adrenal gland (as its main source) **[ F ]**
*Reason: Main sources are corpus luteum/placenta.*
d) Decreases uterine muscle contraction **[ T ]**
*Reason: Progesterone quietens the uterus.*
e) Is the hormone of pregnancy **[ T ]**
*Reason: Progesterone maintains pregnancy.*

### Q11. Thyroxine

a) Is less than 80% bound to plasma proteins **[ F ]**
*Reason: $T_4$ is > 99% protein bound.*

b) Formation requires amino acids **[ T ]**
*Reason: Tyrosine is incorporated in thyroglobulin.*

c) Is essential for skeletal development **[ T ]**
*Reason: Thyroxine is needed for bone growth.*

d) Increases oxygen consumption in the brain **[ F ]**
*Reason: Brain O2 use is unaffected by thyroxine.*

e) Causes increased serum cholesterol in hyperthyroidism **[ F ]**
*Reason: Thyroxine LOWERS cholesterol.*

### Q12. Aldosterone causes

a) A fall in urinary sodium excretion **[ T ]**

b) Weight gain **[ T ]**

c) A fall in serum chloride levels **[ F ]**
*Reason: Aldosterone retains $Na^+$ (and $Cl^-$ follows); serum $Cl^-$ does not fall.*

d) Increased extracellular fluid volume **[ T ]**

e) Increased potassium excretion **[ T ]**

### Q13. Organ of Corti

a) Is the receptor organ of hearing **[ T ]**
*Reason: Organ of Corti is the receptor organ of hearing.*

b) Is the receptor organ of taste **[ F ]**
*Reason: Taste buds are the taste receptors.*

c) Is the receptor organ of smell **[ F ]**
*Reason: Organ of Corti is for hearing; smell receptors are in olfactory epithelium.*

d) Generates nerve impulse **[ T ]**
*Reason: Hair cells generate receptor potentials → impulses.*

e) Lies on Reissner's membrane **[ F ]**
*Reason: Organ of Corti lies on the basilar membrane.*

### Q14. The pheochromocytoma

a) Is a tumour of adrenal medullary tissue **[ T ]**

b) Is benign in the majority of cases **[ T ]**

c) Has its maximum incidence after the menopause **[ F ]**
*Reason: Peak incidence is 3rd–5th decade.*

d) Can be associated with severe episodes of hypertension **[ T ]**

e) Is always found in the gland **[ F ]**
*Reason: ~10% of pheochromocytomas are extra-adrenal.*

### Q15. Growth hormone and IGF-1 cause

a) Decreased insulin sensitivity **[ T ]**
*Reason: GH and IGF-1 are diabetogenic.*

b) $K^+$ retention **[ T ]**
*Reason: GH causes $K^+$ retention.*

c) Protein synthesis **[ T ]**
*Reason: GH increases protein synthesis.*

d) Diaphysis closure **[ F ]**
*Reason: GH causes diabetogenic effects; epiphyseal closure requires puberty.*

e) Lipolysis **[ T ]**
*Reason: GH causes lipolysis.*

**Q16. During the process of fertilization and implantation**

a) Fertilisation occurs in the outer third of the Fallopian tube **[ T ]**
b) The haploid number of chromosomes results **[ F ]**
*Reason: Fertilisation restores the DIPLOID number.*
c) The morula is enclosed by trophoblast **[ F ]**
*Reason: Blastocyst (not morula) has a trophoblast.*
d) The morula actively moves along the oviduct **[ F ]**
*Reason: Morula is moved by cilia/peristalsis.*
e) HCG is produced by the preimplantation embryo **[ F ]**
*Reason: hCG appears after implantation (about day 8–10).*

**Q17. The nerves that carry taste impulses are**

a) Facial nerve **[ T ]**
b) Trigeminal nerve **[ F ]**
*Reason: Taste fibres travel in VII, IX, X; V gives general sensation.*
c) Glossopharyngeal **[ T ]**
d) Vagus nerve **[ T ]**
e) Hypoglossal **[ F ]**
*Reason: Hypoglossal is motor to the tongue.*

**Q18. Lateral spinothalamic tract**

a) Is formed by A alpha fibre **[ F ]**
*Reason: Spinothalamic tract uses Aδ and C fibres.*
b) Carries proprioception **[ F ]**
*Reason: Proprioception travels in dorsal columns/spinocerebellar tracts.*
c) Transmits pain sensation **[ T ]**
d) Transmits thermal sensation **[ T ]**
e) Transmits crude touch sensation **[ F ]**
*Reason: Crude touch travels in the anterior spinothalamic tract.*

**Q19. Basal ganglia**

a) Perform functions in association with the motor control system **[ T ]**
b) Can also perform function by itself **[ F ]**
*Reason: Basal ganglia work with cortex/thalamus, not alone.*
c) Lesion at putamen circuit causes Parkinson's disease **[ F ]**
*Reason: Parkinson's disease increases muscle tone (rigidity).*
d) Damage causes decreased tone of peripheral muscles **[ F ]**
*Reason: Basal ganglia damage increases tone (rigidity).*
e) Controls complex patterns of movements **[ T ]**

**Q20. Short stature is seen in children suffering from**

a) Adrenal insufficiency **[ T ]**
b) Thyroxine deficiency **[ T ]**
c) Growth hormone deficiency **[ T ]**
d) Chronic malnutrition **[ T ]**
e) Castration **[ F ]**
*Reason: Castration delays epiphyseal closure → tall stature.*

# November 2022 — Paper II

### Q1. Substances completely reabsorbed from PCT are

a) $Na^+$ **[ F ]**
*Reason: Only ~65% of $Na^+$ is reabsorbed in PCT.*

b) Glucose **[ T ]**
*Reason: Glucose, amino acids, vitamins, iron are absorbed in upper small intestine.*

c) $K^+$ **[ F ]**
*Reason: ~65% of $K^+$ in PCT; rest later; DCT/CD secrete $K^+$.*

d) Amino acid **[ T ]**
*Reason: Amino acids are completely reabsorbed in PCT.*

e) Acetone **[ F ]**
*Reason: Acetone is not completely reabsorbed.*

### Q2. In presence of ADH, maximum water is reabsorbed in

a) Distal tubules **[ F ]**
*Reason: Maximum fraction of water is reabsorbed in PCT.*

b) PCT **[ T ]**

c) Loop of Henle **[ F ]**
*Reason: Loop reabsorbs ~15% of water.*

d) Cortical collecting ducts **[ T ]**

e) Medullary collecting ducts **[ T ]**

### Q3. During accommodation for near vision

a) The curvature of the cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*

b) Chromatic and spherical aberration is decreased **[ T ]**
*Reason: Small pupil reduces aberrations.*

c) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*

d) The depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*

e) The visual axes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

### Q4. The pituitary gland

a) Lies superior to the sphenoid bone **[ T ]**
*Reason: Pituitary sits in sella turcica of the sphenoid bone.*

b) Is ectodermal **[ T ]**
*Reason: Both parts derive from ectoderm.*

c) Has the optic chiasma anteriorly **[ T ]**
*Reason: Optic chiasma lies above/in front of the pituitary stalk.*

d) Is supplied from the internal carotid artery **[ T ]**
*Reason: Pituitary is supplied by branches of the internal carotid.*

e) Lies inferior to the cavernous sinus **[ F ]**
*Reason: Pituitary lies between the cavernous sinuses (medial to them).*

### Q5. The loop of Henle

a) Is responsible for high medullary osmolarity **[ T ]**
*Reason: Loop of Henle counter-current multiplier builds the gradient.*

b) Returns fluid of low osmolarity to the DCT **[ T ]**
*Reason: Fluid leaving the loop is hypo-osmotic.*

c) Has a high permeability to water in thick ascending limb **[ F ]**
*Reason: Thick ascending limb is impermeable to water.*

d) Actively pumps ions across the epithelium of the descending limb **[ F ]**
*Reason: Descending limb is water-permeable, passive; thick ascending limb transports NaCl actively.*

e) Actively transports $Na^+$ in the thick ascending limb **[ T ]**
*Reason: NKCC2 transports $Na^+$-$K^+$-2Cl- in the thick ascending limb.*

### Q6. Tropic hormones are

a) LH **[ T ]**
b) FSH **[ T ]**
*Reason: FSH is from the anterior pituitary.*
c) Insulin **[ F ]**
*Reason: Insulin increases glucose facilitated diffusion 10–20 fold (GLUT-4).*
d) TSH **[ T ]**
*Reason: TSH acts via cAMP.*
e) Epinephrine **[ F ]**
*Reason: Epinephrine is not a tropic hormone.*

### Q7. Placenta secretes

a) Estrogen **[ T ]**
*Reason: Estrogen causes ductal growth.*
b) Oxytocin **[ F ]**
*Reason: Oxytocin acts through IP3/$Ca^{2+}$.*
c) Progesterone **[ T ]**
d) Prolactin **[ F ]**
*Reason: Placenta secretes hPL, not prolactin.*
e) HCG **[ T ]**

### Q8. Regarding spermatogenesis

a) Normal sperm count is 10–15 million/mL of semen **[ F ]**
*Reason: Normal count ≥ 15–20 million/mL (average ~100 million).*
b) Testosterone helps growth and division of germinal cells **[ T ]**
*Reason: Testosterone supports spermatogenesis.*
c) LH stimulates testosterone secretion by Leydig cells **[ T ]**
*Reason: LH → Leydig cells → testosterone.*
d) FSH stimulates Sertoli cells **[ T ]**
*Reason: FSH acts on Sertoli cells.*
e) Sperms get motile in seminiferous tubule **[ F ]**
*Reason: Sperms become motile in the epididymis.*

### Q9. Human chorionic gonadotrophin (HCG)

a) Has the biological properties of LH **[ T ]**
*Reason: hCG acts on the LH receptor.*
b) Can transform the cyclic corpus luteum into a pseudopregnant corpus luteum **[ T ]**
*Reason: hCG maintains the corpus luteum of pregnancy.*
c) Is responsible for the onset of parturition **[ F ]**
*Reason: hCG is not related to parturition.*
d) Is detectable in maternal serum before implantation of the ovum **[ F ]**
*Reason: hCG secretion begins after implantation.*
e) Is significantly elevated in pre-eclampsia **[ F ]**
*Reason: hCG is not a diagnostic/significantly elevated marker in pre-eclampsia.*

### Q10. Progesterone

a) Is natriuretic **[ T ]**
*Reason: Progesterone is natriuretic (blocks aldosterone).*
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*Reason: Progesterone receptors are induced by estrogen.*
c) Is synthesized in the adrenal gland (as its main source) **[ F ]**
*Reason: Main sources are corpus luteum/placenta.*
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*Reason: $T_4$ is > 99% protein bound.*

b) Formation requires amino acids **[ T ]**
*Reason: Tyrosine is incorporated in thyroglobulin.*

c) Is essential for skeletal development **[ T ]**
*Reason: Thyroxine is needed for bone growth.*

d) Increases oxygen consumption in the brain **[ F ]**
*Reason: Brain O2 use is unaffected by thyroxine.*

e) Leads to increased serum cholesterol when present in excess **[ F ]**
*Reason: Thyroxine LOWERS cholesterol.*

**Q12. Aldosterone causes**

a) A fall in urinary sodium excretion **[ T ]**

b) Weight gain **[ T ]**

c) A fall in serum chloride levels **[ F ]**
*Reason: Aldosterone retains $Na^+$ (and $Cl^-$ follows); serum $Cl^-$ does not fall.*

d) Increased extracellular fluid volume **[ T ]**

e) Increased potassium excretion **[ T ]**

**Q13. Testes can produce**

a) Estradiol **[ T ]**
*Reason: Testes make small amounts of estradiol (aromatisation).*

b) Inhibin **[ T ]**
*Reason: Sertoli cells secrete inhibin.*

c) Fructose **[ F ]**
*Reason: Fructose is made by seminal vesicles.*

d) FSH **[ F ]**
*Reason: FSH is from the anterior pituitary.*

e) Testosterone **[ T ]**
*Reason: Steroid hormones act via intracellular receptors (gene transcription).*

**Q14. Superficial reflexes are**

a) Corneal reflex **[ T ]**

b) Knee jerk **[ F ]**
*Reason: Knee jerk is a deep tendon (stretch) reflex.*

c) Ankle jerk **[ F ]**
*Reason: Ankle jerk is a deep tendon (stretch) reflex.*

d) Cremasteric reflex **[ T ]**

e) Biceps jerk **[ F ]**
*Reason: Biceps jerk is a deep tendon reflex.*

**Q15. Abnormalities of smell are**

a) Anosmia **[ T ]**
*Reason: Anosmia = loss of smell.*

b) Hyposmia **[ T ]**
*Reason: Hyposmia = decreased smell.*

c) Dysosmia **[ T ]**
*Reason: Dysosmia = distorted smell.*

d) Hypogeusia **[ F ]**
*Reason: Hypogeusia = decreased taste.*

e) Ageusia **[ F ]**
*Reason: Ageusia = loss of taste.*

**Q16. During the process of fertilization and implantation**

a) Fertilisation occurs in the outer third of the Fallopian tube **[ T ]**
b) The haploid number of chromosomes results **[ F ]**
*Reason: Fertilisation restores the DIPLOID number.*
c) The morula is enclosed by trophoblast **[ F ]**
*Reason: Blastocyst (not morula) has a trophoblast.*
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*Reason: Morula is moved by cilia/peristalsis.*
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a) Is formed by A alpha fibre **[ F ]**
*Reason: Spinothalamic tract uses Aδ and C fibres.*
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a) Perform functions in association with the motor control system **[ T ]**
b) Can also perform function by itself **[ F ]**
*Reason: Basal ganglia work with cortex/thalamus, not alone.*
c) Lesion at putamen circuit causes Parkinson's disease **[ F ]**
*Reason: Parkinson's disease increases muscle tone (rigidity).*
d) Damage causes decreased tone of peripheral muscles **[ F ]**
*Reason: Basal ganglia damage increases tone (rigidity).*
e) Controls complex patterns of movements **[ T ]**

**Q20. Short stature is seen in children suffering from**

a) Adrenal insufficiency **[ T ]**
b) Thyroxine deficiency **[ T ]**
c) Growth hormone deficiency **[ T ]**
d) Chronic malnutrition **[ T ]**
e) Castration **[ F ]**
*Reason: Castration delays epiphyseal closure → tall stature.*

# May 2022 — Paper I

### Q1. Cell membrane proteins act as

a) Anchors **[ T ]**
*Reason: Membrane proteins anchor cytoskeleton and cells.*

b) Pumps **[ T ]**
*Reason: Membrane proteins act as pumps ($Na^+$-$K^+$ ATPase).*

c) Transporters of ions **[ T ]**
*Reason: Membrane proteins transport ions.*

d) Hormones **[ F ]**
*Reason: Membrane proteins are receptors for hormones; they are not hormones.*

e) Enzymes **[ T ]**
*Reason: Some membrane proteins are enzymes.*

### Q2. Regarding action potential

a) With plateau occurs in some unitary smooth muscle **[ T ]**
*Reason: Unitary smooth muscle shows spikes and plateau APs.*

b) There are two forms in visceral smooth muscle **[ T ]**
*Reason: Visceral smooth muscle: spike potentials and plateau APs.*

c) It sometimes follows the all-or-none law **[ F ]**
*Reason: AP always obeys all-or-none law.*

d) Recording with surface electrodes is biphasic **[ T ]**
*Reason: Extracellular (surface) recording gives a biphasic record.*

e) In cardiac muscle has a long refractory period **[ T ]**
*Reason: Cardiac AP has a long plateau → long refractory period.*

### Q3. Mitochondria

a) Are variable in size and shape **[ T ]**

b) Are the chief sites of protein synthesis **[ F ]**
*Reason: Ribosomes make protein.*

c) Are present in the reticulocyte stage of erythropoiesis **[ T ]**

d) Synthesize ATP in erythrocyte **[ F ]**
*Reason: Mature RBC lacks mitochondria; ATP comes from glycolysis.*

e) Are more in skeletal muscle **[ T ]**

### Q4. Regarding diffusion, the following statements are correct

a) Always uphill movement **[ F ]**
*Reason: Diffusion is always downhill.*

b) Is a passive process **[ T ]**
*Reason: Diffusion/facilitated diffusion needs no ATP.*

c) Is energy dependent **[ F ]**
*Reason: Diffusion is passive.*

d) Is always carrier mediated **[ F ]**
*Reason: Simple diffusion needs no carrier; facilitated diffusion does.*

e) Occurs through protein pores and channels **[ T ]**
*Reason: Diffusion occurs through lipid bilayer or protein channels.*

### Q5. Regarding hemoglobin

a) Hb F has high affinity to O2 **[ T ]**
*Reason: HbF has higher O2 affinity (lower P50).*

b) HbA1 consists of 2α and 2γ chains **[ F ]**
*Reason: HbA1 = 2α2β.*

c) $HbA_{1c}$ measures past three months average blood sugar **[ T ]**
*Reason: $HbA_{1c}$ reflects the previous ~3 months' glucose.*

d) Fetal Hb can bind 2,3-DPG less avidly **[ T ]**
*Reason: HbF binds 2,3-DPG poorly, so its curve is left-shifted.*

e) One hemoglobin molecule carries two O2 molecules **[ F ]**
*Reason: One Hb carries 4 O2 molecules.*

### Q6. Blood platelets

a) Are formed in bone marrow from megakaryocytes **[ T ]**
*Reason: Megakaryocyte fragmentation forms platelets.*

b) Alpha granules contain ADP **[ F ]**
*Reason: ADP is in dense (δ) granules; α granules contain fibrinogen, PDGF, vWF.*

c) Dense bodies contain serotonin **[ T ]**
*Reason: Dense bodies contain ADP, ATP, $Ca^{2+}$, serotonin.*

d) Have a half-life in blood of 8 to 12 days **[ T ]**
*Reason: Platelet life ≈ 8–12 days.*

e) Alter shape when in contact with collagen **[ T ]**
*Reason: Platelets change shape (spiculate) on contact with collagen.*

### Q7. Bleeding disorder

a) Occurs in vitamin K deficiency **[ T ]**
*Reason: Vitamin K deficiency lowers II, VII, IX, X → bleeding.*

b) Is a consequence of thrombocytopenia **[ T ]**
*Reason: Thrombocytopenia → purpura/bleeding.*

c) Is diagnosed by estimating fibrin degradation product **[ T ]**
*Reason: FDP rise in DIC/fibrinolysis.*

d) Is only caused by congenital vascular defect **[ F ]**
*Reason: Bleeding arises from platelet, clotting factor or vascular causes.*

e) Can be screened by BT CT test **[ T ]**
*Reason: BT (platelets) and CT (clotting) are screening tests.*

### Q8. Following statements regarding anemia are correct

a) In severe anemia blood viscosity is increased **[ F ]**
*Reason: Anaemia lowers viscosity.*

b) Decreases the resistance to blood flow **[ T ]**
*Reason: Anaemia lowers viscosity → resistance falls.*

c) Increased 2,3-DPG level **[ T ]**
*Reason: Anaemia/hypoxia raises 2,3-DPG.*

d) Microcytic hypochromic type in acute blood loss **[ F ]**
*Reason: Acute blood loss gives normocytic anaemia; chronic loss gives microcytic.*

e) Present in hypothyroid patient **[ T ]**
*Reason: Hypothyroidism often has anaemia.*

### Q9. Cardiac cycle

a) Time is directly related to heart rate **[ F ]**
*Reason: Cycle time = 60/HR; inversely related.*

b) Duration of systole is much more fixed than diastole **[ T ]**
*Reason: Systole is relatively constant; diastole shortens with rate.*

c) 20% of ventricular filling is active filling **[ T ]**
*Reason: ~80% passive filling, ~20% by atrial systole.*

d) During ejection phase left ventricular pressure is above 80 mmHg **[ T ]**
*Reason: LV pressure exceeds ~80 mmHg (aortic diastolic) when ejection starts.*

e) 70% ejection during the first third of the ejection **[ T ]**
*Reason: ~70% of stroke volume is ejected in the first third.*

### Q10. Regarding cardiac output

a) Is normally expressed as the output of right ventricle **[ F ]**
*Reason: Cardiac output is the output of ONE ventricle (left = right).*

b) Increased in pregnancy **[ T ]**
*Reason: Pregnancy raises CO by ~30–40%.*

c) In resting supine man averages about 7.0 L/min **[ F ]**
*Reason: Resting CO ≈ 5.0–5.6 L/min.*

d) The output per square meter of body surface averages 3.2 L **[ T ]**
*Reason: Cardiac index ≈ 3.0–3.2 L/min/m².*

e) No change during sleep **[ T ]**
*Reason: CO changes little during sleep.*

**Q11. Apoptosis**

a) Is very common during development and adulthood **[ T ]**
*Reason: Apoptosis is physiological throughout life.*

b) Is called programmed cell death **[ T ]**
*Reason: Apoptosis = programmed cell death ('cell suicide').*

c) Is called cell murder **[ F ]**
*Reason: Cell murder describes necrosis, not apoptosis.*

d) Is the same as necrosis **[ F ]**
*Reason: Apoptosis is controlled, no inflammation; necrosis is accidental.*

e) Can be called 'cell suicide' **[ T ]**
*Reason: Apoptosis = programmed cell death.*

**Q12. Following statements regarding sinus node are correct**

a) Have unstable resting membrane potential **[ T ]**
*Reason: SA node cells have unstable RMP.*

b) Distributed mainly by left vagus nerve **[ F ]**
*Reason: Right vagus supplies SA node; left supplies AV node.*

c) Is modified cardiac muscle with fewer striations **[ T ]**
*Reason: SA nodal cells are small, with few striations.*

d) Called reserve pacemaker **[ F ]**
*Reason: AV node is the reserve pacemaker, not SA node.*

e) SA nodal depolarization produces P wave in ECG **[ T ]**
*Reason: SA node/atrial depolarisation produces the P wave.*

**Q13. The second heart sound**

a) Is higher in frequency **[ T ]**
*Reason: S2 is shorter, higher pitched.*

b) Is best heard in mitral area **[ F ]**
*Reason: S2 is best heard at the base (aortic/pulmonary).*

c) Coincides with apex beat **[ F ]**
*Reason: Apex beat coincides with S1.*

d) Occurs at the end of T wave **[ T ]**
*Reason: S2 occurs at the end of T wave.*

e) Is heard just after carotid pulse **[ T ]**
*Reason: Carotid pulse follows S1; S2 follows the pulse.*

**Q14. Respiratory dead space**

a) Air is about 350 mL in young adult **[ F ]**
*Reason: Dead space ≈ 150 mL.*

b) Air is not useful for gas exchange **[ T ]**
*Reason: Dead space air does not take part in exchange.*

c) Decreases during a deep inspiration **[ F ]**
*Reason: Dead space increases with deep inspiration (airway traction).*

d) Saturates inspired air with water vapour **[ T ]**
*Reason: Dead space humidifies and warms inspired air.*

e) Is approximately equal to the body weight in pound **[ T ]**
*Reason: Dead space (mL) ≈ body weight (lb).*

**Q15. Peripheral resistance increases due to**

a) Anaemia **[ F ]**
*Reason: Anaemia lowers viscosity and resistance.*

b) Dehydration **[ T ]**
*Reason: Dehydration raises viscosity and resistance.*

c) Vasoconstriction **[ T ]**
*Reason: Vasoconstriction raises resistance (radius).*

d) Increase in distance of a blood vessel from heart **[ F ]**
*Reason: Resistance depends on vessel length, radius, viscosity, not distance from heart.*

e) Increase in viscosity of blood **[ T ]**
*Reason: Resistance ∝ viscosity.*

### Q16. Hypovolemic shock

a) Is characterized by hypertension **[ F ]**
*Reason: Hypovolaemic shock → hypotension.*

b) Is common in fracture neck femur **[ T ]**
*Reason: Haemorrhage in fracture can cause hypovolaemia.*

c) Results from widespread vasodilatation **[ F ]**
*Reason: Vasodilatation causes distributive (septic/neurogenic) shock.*

d) Is managed by fluid/blood transfusion **[ T ]**
*Reason: Volume replacement treats hypovolaemic shock.*

e) Is also called 'cold shock' **[ T ]**
*Reason: Hypovolaemic shock: cold, clammy skin (peripheral vasoconstriction).*

### Q17. Lung collapse is prevented by

a) Tidal volume **[ F ]**
*Reason: Tidal volume does not prevent collapse.*

b) Residual volume **[ T ]**
*Reason: Residual volume prevents total collapse.*

c) Negative intrapleural pressure **[ T ]**
*Reason: Negative intrapleural pressure holds lungs expanded.*

d) Positive intra-alveolar pressure **[ F ]**
*Reason: Positive alveolar pressure is transient in expiration.*

e) Surfactant **[ T ]**
*Reason: Surfactant prevents collapse.*

### Q18. Apneusis

a) Is the temporary cessation of breathing **[ F ]**
*Reason: Apnoea = cessation; apneusis = prolonged inspiratory cramp.*

b) Means prolonged inspiration with short expiration **[ T ]**
*Reason: Apneustic breathing follows pneumotaxic damage with vagi cut.*

c) Occurs in lesions of pneumotaxic center **[ T ]**
*Reason: Pneumotaxic centre lesion → apneusis.*

d) Occurs in periodic breathing **[ F ]**
*Reason: Cheyne–Stokes is periodic breathing, not apneusis.*

e) Is rapid shallow breathing **[ F ]**
*Reason: Rapid shallow = tachypnoea.*

### Q19. The stomach

a) Juice is secreted from G cells **[ F ]**
*Reason: G cells secrete gastrin; juice is from parietal/chief/mucous cells.*

b) Peristaltic contractions start from the pyloric region **[ F ]**
*Reason: Gastric waves begin in the mid-body and move to the antrum.*

c) Motility increases when fat enters the duodenum **[ F ]**
*Reason: Fat in duodenum inhibits gastric motility (CCK).*

d) Ferric iron is reduced to ferrous iron by HCl **[ T ]**
*Reason: HCl and vitamin C reduce $Fe^{3+}$ to $Fe^{2+}$.*

e) Juice inactivates the digestive enzymes secreted with saliva **[ T ]**
*Reason: Gastric acid inactivates salivary amylase.*

### Q20. Regarding enteric nervous system

a) Is essential for intestinal movement **[ T ]**
*Reason: ENS mediates peristalsis.*

b) Is a special autonomic nervous system **[ T ]**
*Reason: ENS is a division of the autonomic system.*

c) Its developmental defect causes megacolon **[ T ]**
*Reason: Absent ENS ganglia (Hirschsprung) → megacolon.*

d) It maintains gastric secretion only **[ F ]**
*Reason: ENS controls both motility and secretion.*

e) Maintains both gastrointestinal movement and secretion **[ T ]**
*Reason: ENS controls both motility and secretion.*

# May 2022 — Paper II

### Q1. Following statements are correct

a) Aldosterone hormone controls urine concentration **[ T ]**
*Reason: Aldosterone controls $Na^+/K^+$; ADH controls concentration (partly).*

b) Fluid leaving the early DCT is hypo-osmotic **[ T ]**
*Reason: Fluid leaving early DCT is hypo-osmotic.*

c) In malnutrition impairment of concentrating ability occurs **[ T ]**
*Reason: Protein malnutrition reduces medullary urea → poor concentration.*

d) Normally the amount of protein in urine is < 100 mg/d **[ F ]**
*Reason: Normal urinary protein < 150 mg/day.*

e) ADH secretion causes fall in urinary pH **[ F ]**
*Reason: ADH does not affect urinary pH.*

### Q2. Micturition

a) Reflex is an autonomic spinal cord reflex **[ T ]**
*Reason: Micturition is a spinal reflex with higher control.*

b) Is finally controlled by pons and cerebral cortex **[ T ]**
*Reason: Pontine/cortical centres control voiding.*

c) Reflex remains if sensory nerve is destroyed **[ F ]**
*Reason: Destruction of sensory nerve abolishes the reflex.*

d) Reflex is suppressed during spinal shock **[ T ]**
*Reason: Spinal shock → atonic bladder.*

e) Is initiated by sensory stretch receptors in the kidney **[ F ]**
*Reason: Stretch receptors are in the bladder wall.*

### Q3. Vitamin D

a) Decreases the intestinal absorption of calcium **[ F ]**
*Reason: Vitamin D increases $Ca^{2+}$ absorption.*

b) Is essential for normal calcification in childhood **[ T ]**

c) Requires hepatic modification for activation **[ T ]**

d) Cannot be formed in the body **[ F ]**
*Reason: Vitamin D is made in skin by sunlight.*

e) Deficiency may result in hypoparathyroidism **[ F ]**
*Reason: Vitamin D deficiency → secondary hyperparathyroidism.*

### Q4. Features of hyperadrenalism

a) Weakness **[ T ]**
*Reason: Hypercortisolism → muscle weakness.*

b) Hypoglycaemia **[ F ]**
*Reason: Cortisol excess causes hyperglycaemia.*

c) Suppressed immune system **[ T ]**
*Reason: Cortisol is immunosuppressive.*

d) Hypokalaemia **[ T ]**
*Reason: Cortisol has mineralocorticoid effect.*

e) Hypernatraemia **[ T ]**
*Reason: Mild $Na^+$ retention.*

### Q5. Shell temperature

a) Is the temperature of deep tissues of the body **[ F ]**
*Reason: Core temperature is that of deep tissues; shell temperature is skin.*

b) Rises and falls with the temperature of surroundings **[ T ]**
*Reason: Shell temperature varies with the environment.*

c) Is generally more than core temperature **[ F ]**
*Reason: Shell temperature is lower than core.*

d) Is almost constant **[ F ]**
*Reason: Shell temperature varies with surroundings.*

e) Is less accurate **[ T ]**
*Reason: Shell temperature is a poor index of body temperature.*

**Q6. Muscle tone is reduced in**

a) Parkinson's disease **[ F ]**
*Reason: Parkinson's disease increases muscle tone (rigidity).*
b) Lower motor neuron lesions **[ T ]**
c) Cerebellar lesions **[ T ]**
d) Gamma efferent impulses to muscle spindles **[ F ]**
*Reason: Increased γ-discharge increases tone.*
e) In pyramidal tract lesion **[ F ]**
*Reason: Pyramidal (UMN) lesion → spasticity (increased tone).*

**Q7. Chemoreceptors detect**

a) Damage in tissues **[ F ]**
*Reason: Nociceptors detect tissue damage.*
b) Oxygen level in arterial blood **[ T ]**
c) Osmolality of body fluids **[ F ]**
*Reason: Osmoreceptors sense osmolality; chemoreceptors sense O2/$CO_2$/$H^+$.*
d) Smell in nose **[ T ]**
e) $CO_2$ concentration in blood **[ T ]**

**Q8. In lower motor neuron lesion**

a) Tendon reflexes are exaggerated **[ F ]**
*Reason: LMN lesion → hypo/areflexia.*
b) Babinski's sign is negative **[ T ]**
*Reason: Babinski's sign is positive in UMN lesion.*
c) Spastic type of paralysis occurs **[ F ]**
*Reason: LMN lesion → flaccid paralysis; spastic = UMN.*
d) Muscle wasting occurs **[ T ]**
*Reason: LMN lesion → neurogenic atrophy.*
e) Individual muscles are affected **[ T ]**
*Reason: LMN lesion affects individual muscles; UMN affects movements.*

**Q9. Nerve deafness occurs due to**

a) Otitis media **[ F ]**
*Reason: Otitis media causes conduction deafness.*
b) Otosclerosis **[ F ]**
*Reason: Otosclerosis causes conduction deafness.*
c) Tumors in brain **[ T ]**
*Reason: Central lesions cause nerve deafness.*
d) Damage to hair cells **[ T ]**
*Reason: Hair cell damage causes sensorineural deafness.*
e) Vascular damage in medulla **[ T ]**
*Reason: Cochlear nuclei lesion → nerve deafness.*

**Q10. Regarding cones**

a) Found in highest concentration in optic disc **[ F ]**
*Reason: Optic disc has no photoreceptors (blind spot); cones are densest in fovea.*
b) Sensitive to light **[ T ]**
*Reason: Cones respond to bright light.*
c) Destruction causes complete blindness **[ F ]**
*Reason: Loss of cones leaves rod (scotopic) vision.*
d) Responsible for photopic vision **[ T ]**
*Reason: Cones mediate photopic (colour) vision.*
e) Helps in accommodation reflex **[ F ]**
*Reason: Accommodation is a lens/ciliary muscle reflex, not a cone function.*

**Q11. Major levels of central nervous system function include**

a) Spinal cord level **[ T ]**
b) Dorsal root ganglion **[ F ]**
*Reason: DRG is peripheral, not a CNS level.*
c) Lower brain or subcortical level **[ T ]**
d) Higher brain or cortical level **[ T ]**
e) Spinal nerves **[ F ]**
*Reason: Spinal nerves are peripheral nerves, not a CNS level.*

**Q12. Effect of autonomic nervous system**

a) Pupil is constricted by sympathetic stimulation **[ F ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*
b) Bronchi are constricted by sympathetic stimulation **[ F ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*
c) Penile erection is caused by sympathetic stimulation **[ F ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*
d) Heart rate is increased by sympathetic stimulation **[ T ]**
*Reason: Saliva is stimulated by both divisions; parasympathetic gives watery saliva.*
e) Force of contraction of muscle is increased by parasympathetic stimulation **[ F ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*

**Q13. Muscle spindle**

a) Is a receptor **[ T ]**
b) Helps to maintain muscle tone **[ T ]**
c) Alpha motor efferent fiber supplies the intrafusal fibers **[ F ]**
*Reason: γ-motor neurons supply intrafusal fibres; α supply extrafusal.*
d) Nuclear chain fiber is supplied by both primary and secondary afferent fibers **[ T ]**
e) Gamma motor efferent fiber supplies the extrafusal fibers **[ F ]**
*Reason: α-motor neurons supply extrafusal fibres.*

**Q14. Regarding hormones**

a) Sex hormones and adrenocortical hormones are steroid hormones **[ T ]**
*Reason: GH is a protein hormone.*
b) The receptors of $T_3$ hormone are located in the cytoplasm **[ F ]**
*Reason: $T_3$ receptors are nuclear.*
c) Epinephrine binds to the receptor on DNA **[ F ]**
*Reason: Epinephrine binds membrane (adrenergic) receptors.*
d) Norepinephrine derives from amino acid tyrosine **[ T ]**
e) Aldosterone binds with the receptor on the cell membrane **[ F ]**
*Reason: Steroid hormones bind intracellular receptors.*

**Q15. Urine formation results from**

a) Glomerular filtration **[ T ]**
b) Reabsorption from renal tubule **[ T ]**
c) Secretion from peritubular capillary **[ T ]**
d) Ultrafiltration of afferent arteriole **[ F ]**
*Reason: Ultrafiltration occurs across the glomerular capillary into Bowman's space.*
e) Secretion and reabsorption from ureter **[ F ]**
*Reason: The ureter only conducts urine.*

**Q16. A reflex action**

a) Is initiated at a sensory receptor organ **[ T ]**
*Reason: A reflex starts at a receptor.*
b) May result in endocrine secretion **[ T ]**
*Reason: Reflexes may cause hormone secretion (milk ejection).*
c) Involves transmission across at least 2 synapses **[ F ]**
*Reason: Monosynaptic reflexes have only one synapse.*
d) May be excitatory or inhibitory **[ T ]**
*Reason: Reflexes may excite or inhibit.*
e) Is independent of higher centers **[ F ]**
*Reason: Reflexes are modified by higher centres.*

**Q17. The functions of CNS at spinal cord level are**

a) Control of equilibrium **[ F ]**
*Reason: Equilibrium is controlled by brainstem/cerebellum.*

b) Walking movement **[ T ]**
*Reason: Walking patterns are generated in spinal cord circuits.*

c) Regulation of respiration **[ F ]**
*Reason: Respiration is controlled by medulla and pons.*

d) Reaction to excitement **[ F ]**
*Reason: Excitement reaction involves hypothalamus/limbic system.*

e) Withdrawal reflex **[ T ]**
*Reason: Withdrawal is a spinal reflex.*

**Q18. Estrogen**

a) Is a hormone of female **[ T ]**
*Reason: Estrogen is the female hormone.*

b) Is a hormone of pregnancy **[ T ]**
*Reason: Progesterone maintains pregnancy.*

c) Is a hormone of lactation **[ F ]**
*Reason: Prolactin is the hormone of lactation.*

d) Helps spermiogenesis **[ F ]**
*Reason: Spermiogenesis is Sertoli/testosterone/FSH-driven.*

e) Helps in LH surge for ovulation **[ T ]**
*Reason: Estrogen positive feedback causes LH surge.*

**Q19. EPSP is due to**

a) Opening of sodium channel **[ T ]**
*Reason: EPSP: $Na^+$ ($Ca^{2+}$) channel opening or $K^+/Cl^-$ conductance decreasing.*

b) Opening of potassium channel **[ F ]**
*Reason: $K^+$ channel opening causes IPSP.*

c) Opening of chloride channel **[ F ]**
*Reason: $Cl^-$ channel opening causes IPSP.*

d) Increased conductance of $K^+$ ion **[ F ]**
*Reason: $\uparrow K^+$ conductance → IPSP.*

e) Decreased conductance of $K^+$ ion **[ T ]**
*Reason: $\downarrow K^+$ conductance depolarises (EPSP).*

**Q20. Life saving hormones are**

a) Thyroid hormone **[ F ]**
*Reason: Thyroid hormone is not life-saving; aldosterone and cortisol are.*

b) Aldosterone **[ T ]**
*Reason: Aldosterone retains $Na^+$ and water.*

c) Growth hormone **[ F ]**
*Reason: GH contributes to breast development.*

d) Parathyroid hormone **[ T ]**
*Reason: Thyroid hormone is not life-saving; aldosterone and cortisol are.*

e) ACTH **[ F ]**
*Reason: ACTH acts on adrenal cortex; it does not regulate calcium.*

# November 2021 — Paper I

### Q1. Cell membrane proteins

a) Can act as antigen **[ T ]**
*Reason: Membrane glycoproteins act as antigens (blood groups, HLA).*

b) Constitute about 50% of cell membrane **[ T ]**
*Reason: Proteins form ~55% of membrane mass.*

c) Can act as carrier protein **[ T ]**
*Reason: Globulins carry hormones, iron, lipids.*

d) Are mostly lipoprotein **[ F ]**
*Reason: Membrane proteins are mostly integral glycoproteins.*

e) Can act as second messenger **[ F ]**
*Reason: Second messengers (cAMP, IP3) are small molecules, not membrane proteins.*

### Q2. Carrier-mediated facilitated diffusion

a) Uses cellular ATP **[ F ]**
*Reason: Facilitated diffusion does not use ATP.*

b) Is used for cellular uptake of blood glucose **[ T ]**
*Reason: GLUT carriers take up glucose by facilitated diffusion.*

c) Is a form of active transport **[ F ]**
*Reason: Facilitated diffusion is passive.*

d) Transports molecules against their concentration gradient **[ F ]**
*Reason: Facilitated diffusion goes only down the gradient.*

e) Transports substance by changing molecular conformation of carrier protein **[ T ]**
*Reason: Carrier changes shape to move the substrate.*

### Q3. Multi-unit smooth muscle is present in the

a) Ciliary muscle of the lens **[ T ]**
*Reason: Ciliary muscle and iris are multi-unit smooth muscle.*

b) Iris (eye) **[ T ]**

c) Ureter **[ F ]**
*Reason: Ureters are single-unit smooth muscle.*

d) Urinary bladder **[ F ]**
*Reason: Bladder is unitary (single-unit) smooth muscle.*

e) Vas deferens **[ T ]**
*Reason: Vas deferens is multi-unit smooth muscle.*

### Q4. Lymphocytes

a) Are produced in lymphoid tissue **[ T ]**
*Reason: Lymphocytes are made in marrow and lymphoid tissue.*

b) Are produced in bone marrow **[ T ]**
*Reason: Lymphocytes originate in bone marrow.*

c) Induce passive immunity **[ F ]**
*Reason: Lymphocytes induce ACTIVE immunity.*

d) Are concerned with cell-mediated immunity **[ T ]**
*Reason: T lymphocytes mediate cellular immunity.*

e) Produce interleukin **[ T ]**
*Reason: Lymphocytes (helper T) secrete interleukins.*

### Q5. Hemoglobin

a) Is a globular protein **[ T ]**
*Reason: Hb is a globular tetrameric protein.*

b) Is absent in reticulocyte **[ F ]**
*Reason: Reticulocytes contain Hb.*

c) Is responsible for oxygen transport in blood **[ T ]**
*Reason: Hb carries most blood O2.*

d) Is synthesized in the nucleus of the cell **[ F ]**
*Reason: Hb is made in cytoplasm (and mitochondria for heme).*

e) Deficiency causes polycythemia **[ F ]**
*Reason: Hb deficiency → anaemia.*

### Q6. In anemic hypoxia

a) Decreased $PCO_2$ in arterial blood occurs **[ F ]**
b) Decreased PO2 in arterial blood occurs **[ F ]**
*Reason: Arterial PO2 is normal in anaemic hypoxia.*
c) There is tachycardia **[ T ]**
d) There is increased 2,3-DPG in the blood **[ T ]**
*Reason: ↑2,3-DPG, ↑$H^+$, ↑$PCO_2$, ↑temperature shift ODC right.*
e) Oxygen carrying capacity of hemoglobin is reduced **[ T ]**
*Reason: Anaemia reduces O2 carrying capacity.*

### Q7. Serum globulin

a) Is synthesized in the liver **[ T ]**
*Reason: α/β globulins are hepatic; γ-globulins from plasma cells.*
b) Acts as hormone **[ F ]**
*Reason: Globulins are carrier/immune proteins, not hormones.*
c) Is increased in inflammatory condition **[ T ]**
*Reason: Acute-phase globulins and immunoglobulins rise.*
d) Contributes to viscosity **[ T ]**
*Reason: Plasma proteins add to viscosity.*
e) Acts as carrier protein **[ T ]**
*Reason: Globulins carry hormones, iron, lipids.*

### Q8. Pulse pressure is

a) The highest pressure measured in the arteries **[ F ]**
*Reason: Systolic is the highest; pulse pressure is the difference.*
b) The difference between the systolic and diastolic pressure **[ T ]**
*Reason: Pulse pressure = systolic – diastolic.*
c) Measured only during the diastole **[ F ]**
*Reason: Pulse pressure uses both phases.*
d) Determined by stroke volume **[ T ]**
*Reason: Stroke volume and compliance determine pulse pressure.*

### Q9. Clotting time is

a) Increased in thrombocytopenia **[ F ]**
*Reason: Clotting time is normal in thrombocytopenia (BT is prolonged).*
b) Decreased in hemophilia **[ F ]**
*Reason: Haemophilia prolongs clotting time.*
c) The time required for activation of clotting factors **[ F ]**
*Reason: Clotting time = time for blood to clot in vitro.*
d) A test for bleeding disorder **[ T ]**
*Reason: Clotting time is a screening test.*
e) 15 minutes in a normal person **[ F ]**
*Reason: Normal clotting time is about 4–10 minutes.*

### Q10. Cardiac muscles

a) Have many tight junctions **[ F ]**
*Reason: Cardiac muscle has gap junctions at intercalated discs.*
b) Act as a single syncytium **[ T ]**
c) Have intercalated discs **[ T ]**
d) Are interconnected in latticework fashion **[ T ]**
e) Have refractory period **[ T ]**
*Reason: Cardiac muscle has a long refractory period (plateau), so it cannot be tetanised; summation is absent.*

### Q11. Salivary secretions are abolished

a) By activation of the sympathetic nerve **[ F ]**
*Reason: Sympathetic stimulation gives scanty, thick saliva; it does not abolish saliva.*

b) When weak acid is present in the mouth **[ F ]**
*Reason: Acid increases salivary secretion.*

c) At the onset of vomiting **[ F ]**
*Reason: Salivation increases before vomiting.*

d) By the administration of anticholinergic agents **[ T ]**
*Reason: Atropine abolishes salivary secretion.*

e) By thinking of appetizing food **[ F ]**
*Reason: Conditioned reflex increases saliva.*

### Q12. Secondary (specific) granules of neutrophil contain

a) Alkaline phosphatase **[ T ]**
*Reason: Specific granules contain alkaline phosphatase, lactoferrin, lysozyme, collagenase.*

b) Lactoferrin **[ T ]**
*Reason: Lactoferrin is in specific granules.*

c) Lysosomal enzyme **[ F ]**
*Reason: Lysosomal enzymes/myeloperoxidase are in azurophilic (primary) granules.*

d) Myeloperoxidase **[ F ]**
*Reason: Myeloperoxidase is in azurophilic granules.*

e) Vitamin $B_{12}$ binding protein **[ T ]**
*Reason: $B_{12}$-binding protein is in specific granules.*

### Q13. Factors that inhibit stomach emptying include

a) Gastrocolic reflex **[ F ]**
*Reason: Gastrocolic reflex = food in stomach → colonic mass movement; enterogastric reflex inhibits emptying.*

b) Enterogastric reflex **[ T ]**
*Reason: Enterogastric reflex slows gastric emptying.*

c) Gastrin **[ F ]**
*Reason: Gastrin stimulates acid secretion and mildly motility; it does not inhibit emptying.*

d) CCK **[ T ]**
*Reason: CCK delays gastric emptying, stimulates gallbladder contraction and enzyme secretion.*

e) Decreased acidity of duodenal chyme **[ F ]**
*Reason: Increased duodenal acidity inhibits emptying.*

### Q14. Cholecystokinin (CCK)

a) Inhibits gastric emptying **[ T ]**
*Reason: CCK slows gastric emptying to allow fat digestion.*

b) Inhibits pancreatic $HCO_3^-$ secretion **[ F ]**
*Reason: CCK potentiates enzyme secretion; secretin stimulates $HCO_3^-$.*

c) Stimulates digestive enzyme secretion **[ T ]**
*Reason: CCK stimulates pancreatic enzyme secretion.*

d) Inhibits contraction of the gall bladder **[ F ]**
*Reason: CCK CONTRACTS the gallbladder.*

e) Causes relaxation of the sphincter of Oddi **[ T ]**
*Reason: CCK relaxes the sphincter of Oddi.*

### Q15. Liver stores

a) Iron **[ T ]**
*Reason: Liver stores iron, vitamins A, D, $B_{12}$, glycogen.*

b) Plasma protein **[ F ]**
*Reason: Liver SYNTHESISES plasma proteins; it does not store them.*

c) Vitamin D **[ T ]**
*Reason: Liver stores vitamins A, D, $B_{12}$.*

d) Vitamin C **[ F ]**
*Reason: Vitamin C is not stored in the liver.*

e) Bile salt **[ F ]**
*Reason: Bile salts are stored in the gallbladder.*

### Q16. Slow waves of GIT smooth muscle

a) Are true action potentials **[ F ]**
*Reason: Slow waves are not APs; spikes riding on them are.*

b) Determine the basic rhythm of GIT **[ T ]**
*Reason: Slow waves set the basic electrical rhythm.*

c) Are caused by interstitial cells of Cajal **[ T ]**
*Reason: ICC are the pacemakers of gut slow waves.*

d) Have an intensity of 50 to 150 mV **[ F ]**
*Reason: Slow waves are 5–15 mV.*

e) Directly cause muscle contraction **[ F ]**
*Reason: Contraction needs spikes on top of slow waves.*

### Q17. Compliance of the lungs is greater

a) In adults than in infants **[ T ]**
*Reason: Compliance rises with lung size.*

b) When expanded above normal tidal volume **[ F ]**
*Reason: Compliance falls at high volumes.*

c) In COPD patients **[ T ]**
*Reason: Emphysema loses elastic recoil → compliance increases.*

d) In standing than in recumbent position **[ T ]**

e) Than compliance of lungs and thorax together **[ T ]**
*Reason: Lung alone 200 mL/cm$H_2O$; lung + thorax 110.*

### Q18. Lack of surfactant is associated with

a) Immature infant **[ T ]**
*Reason: Premature infants lack surfactant → RDS.*

b) Pulmonary edema **[ T ]**
*Reason: Lack of surfactant → fluid transudation into alveoli.*

c) Lung collapse **[ T ]**
*Reason: ↑surface tension → atelectasis.*

d) Pulmonary fibrosis **[ F ]**
*Reason: Fibrosis is not due to surfactant lack.*

e) Prolonged oxygen inhalation **[ F ]**
*Reason: Prolonged high O2 damages lung but isn't caused by lack of surfactant.*

### Q19. The ventilation–perfusion ratio

a) Is about 0.8 **[ T ]**
*Reason: Normal V/Q ≈ 0.8.*

b) Is more in apex than in base of the lungs in erect posture **[ T ]**
*Reason: Apex: V/Q high (≈ 3); base low (≈ 0.6).*

c) Is decreased with increased physiological dead space **[ F ]**
*Reason: ↑ physiological dead space → V/Q ↑.*

d) Is decreased in exercise **[ F ]**
*Reason: Exercise makes V/Q more uniform (≈ 1).*

e) Is increased by hemorrhage **[ T ]**
*Reason: ↓ perfusion → V/Q rises.*

### Q20. Respiratory center

a) Is situated in medulla oblongata **[ T ]**
*Reason: Respiratory centre: medulla + pons.*

b) Maintains rhythmic ventilation **[ T ]**
*Reason: Dorsal group sets the inspiratory rhythm.*

c) Is sensitive to alteration of pH in blood **[ F ]**
*Reason: The chemosensitive area (not the respiratory centre) responds to $H^+$.*

d) Responds to cerebral impulse **[ T ]**
*Reason: Cortex can override.*

e) Is controlled by phrenic nerve **[ F ]**
*Reason: Respiratory centre controls the phrenic nerve, not vice versa.*

# November 2021 — Paper II

**Q1. In collecting duct**

a) Water reabsorption is active **[ F ]**
*Reason: Water moves passively through aquaporins.*
b) Reabsorption is critically related to tubular transit time **[ T ]**
c) Reabsorption is complete **[ F ]**
*Reason: Collecting duct reabsorption is variable (ADH-dependent).*
d) ADH helps in urea reabsorption **[ T ]**
e) Water is impermeable (without ADH) **[ T ]**

**Q2. Posterior column damage in the spinal cord may impair**

a) Vibration sense **[ T ]**
b) Pain sensation **[ F ]**
*Reason: Posterior column carries fine touch, vibration, proprioception — not pain.*
c) The flexor plantar response to stimulation of the sole **[ F ]**
*Reason: Plantar reflex tests pyramidal tract, not the dorsal column.*
d) Tactile sensation **[ T ]**
e) The ability to stand steadily with eyes closed **[ T ]**

**Q3. During accommodation for near vision**

a) The curvature of the cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*
b) Chromatic and spherical aberration is decreased **[ T ]**
*Reason: Small pupil reduces aberrations.*
c) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*
d) The depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*
e) The visual axes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

**Q4. The pituitary gland**

a) Lies superior to the sphenoid bone **[ T ]**
*Reason: Pituitary sits in sella turcica of the sphenoid bone.*
b) Is all ectodermal **[ T ]**
*Reason: Anterior pituitary from Rathke's pouch (ectoderm), posterior from neuroectoderm.*
c) Has the optic chiasma anteriorly **[ T ]**
*Reason: Optic chiasma lies above/in front of the pituitary stalk.*
d) Is supplied from the internal carotid artery **[ T ]**
*Reason: Pituitary is supplied by branches of the internal carotid.*
e) Lies inferior to the cavernous sinus **[ F ]**
*Reason: Pituitary lies between the cavernous sinuses (medial to them).*

**Q5. The loop of Henle**

a) Is responsible for high medullary osmolarity **[ T ]**
*Reason: Loop of Henle counter-current multiplier builds the gradient.*
b) Returns fluid of low osmolarity to the DCT **[ T ]**
*Reason: Fluid leaving the loop is hypo-osmotic.*
c) Has a high permeability to water in thick ascending limb **[ F ]**
*Reason: Thick ascending limb is impermeable to water.*
d) Is parallel to efferent arteriole **[ F ]**
*Reason: Vasa recta (from efferent arterioles) run parallel to the loop.*
e) Actively transports $Na^+$ in the thick ascending limb **[ T ]**
*Reason: NKCC2 transports $Na^+$-$K^+$-2Cl- in the thick ascending limb.*

### Q6. The cerebellum

a) Is essential for finely coordinated movements **[ T ]**
b) Does not modify the discharge of spinal motor neurons **[ F ]**
*Reason: Cerebellum modifies spinal motor neuron discharge via descending tracts.*
c) Has afferent input from the motor cortex **[ T ]**
*Reason: Cerebellum receives corticopontocerebellar input from motor cortex.*
d) Has afferent input from the muscle proprioceptors **[ T ]**
*Reason: Spinocerebellar tracts carry proprioception.*
e) Has no afferent input from the vestibular system **[ F ]**
*Reason: Flocculonodular lobe receives vestibular input.*

### Q7. Parathormone

a) Is produced by the cells of the thyroid gland **[ F ]**
*Reason: PTH is made by chief cells of parathyroids.*
b) Decreases urinary excretion of calcium **[ T ]**
c) Blood level increases when serum calcium falls **[ T ]**
d) Depresses activity of the anterior pituitary **[ F ]**
*Reason: PTH does not affect the anterior pituitary.*
e) Is independent of magnesium level **[ F ]**
*Reason: Low $Mg^{2+}$ impairs PTH secretion and action.*

### Q8. During normal menstrual cycle

a) Oestrogen is predominant during the proliferative phase **[ T ]**
*Reason: Proliferative phase = estrogen.*
b) Menstrual blood loss averages 150 mL **[ F ]**
*Reason: Menstrual loss averages 35–50 mL.*
c) Cervical mucus becomes more viscous at the time of ovulation **[ F ]**
*Reason: Cervical mucus is thin, watery and elastic (ferning) at ovulation.*
d) Menstruation is preceded by a rise in plasma progesterone levels **[ F ]**
*Reason: Menstruation follows a FALL in progesterone.*
e) Menstruation is associated with dilatation of basal segment of spiral arteries within endometrium **[ F ]**

### Q9. Human chorionic gonadotrophin (HCG)

a) Has the biological properties of LH **[ T ]**
*Reason: hCG acts on the LH receptor.*
b) Can transform the cyclic corpus luteum into a pseudopregnant corpus luteum **[ T ]**
*Reason: hCG maintains the corpus luteum of pregnancy.*
c) Is responsible for the onset of parturition **[ F ]**
*Reason: hCG is not related to parturition.*
d) Is detectable in serum before implantation of the ovum **[ F ]**
*Reason: hCG secretion begins after implantation.*
e) Is significantly elevated in pre-eclampsia **[ F ]**
*Reason: hCG is not a diagnostic/significantly elevated marker in pre-eclampsia.*

### Q10. Progesterone

a) Is natriuretic **[ T ]**
*Reason: Progesterone is natriuretic (blocks aldosterone).*
b) Requires estrogen priming before its action **[ T ]**
*Reason: Progesterone receptors are induced by estrogen.*
c) Is synthesized in the adrenal gland (as its main source) **[ F ]**
*Reason: Main sources are corpus luteum/placenta.*
d) Decreases uterine muscle contraction **[ T ]**
*Reason: Progesterone quietens the uterus.*
e) Is the hormone of pregnancy **[ T ]**
*Reason: Progesterone maintains pregnancy.*

### Q11. Thyroxine

a) Is less than 80% bound to plasma proteins **[ F ]**
*Reason: $T_4$ is > 99% protein bound.*

b) Formation requires amino acids **[ T ]**
*Reason: Tyrosine is incorporated in thyroglobulin.*

c) Is essential for skeletal development **[ T ]**
*Reason: Thyroxine is needed for bone growth.*

d) Increases oxygen consumption in the brain **[ F ]**
*Reason: Brain O2 use is unaffected by thyroxine.*

e) Leads to hyperlipidaemia when present in excess **[ F ]**
*Reason: Excess thyroxine lowers lipids.*

### Q12. Aldosterone causes

a) A fall in urinary sodium excretion **[ T ]**

b) Weight gain **[ T ]**

c) A fall in serum chloride levels **[ F ]**
*Reason: Aldosterone retains $Na^+$ (and $Cl^-$ follows); serum $Cl^-$ does not fall.*

d) Increased extracellular fluid volume **[ T ]**

e) Increased potassium excretion **[ T ]**

### Q13. Organ of Corti

a) Is the receptor organ of hearing **[ T ]**
*Reason: Organ of Corti is the receptor organ of hearing.*

b) Is the receptor organ of taste **[ F ]**
*Reason: Taste buds are the taste receptors.*

c) Is the receptor organ of smell **[ F ]**
*Reason: Organ of Corti is for hearing; smell receptors are in olfactory epithelium.*

d) Generates nerve impulse **[ T ]**
*Reason: Hair cells generate receptor potentials → impulses.*

e) Lies on Reissner's membrane **[ F ]**
*Reason: Organ of Corti lies on the basilar membrane.*

### Q14. The pheochromocytoma

a) Is a tumor of adrenal medullary tissue **[ T ]**

b) Is benign in the majority of cases **[ T ]**

c) Has its maximum incidence after the menopause **[ F ]**
*Reason: Peak incidence is 3rd–5th decade.*

d) Can be associated with severe episodes of hypertension **[ T ]**

e) Is always found in the gland **[ F ]**
*Reason: ~10% of pheochromocytomas are extra-adrenal.*

### Q15. Statements relating to thyroid function are correct

a) TRH is a decapeptide **[ F ]**
*Reason: TRH is a tripeptide.*

b) Placental thyrotrophin is biochemically identical to pituitary TSH **[ F ]**
*Reason: hCG has TSH-like effect, but placental thyrotrophin is not identical.*

c) The metabolic activity of $T_3$ is about four times that of $T_4$ **[ T ]**
*Reason: $T_3$ is about 4× more potent than $T_4$.*

d) 5% of the circulating $T_4$ is free **[ F ]**
*Reason: Free $T_4$ is ~0.03%.*

e) Circulating TBG increases under the influence of oestrogens **[ T ]**
*Reason: Oestrogen raises TBG.*

**Q16. During the process of fertilization and implantation**

a) Fertilisation occurs in the outer third of the Fallopian tube **[ T ]**
b) The haploid number of chromosomes results **[ F ]**
*Reason: Fertilisation restores the DIPLOID number.*
c) The morula is enclosed by trophoblast **[ F ]**
*Reason: Blastocyst (not morula) has a trophoblast.*
d) The morula actively moves along the oviduct **[ F ]**
*Reason: Morula is moved by cilia/peristalsis.*
e) HCG is produced by the preimplantation embryo **[ F ]**
*Reason: hCG appears after implantation (about day 8–10).*

**Q17. The nerves that carry taste impulses are**

a) Facial nerve **[ T ]**
b) Trigeminal nerve **[ F ]**
*Reason: Taste fibres travel in VII, IX, X; V gives general sensation.*
c) Glossopharyngeal **[ T ]**
d) Vagus nerve **[ T ]**
e) Hypoglossal **[ F ]**
*Reason: Hypoglossal is motor to the tongue.*

**Q18. Lateral spinothalamic tract**

a) Is formed by A alpha fibre **[ F ]**
*Reason: Spinothalamic tract uses Aδ and C fibres.*
b) Carries proprioception **[ F ]**
*Reason: Proprioception travels in dorsal columns/spinocerebellar tracts.*
c) Transmits pain sensation **[ T ]**
d) Transmits thermal sensation **[ T ]**
e) Transmits crude touch sensation **[ F ]**
*Reason: Crude touch travels in the anterior spinothalamic tract.*

**Q19. Basal ganglia**

a) Functions in association with the motor control system **[ T ]**
b) Can also perform function by itself **[ F ]**
*Reason: Basal ganglia work with cortex/thalamus, not alone.*
c) Damage causes decreased tone of peripheral muscles **[ F ]**
*Reason: Basal ganglia damage increases tone (rigidity).*
d) Lesion at putamen circuit causes Parkinson's disease **[ F ]**
*Reason: Parkinson's disease increases muscle tone (rigidity).*
e) Controls complex patterns of movements **[ T ]**

**Q20. Short stature is seen in children suffering**

a) Adrenal insufficiency **[ T ]**
b) Thyroxine deficiency **[ T ]**
c) Growth hormone deficiency **[ T ]**
d) Chronic malnutrition **[ T ]**
e) Castration **[ F ]**
*Reason: Castration delays epiphyseal closure → tall stature.*

# May 2021 — Paper I

### Q1. Cyclic AMP

a) Is an important 1st messenger **[ F ]**
*Reason: cAMP is a SECOND messenger.*

b) Acts via $Ca^{2+}$ **[ F ]**
*Reason: cAMP is a second messenger acting via protein kinase A.*

c) Is formed from ATP **[ T ]**

d) Activates protein kinases **[ T ]**

e) Is metabolized by adenylyl cyclase **[ F ]**
*Reason: cAMP is formed by adenylyl cyclase and broken down by phosphodiesterase.*

### Q2. Cell membrane acts as

a) Transporters of ions **[ T ]**
*Reason: Membrane proteins transport ions.*

b) Cell adhesion molecule **[ T ]**
*Reason: Adhesion molecules are membrane proteins.*

c) Pumps **[ T ]**
*Reason: Membrane proteins act as pumps ($Na^{+}$-$K^{+}$ ATPase).*

d) Receptors for hormones **[ T ]**
*Reason: Membrane receptors bind hormones.*

e) Cellular defense system **[ T ]**
*Reason: Membrane antigens/immune molecules take part in defence.*

### Q3. Blood

a) Makes up about 7% of body weight **[ T ]**
*Reason: Blood is ~7% (8%) of body weight.*

b) Forms a higher percentage of body weight in fat than in thin people **[ F ]**
*Reason: Blood volume per kg is lower in obese persons (fat is poorly vascular).*

c) Volume can be calculated if hemoglobin concentration is known **[ F ]**
*Reason: Volume is measured by dilution (Evans blue, $^{131}$I-albumin).*

d) Volume rises after water is drunk **[ F ]**
*Reason: Drinking water is quickly redistributed; blood volume rises little.*

e) Expresses serum when it clots **[ T ]**
*Reason: Clot retraction expresses serum.*

### Q4. Reaction occurs in transfusion of blood group

a) A to group B person **[ T ]**
*Reason: A cells + anti-A in B plasma → agglutination.*

b) O to a group AB person **[ F ]**
*Reason: O cells have no antigens; AB has no antibodies → safe.*

c) A to a group O person **[ T ]**
*Reason: O plasma has anti-A → agglutination.*

d) A to a group AB group **[ F ]**
*Reason: AB has no antibodies → compatible.*

e) O Rh-negative to a group AB Rh-positive person **[ F ]**
*Reason: Compatible (Rh-negative to Rh-positive).*

### Q5. Intercellular communications are

a) Neural **[ T ]**

b) Endocrine **[ T ]**

c) Autocrine and paracrine **[ T ]**

d) ATP-dependent process **[ F ]**
*Reason: Intercellular communication (paracrine/endocrine) is via chemical messengers.*

e) Carrier-mediated process **[ F ]**
*Reason: Intercellular communication uses messengers/gap junctions, not carriers.*

**Q6. Action potential**

a) Is a long-lasting event **[ F ]**
*Reason: AP lasts only ~1 ms in nerve.*

b) Occurs in adipose tissue cells **[ F ]**
*Reason: Adipocytes are non-excitable.*

c) Is generated by voltage-gated ion channels **[ T ]**
*Reason: Voltage-gated $Na^+/K^+/Ca^{2+}$ channels generate AP.*

d) Occurs in endocrine pituitary cells **[ T ]**
*Reason: Some endocrine cells generate APs (e.g., pituitary cells).*

e) Occurs in neuron **[ T ]**
*Reason: Neurons conduct APs.*

**Q7. Plasmin**

a) Is involved in the extrinsic clotting system **[ F ]**
*Reason: Plasmin lyses fibrin (fibrinolysis).*

b) Is a hormone **[ F ]**
*Reason: Vitamin/ prostaglandin...; local factor, not a hormone.*

c) Is responsible for fibrinolysis **[ T ]**
*Reason: Plasmin digests fibrin.*

d) Is responsible for delayed healing **[ F ]**
*Reason: Plasmin does not delay healing.*

e) Is responsible for promoting the formation of emboli **[ F ]**
*Reason: Plasmin dissolves clots, preventing emboli.*

**Q8. In hypokalemia (plasma $K^+$ ≈ 3.5 mEq/L)**

a) PR interval is lengthened to 0.32 sec **[ T ]**

b) ST segment is depressed **[ T ]**

c) QRS interval is 0.06 sec **[ F ]**
*Reason: Hypokalaemia may slightly widen QRS.*

d) T waves remain tall and slender **[ F ]**
*Reason: Hypokalaemia flattens T waves.*

e) A prominent U wave is present following T wave **[ T ]**

**Q9. Pacemaker potential is**

a) –50 mV **[ T ]**
*Reason: SA node pacemaker begins about –55 to –60 mV.*

b) More slope with vagal stimulation **[ F ]**
*Reason: Vagal stimulation DEcreases the slope (more hyperpolarised, slower rate).*

c) Due to leakiness of the membrane to sodium **[ T ]**
*Reason: $Na^+$ leak (funny current) drives pacemaker depolarisation.*

d) Due to increase in potassium influx **[ F ]**
*Reason: Pacemaker potential is due to reduced $K^+$ efflux, not $K^+$ influx.*

e) Completed by closing of calcium transient channel **[ F ]**
*Reason: Pacemaker potential ends with OPENING of $Ca^{2+}$ channels.*

**Q10. Gap junction**

a) Is absent in cardiac muscle **[ F ]**
*Reason: Cardiac muscle has abundant gap junctions at intercalated discs.*

b) Connects sarcotubular system to skeletal muscle cells **[ F ]**
*Reason: Gap junctions link cardiac/smooth cells, not sarcotubular system.*

c) Provides pathway for rapid spread of excitation from one cardiac muscle fiber to another **[ T ]**
*Reason: Gap junctions link cardiac fibres as a syncytium.*

d) Is present in smooth muscle cell **[ T ]**
*Reason: Unitary smooth muscle has gap junctions.*

e) Allows slow diffusion of ions **[ F ]**
*Reason: Gap junctions allow rapid ion flow.*

### Q11. Stimuli that increase gastrin secretion

a) Acid **[ F ]**
*Reason: Antral acid INHIBITS gastrin.*

b) Calcium **[ T ]**
*Reason: $Ca^{2+}$ stimulates gastrin.*

c) Calcitonin **[ F ]**
*Reason: Calcitonin is not a physiological gastrin stimulus.*

d) Peptides **[ T ]**
*Reason: Peptides/amino acids stimulate gastrin.*

e) GIP **[ F ]**
*Reason: GIP inhibits gastric motility and acid; stimulates insulin.*

### Q12. Bile

a) Is synthesized from gallbladder **[ F ]**
*Reason: Bile is made by hepatocytes.*

b) Is stored in gallbladder **[ T ]**
*Reason: Gallbladder stores/concentrates bile (5–20×).*

c) Plays an important role in protein digestion **[ F ]**
*Reason: Bile has no enzymes.*

d) Plays a role in fat absorption **[ T ]**
*Reason: Bile salts form micelles for fat absorption.*

e) Can be concentrated up to a maximum of 20-fold **[ T ]**
*Reason: Gallbladder concentrates bile up to 5–20-fold.*

### Q13. Saliva

a) Is primarily under hormonal control **[ F ]**
*Reason: Salivary secretion is purely nervous.*

b) Secretion is increased by parasympathetic stimulation **[ T ]**
*Reason: Sympathetic does not increase thyroid hormone secretion.*

c) Has a larger proportion of $Na^+$ in response to aldosterone administration **[ F ]**
*Reason: Aldosterone increases salivary $K^+$ and reduces $Na^+$.*

d) Is necessary for taste sensation **[ T ]**
*Reason: Saliva dissolves tastants.*

e) Is necessary for antisepsis in mouth **[ T ]**

### Q14. Vasodilator agents are

a) Histamine **[ T ]**

b) Angiotensin II **[ F ]**
*Reason: Angiotensin II is the vasoconstrictor/aldosterone stimulator; it is a potent but second $Na^+$-retaining agent.*

c) Vasopressin **[ F ]**
*Reason: Vasopressin is a vasoconstrictor.*

d) Bradykinin **[ T ]**

e) Norepinephrine **[ F ]**
*Reason: NE is a vasoconstrictor, not a vasodilator.*

### Q15. Chemosensitive area is

a) Located in medulla **[ T ]**

b) Sensitive to low PO2 **[ F ]**
*Reason: The chemosensitive area is insensitive to O2; peripheral chemoreceptors respond to low PO2.*

c) Highly sensitive to low $PCO_2$ **[ F ]**
*Reason: Chemosensitive area is stimulated by HIGH $PCO_2$ ($\uparrow H^+$).*

d) Sensitive to decrease in pH **[ T ]**

e) A pressure sensitive area **[ F ]**
*Reason: The chemosensitive area responds to $H^+/CO_2$, not pressure.*

### Q16. Ventilation is increased

a) At high altitude **[ T ]**

b) When plasma $CO_2$ level is reduced **[ F ]**
*Reason: Low $CO_2$ reduces ventilation.*

c) During muscular exercise **[ T ]**

d) In hypoxia **[ T ]**

e) In fever **[ T ]**

### Q17. Iron deficiency

a) Is more common in men than in women **[ F ]**
*Reason: Women (menses, pregnancy) are more affected.*
b) Frequently follows loss of blood from body **[ T ]**
c) Is commonly seen in pregnancy **[ T ]**
d) Causes anemia by limiting rate of Hb synthesis **[ T ]**
e) May cause large pale erythrocytes to appear in blood **[ F ]**
*Reason: Iron deficiency → small pale (microcytic hypochromic) cells.*

### Q18. Hyperpnoea during exercise is caused by

a) Elevated $CO_2$ **[ F ]**
*Reason: Exercise $PCO_2$ and pH stay near normal; hyperpnoea is neurogenic.*
b) Decreased pH **[ F ]**
*Reason: Arterial pH changes little in moderate exercise.*
c) Direct impulses from motor cortex **[ T ]**
*Reason: Collateral from motor cortex drives exercise hyperpnoea.*
d) Impulses from body parts which are moving **[ T ]**
*Reason: Proprioceptive impulses from moving limbs drive exercise hyperpnoea.*
e) Elevated O2 **[ F ]**
*Reason: O2 does not rise in exercise.*

### Q19. Respiratory dead space

a) Decreases during a deep inspiration **[ F ]**
*Reason: Dead space increases with deep inspiration (airway traction).*
b) Decreases during a cough **[ T ]**
*Reason: Forceful expiration/cough reduces dead space.*
c) Decreases when adrenaline level rises **[ F ]**
*Reason: Adrenaline dilates bronchi → increases dead space.*
d) Is being influenced by higher center **[ T ]**
*Reason: Bronchomotor tone is under higher control.*
e) Air is not useful for gas exchange **[ T ]**
*Reason: Dead space air does not take part in exchange.*

### Q20. Acclimatization to high altitude causes

a) A small increase in P50 **[ T ]**
*Reason: Acclimatisation raises 2,3-DPG → P50 increases slightly.*
b) Decreases erythropoietin secretion **[ F ]**
c) Increase in number of mitochondria **[ T ]**
*Reason: Acclimatisation increases mitochondria.*
d) Decrease in myoglobin **[ F ]**
*Reason: Myoglobin increases at altitude.*
e) Decrease in tissue content of cytochrome oxidase **[ F ]**
*Reason: Tissue cytochrome oxidase increases.*

# May 2021 — Paper II

### Q1. Proximal tubule

a) Reabsorbs 67% of $Na^+$ and $H_2O$ **[ T ]**
*Reason: PCT reabsorbs ~65% of filtered $Na^+$ and water.*

b) Is the site of glomerulotubular balance **[ T ]**

c) Fluid is slightly iso-osmotic **[ T ]**
*Reason: PCT fluid stays iso-osmotic.*

d) Reabsorbs 50% of filtered glucose **[ F ]**
*Reason: PCT reabsorbs 100% of filtered glucose below threshold.*

e) Secretes $H^+$ and $Cl^-$ **[ F ]**
*Reason: PCT secretes $H^+$; $Cl^-$ is passively reabsorbed.*

### Q2. The renal clearance of

a) Bicarbonate is similar to that of glucose **[ T ]**
*Reason: Both are completely reabsorbed at normal plasma levels → clearance ≈ 0.*

b) PAH is nearer to 600 mL/min **[ T ]**
*Reason: PAH clearance ≈ RPF (~625 mL/min).*

c) Glucose is more than threshold **[ F ]**
*Reason: Glucose clearance is zero below threshold.*

d) Inulin is the same as GFR **[ T ]**
*Reason: Inulin is filtered, not reabsorbed or secreted.*

e) Creatinine is lower than urea **[ F ]**
*Reason: Creatinine clearance (≈ GFR) exceeds urea clearance (≈ 60–70).*

### Q3. GFR is

a) Relatively constant **[ T ]**
*Reason: Autoregulation keeps GFR constant.*

b) About 20% of the renal plasma flow **[ T ]**
*Reason: Filtration fraction ≈ 20%.*

c) Increased by angiotensin II stimulation **[ T ]**
*Reason: Ang II constricts efferent arteriole → maintains GFR.*

d) Altered by mesangial cell contraction **[ T ]**
*Reason: Mesangial contraction lowers Kf and GFR.*

e) 125 L/min **[ F ]**
*Reason: GFR is 125 mL/min (≈ 180 L/day).*

### Q4. Administration of ADH

a) Decreases water loss by lungs **[ F ]**
*Reason: ADH acts on renal collecting ducts.*

b) Decreases water loss by kidney **[ T ]**

c) Increases perspiration **[ F ]**
*Reason: ADH does not affect sweating.*

d) Decreases reabsorption of $Na^+$ by PCT **[ F ]**
*Reason: ADH does not act on PCT.*

e) Increases reabsorption of water by distal tubule **[ T ]**
*Reason: ADH acts on late DCT and collecting duct.*

### Q5. Hormones acting via gene transcription are

a) Cortisol **[ T ]**
*Reason: Cortisol (glucocorticoid) supports breast development.*

b) Aldosterone **[ T ]**
*Reason: Aldosterone retains $Na^+$ and water.*

c) Insulin **[ F ]**
*Reason: Insulin increases glucose facilitated diffusion 10–20 fold (GLUT-4).*

d) Progesterone **[ T ]**

e) TSH **[ F ]**
*Reason: TSH acts via cAMP.*

**Q6. The following are sensory receptors with their respective sensations**

a) Free nerve endings: pain **[ T ]**
b) Pacinian corpuscles: cold sensation **[ F ]**
*Reason: Pacinian corpuscles detect vibration/pressure; cold uses cold receptors.*
c) Osmoreceptors: blood $PCO_2$ **[ F ]**
*Reason: Osmoreceptors detect osmolality; chemoreceptors detect $PCO_2$.*
d) Olfactory epithelium: smell **[ T ]**
e) Baroreceptor: blood pressure **[ T ]**

**Q7. Aldosterone on renal tubule normally reabsorbs**

a) All filtered glucose **[ F ]**
*Reason: Aldosterone does not reabsorb glucose; PCT does.*
b) All filtered plasma proteins **[ F ]**
*Reason: Liver SYNTHESISES plasma proteins; it does not store them.*
c) About 5–10% of filtered water **[ T ]**
d) More $K^+$ ion than $Cl^-$ ion **[ F ]**

**Q8. Aldosterone causes**

a) A fall in urinary sodium excretion **[ T ]**
b) Weight gain **[ T ]**
c) A fall in serum chloride level **[ F ]**
*Reason: Aldosterone retains $Na^+$ (and $Cl^-$ follows); serum $Cl^-$ does not fall.*
d) Increased intracellular fluid volume **[ F ]**
*Reason: Aldosterone expands ECF, not ICF.*
e) Increased potassium excretion **[ T ]**

**Q9. Ovulation occurs**

a) Before the rise in temperature **[ T ]**
*Reason: BBT rises after ovulation.*
b) Before the LH surge **[ F ]**
*Reason: Ovulation follows the LH surge by ~16–24 h.*
c) Following follicular ripening by FSH **[ T ]**
*Reason: FSH ripens follicles; LH surge triggers ovulation.*
d) Infrequently in women with amenorrhoea **[ T ]**
*Reason: Hypopituitarism → loss of FSH/LH → amenorrhoea.*
e) After the disappearance of cervical mucus ferning **[ F ]**
*Reason: Ferning is maximal at ovulation.*

**Q10. Human growth hormone**

a) Comes from lactotropes **[ F ]**
*Reason: GH comes from somatotropes.*
b) Is a steroid hormone **[ F ]**
*Reason: GH is a protein hormone.*
c) Acts as an enzyme **[ F ]**
*Reason: GH is a hormone, not an enzyme.*
d) Acts as a protein sparer **[ T ]**
*Reason: GH spares protein.*
e) Causes diabetes insipidus **[ F ]**
*Reason: GH excess can cause diabetes mellitus (diabetogenic), not DI.*

**Q11. The cerebellum**

a) Initiates all movements **[ F ]**
*Reason: Cerebellum coordinates/ modifies; it does not initiate movement.*
b) Is essential for finely coordinated movements **[ T ]**
c) Has afferent input from the visual cortex **[ F ]**
*Reason: Cerebellum receives visual info indirectly via pons.*
d) Has afferent input from the muscle proprioceptors **[ T ]**
*Reason: Spinocerebellar tracts carry proprioception.*
e) Controls contralateral muscle tone **[ F ]**
*Reason: Each cerebellar hemisphere controls ipsilateral muscles.*

**Q12. A reflex action**

a) Is initiated at a sensory receptor organ **[ T ]**
*Reason: A reflex starts at a receptor.*

b) May result in endocrine secretion **[ T ]**
*Reason: Reflexes may cause hormone secretion (milk ejection).*

c) Involves transmission across at least two central nervous synapses **[ F ]**
*Reason: Monosynaptic reflexes have only one synapse.*

d) May be excitatory or inhibitory **[ T ]**
*Reason: Reflexes may excite or inhibit.*

e) Is independent of higher center in the brain **[ F ]**
*Reason: Reflexes are modified by higher centres.*

**Q13. During accommodation for near vision**

a) More light enters the eye **[ F ]**
*Reason: Pupillary constriction reduces light entry.*

b) The curvature of the cornea increases **[ F ]**
*Reason: Accommodation changes the LENS curvature, not cornea.*

c) Chromatic and spherical aberration is decreased **[ T ]**
*Reason: Small pupil reduces aberrations.*

d) The depth of focus increases **[ T ]**
*Reason: Constricted pupil increases depth of focus.*

e) The visual axes of the two eyes converge **[ T ]**
*Reason: Convergence accompanies near vision.*

**Q14. Growth hormone**

a) Promotes positive nitrogen and phosphorus balance **[ T ]**
*Reason: GH is anabolic.*

b) Secretion is under hypothalamic control **[ T ]**
*Reason: GHRH/GHIH regulate GH.*

c) Levels in blood are higher in children than in adults **[ T ]**
*Reason: GH is highest in childhood/puberty.*

d) Secretion surges during sleep **[ T ]**
*Reason: GH peaks in slow-wave sleep.*

e) Stimulates liver to secrete somatomedins **[ T ]**
*Reason: GH stimulates liver IGF-1 (somatomedins).*

**Q15. Thyroxine**

a) Is stored in the follicular cells as thyroglobulin **[ T ]**
*Reason: Hormone is stored as thyroglobulin in the colloid.*

b) Increases the resting rate of carbon dioxide production **[ T ]**
*Reason: Thyroxine raises BMR/$CO_2$ production.*

c) Is essential for normal development of the brain **[ T ]**
*Reason: Thyroxine is essential for brain development.*

d) Is essential for normal red cell production **[ F ]**
*Reason: Thyroxine has minor effects on erythropoiesis.*

e) Acts more rapidly than triiodothyronine **[ F ]**
*Reason: $T_3$ acts faster than $T_4$.*

**Q16. In the normal menstrual cycle**

a) Blood loss during menstruation averages around 100 mL **[ F ]**
*Reason: Menstrual loss averages 35–50 mL.*

b) The proliferative phase depends on estrogen secretion **[ T ]**
*Reason: Proliferative phase depends on estrogen.*

c) Cervical mucus becomes more fluid around the time of ovulation **[ T ]**
*Reason: Cervical mucus becomes thin around ovulation.*

d) Ovulation is followed by LH surge **[ F ]**
*Reason: LH surge precedes ovulation.*

e) Basal body temperature is higher after ovulation **[ T ]**
*Reason: Progesterone raises BBT by ~0.5 °C.*

**Q17. Superficial reflexes are**

a) Corneal reflex **[ T ]**
b) Knee jerk **[ F ]**
*Reason: Knee jerk is a deep tendon (stretch) reflex.*
c) Ankle jerk **[ F ]**
*Reason: Ankle jerk is a deep tendon (stretch) reflex.*
d) Cremasteric reflex **[ T ]**
e) Biceps jerk **[ F ]**
*Reason: Biceps jerk is a deep tendon reflex.*

**Q18. The fovea centralis**

a) Is not crossed by any major blood vessels **[ T ]**
*Reason: Fovea is avascular.*
b) Is the thickest part of retina **[ F ]**
*Reason: Fovea is the THINNEST part of the retina.*
c) Has higher visual acuity than other parts of retina **[ T ]**
*Reason: Fovea has the highest acuity.*
d) Lies on the temporal side of the optic disc **[ T ]**
*Reason: Fovea is on the temporal side.*
e) Lies at the center of the optic disc **[ F ]**
*Reason: The optic disc is the blind spot, nasal to the fovea.*

**Q19. Posterior column damage may impair**

a) Vibration sense **[ T ]**
b) Pain sense **[ F ]**
*Reason: Pain travels in the spinothalamic tract.*
c) The flexor plantar response to stimulation **[ F ]**
*Reason: Plantar reflex tests pyramidal tract, not the dorsal column.*
d) Touch sensation **[ T ]**
e) The ability to stand steadily with the eyes closed **[ T ]**

**Q20. Sympathetic stimulation causes**

a) Increased heart rate **[ T ]**
b) Increased sweating **[ T ]**
c) Dilatation of bronchi **[ T ]**
d) Constriction of pupil **[ F ]**
e) Glycogenolysis **[ T ]**